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1 Commutative Algebra

1.1 Basics

- Let $A \rightarrow B$ ring map, M, N be B -modules. We have a coequalizer diagram:

$$M \otimes_A B \otimes_A N \rightrightarrows M \otimes_A N \longrightarrow M \otimes_B N$$

This gives another interpretation for tensor products of modules over an A -algebra.

- (Nakayama). Let M f.g. R -module, if $J(R)M = M$, then $M = 0$.
- (In proving Nakayama): for a matrix B , there exists $\text{Adj } B$ such that $\text{Adj } B \cdot B = \det B \cdot I$.
- If there is a prime factorization for an element in the ring, then it is unique. The same holds for prime ideal factorizations. The converse is true in UFDs (former) or Dedekind domains (latter).
 - Geometrically, such decomposition gives a decomposition of $V(I)$ to irreducible components. Such a decomposition corresponds to $I = \bigcap \mathfrak{p}_i$ but not $I = \prod \mathfrak{p}_i$. It corresponds to a prime ideal factorization if any pair of prime ideals are coprime (which is true in a Dedekind domain).
- An ideal $I \subset A$ is primary if the zero-divisors A/I are all nilpotent. I primary implies that $\sqrt{I}$ is prime.
- One can show that prime ideals and primary ideals in $S^{-1}A$ correspond to those in A that do not intersect S . Also, one can show that all ideals in $S^{-1}A$ come from extension.
- We can define the maximal $\tilde{S}$ multiplicative closed, such that $\tilde{S}^{-1}A = S^{-1}A$: we define $\tilde{S} = \{a \in A, ab \in S \text{ for some } b\} = \{a \in A, \frac{a}{1} \text{ unit in } S^{-1}A\}$.
- The residue field $\kappa(\mathfrak{p}) = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} = A/\mathfrak{p}A_{\mathfrak{p}} \setminus 0 = QF(A/\mathfrak{p})$, which follows because localization commutes with taking quotients.
- Localization at a larger set “overloads” the localization at a smaller set. For example, $S \subset T$ implies $(A_S)_T = A_T$. In particular this applies to localization at prime ideals.
- $\text{Supp } M = V(\text{ann } M)$ in $\text{Spec } A$ if M is finitely generated:

- Localization commutes with finite intersections but not intersections in general; in particular, $(\text{ann } M)_{\mathfrak{p}} = (\bigcap \text{ann } m_i)_{\mathfrak{p}} = \bigcap (\text{ann } m_i)_{\mathfrak{p}} = \bigcap (\text{ann } (m_i)_{\mathfrak{p}}) = \text{ann } M_{\mathfrak{p}}$.
- Therefore, $M_{\mathfrak{p}} = 0$ iff $\text{ann } M_{\mathfrak{p}} = A_{\mathfrak{p}}$, iff $(\text{ann } M)_{\mathfrak{p}} = A_{\mathfrak{p}}$, iff $\text{ann } M \not\subset \mathfrak{p}$.
- This is false in general. For example, consider $A = k[x_1, x_2, \dots]$, $M = \bigoplus A/(x_i)$, then $M_{(0)} = 0$, although $\text{ann } M = 0$ as well.
- In particular, if $x \neq 0$ in $A_{\mathfrak{p}}$, then $\text{ann } x \subset \mathfrak{p}$.
- Recall that a f.g. A -module can be viewed as a quasi-coherent sheaf that is locally of finite type over $\text{Spec } A$, and it is zero iff every fiber is zero.
 - Let $f : A \rightarrow B$ ring map. M finite B -module. If $M \otimes \kappa(\mathfrak{p}) = 0$ for every $\mathfrak{p} \in \text{Spec } f(\text{Spec } B)$, then $M = 0$.
 - $M \otimes \kappa(\mathfrak{p})$ depicts the fiber over $\mathfrak{p}$ of the pushforward of M . Geometrically, the locally finite type quasi-coherent sheaf itself is zero iff every fiber of the pushforward is zero.
 - In general this does not hold for arbitrary quasi-coherent sheaf. Also in general this does not hold if we restrict to only the maximal ideals $\mathfrak{p}$.
- Maximal ideals in $k[x_1, \dots, x_n]$ are in the form of $\mathfrak{m} \cap k[x_1, \dots, x_n]$, where $\mathfrak{m}$ is maximal in $\bar{k}[x_1, \dots, x_n]$. In general, if $A \rightarrow B$ is integral, then the induced map on spectrum is surjective for maximal ideals.
 - For example, maximal ideals in $\mathbb{R}[x, y]$ are in the form $(x - a, y - b)$ or $(x - a, y^2 + by + c)$ where each quadratic is irreducible over $\mathbb{R}$.
 - Note that for example $(x^2 + 1, y^2 + 1)$ is not maximal, since $\mathbb{R}[x, y]/(x^2 + 1, y^2 + 1) = \mathbb{C}[y]/(y^2 + 1) = \mathbb{C} \times \mathbb{C}$ is not a field.
- An element algebraic over $k = QF(A)$ does not imply it is integral over A . For example, in $k[x, x^{-1}]$, x^{-1} is algebraic over $k(x)$ but is not integral over $k[x]$.

1.2 Flatness

- Flatness is detected locally: if M is a B -module, B an A -algebra, then M is flat over A iff $M_{\mathfrak{p}}$ is flat over $A_{\mathfrak{p}}$, where $P \cap A = \mathfrak{p}$. In particular we can apply to when $B = A$.
- Let $f : A \rightarrow B$, the fibre of $\text{Spec } f$ over $\mathfrak{p} \in \text{Spec } A$ corresponds to $\text{Spec } B \otimes \kappa(\mathfrak{p})$.
 - Note that $B \otimes \kappa(\mathfrak{p}) = B \otimes A/\mathfrak{p} \otimes A_{\mathfrak{p}} = (B/\mathfrak{p}B)_{\mathfrak{p}} = (B/\mathfrak{p}B)_{f(A \setminus \mathfrak{p})}$. Therefore, $\text{Spec } B \otimes \kappa(\mathfrak{p})$ contains exactly those prime ideals that contain $f(\mathfrak{p})$ and does not intersect $f(A \setminus \mathfrak{p})$.
- M is flat iff $I \otimes M \rightarrow IM$ is an isomorphism for each ideal I .
 - This implies that over a PID A , a module is flat iff it is torsion-free:

$$\begin{array}{ccc} A \otimes M & \longrightarrow & I \otimes M \\ \downarrow & \swarrow & \\ IM & & \end{array}$$

- Submodules of flat modules are not necessarily flat. For example, $I \rightarrow A$ is always injective, but $I \otimes_A I \rightarrow I \otimes A = I$ might not be.
- Let M flat, then tensoring with M preserves kernel and image: consider tensoring M with

$$0 \longrightarrow \ker f \longrightarrow N \longrightarrow N' \longrightarrow \text{coker } f \longrightarrow 0$$

we then get $\ker f \otimes M = \ker f_M$ and $\text{coker } f \otimes M = \text{coker } f_M$ (which implies $\text{im } f \otimes M = \text{im } f_M$).

- A Dedekind domain is an ID where localizations at prime ideals are PIDs. This and the localness of flatness implies that an A -module is flat iff it is torsion-free.
 - Torsion-freeness is also a local property.
- (Faithfully Flatness): A flat A -module M is faithfully flat if one of the followings holds:
 - $M \neq \mathfrak{m}M$ for every $\mathfrak{m}$.
 - $M \otimes N = 0$ implies $N = 0$.
 - $N_1 \otimes M \rightarrow N_2 \otimes M$ isomorphism implies $N_1 \rightarrow N_2$ isomorphism.
- Examples: A_S is flat over A but not faithfully flat, unless S contains only units. $k[x]$ is not flat over $k[x^2, x^3]$.
- A ring map is faithfully flat iff it induces surjection on Spec or mSpec.
- Another property for flat modules: If B flat A -module, M f.p. A -module, N A -module, then

$$\mathrm{Hom}(M, N) \otimes B \rightarrow \mathrm{Hom}(M \otimes B, N \otimes B)$$

is an isomorphism.

- Minimal basis and basis are in general different (a basis has to be linearly independent). However, if A is local and M is finite flat A -module, then minimal basis in M corresponds to basis in $M/\mathfrak{m}M = k^n$, and is itself a basis. Therefore over a local ring free, flat, and faithfully flat are the same if the module is finitely generated.
- A f.p. module is flat iff it is projective.

1.2.1 Pure Submodules

- $N \subset M$ is a pure submodule if $N \otimes E \rightarrow M \otimes E$ is injective for every A -module E . Equivalently, if a system of linear equations $\sum a_{ij}x_j = n_i$ has solutions in M , then it also has solutions in N .
- Geometrically, N is a pure submodule of M if not only as quasi-coherent sheaves the fibres are subsets, but also this holds up to any base change.

1.3 Chain Conditions

- For modules, Noetherian and Artinian are not related ($W = \{\frac{a}{p^n} \in \mathbb{Q}\}$ is not Noetherian but is Artinian). For rings, Artinian implies Noetherian:
 - An Artinian ring has a finite number of maximal ideals. Let $I = \mathfrak{p}_1 \dots \mathfrak{p}_n$, we have a composition series: $A \supset \mathfrak{p}_1 \supset \dots \supset I \supset I\mathfrak{p}_1 \supset \dots \supset I^s = 0$.
 - An Artinian ring is a dim 0 Noetherian ring, where we show the local case first.
- In fact, if an A -module M is Noetherian, then $A/\mathrm{ann}(M)$ is a Noetherian ring.
 - WLOG assume M faithful. Assume $M = Ax_1 + \dots + Ax_n$. $A = \{(ax_1, \dots, ax_n)\} \subset M^n$ is a submodule and is Noetherian.

1.4 Completion

- Given an Abelian group G , we can make G a topological group by considering filtration (G_n) , where $G_0 \supset G_1 \supset \dots$
 - Note that to turn G into a topological group, it suffices to define a basis at 0 and translate it to every $g \in G$, where the basis satisfies that for every V , there exists $V_1 - V_2 \subset V$ in the basis.
 - Two filtration define the same topology if they are “included” in each other.
 - We can do the same for rings and modules. In particular, we have I -adic topology on A and M by considering (I^n) and $(I^n M)$.
 - G is Hausdorff iff $\bigcap G_n = 0$.
- The completion of G to be $\hat{G} = \lim G/G_n$, where $G/G_n \rightarrow G/G_{n-1}$ are naturally induced maps. Let $\hat{G}_n = \{(g_m), g_m = 0 \text{ for } m \leq n\}$, then $G/G_n = \hat{G}/\hat{G}_n$. (This implies that the completion of completion does not do anything.)
 - A completion is complete and Hausdorff, and uniquely factors through any map $G \rightarrow K$, K complete and Hausdorff.
 - Examples include: p -adic completion $\mathbb{Z}_p$; $\mathfrak{n}$ -adic completion of $A[x_1, \dots, x_n] = A[[x_1, \dots, x_n]]$.
- Let $H \subset G$ be a subgroup. In general $\hat{H} = \overline{\phi(H)}$, and it is not true that $\hat{G} \rightarrow \widehat{G/H}$ is surjective. However, this is true if the index set is $\mathbb{Z}_+$.
- (Hensel’s lemma): Suppose $(A, \mathfrak{m}, k)$ is $\mathfrak{m}$ -adic complete. Suppose $F \in A[x]$ can be factored into gh in $k[x]$, where $(g, h) = 1$, then there exists $G, H \in A[x]$ such that $F = GH$, $\bar{G} = g$, $\bar{H} = h$.
 - We sort of construct G_n, H_n inductively such that $F - G_n H_n \in \mathfrak{m}^n$. Now, completeness guarantees that $G = \lim G_n$ and $H = \lim H_n$ exist.
- (Artin-Rees): Let A Noetherian, M f.g., $N \subset M$, then there exists n_0 such that $n \geq n_0$ implies:

$$(I^{n+1}M \cap N) = I(I^n M \cap N).$$

- This follows from the more general fact that the restriction of a stable I -filtration is still stable (given finiteness conditions).
- A corollary: given the finiteness conditions, M is complete iff A is complete w.r.t. the I -adic topology.
- Let A Noetherian, the I -adic completion $\hat{A}$ is flat over A .
- A special example: Let A be the ring of real analytic functions at 0, then the (x) -completion of A is $\mathbb{R}[[x]]$.
 - Note that $A/(x^n) = \mathbb{R}[x]/(x^n)$, and $\mathbb{R}[[x]]$ is exactly the (x) -completion of $\mathbb{R}[x]$.
- Suppose that M is an A -module with I -adic topology. A submodule N is closed if $N = \bigcap (N + I^n M)$. Indeed, the right hand side depicts points that are arbitrarily close to N , i.e., in the closure of N .

1.5 Dimension Theory

- If $A = k[\alpha_1, \dots, \alpha_n]$ satisfies that $\text{tr. deg}_k A > 0$, then A is not a field. This implies that $k[x_1, \dots, x_n]/\mathfrak{m}$ is an algebraic extension of k .
 - This implies that we have an embedding $\theta : k[x_1, \dots, x_n]/\mathfrak{m} \rightarrow \bar{k}$.
- Let $(A, \mathfrak{m})$ be Noetherian local, $f \in \mathfrak{m}$, then $\dim A/fA = \dim A - 1$ if f is not in any minimal prime ideal (the closed irreducible sets are not altered by modulo f).
- Over Noetherian rings, transcendence degree behaves well w.r.t. dimension. i.e., $\text{ht } \mathfrak{n} = \text{ht } \mathfrak{m} + n$ if $\mathfrak{n}$ is a maximal ideal of $A[T_1, \dots, T_n]$ whose restriction to A is $\mathfrak{m}$. In particular, $\dim A[T_1, \dots, T_n] = \dim A + n$.
- If $A \rightarrow B$ is an integral extension, then $\dim A = \dim B$. In particular, for f.g. ID over k , this shows $\text{tr. deg } QF(A) = \dim A$.
- For f.g. ID A over k , $\dim A/\mathfrak{p} = \dim A - 1$ for $\mathfrak{p} \in \text{Spec } A$ of height 1.
- Let $\text{gr } A = \bigoplus I^n/I^{n+1}$. If A is f.g., then $\text{gr } A = (A/I)[\xi_1, \dots, \xi_n]$ which is f.g..
- Recall that if M_n is a f.g. R_0 -module, with R_0 Artinian, then M_n has finite length. We define the Hilbert series to be

$$P(M, t) = \sum l(M_n)t^n.$$

If R Noetherian with r generators over R_0 , we can write $P(M, t) = \frac{f(t)}{\prod_{i=1}^r (1-t^{d_i})}$ where a generator x_i is of degree d_i .

- A corollary, if $R = R[x_i]$ with $\deg x_i = 1$, we can express $l(M_n)$ in some polynomial $\varphi_M(n)$ for all large enough n . φ_M is the Hilbert polynomial, and $d(M) = \deg \varphi_M + 1$.
- (Samuel Function). Essentially the same as Hilbert function, except that we are working over a Noetherian semilocal ring and we pass to the graded ring and module $\text{gr}_I(A)$ and $\text{gr}_I(M)$ so that the generators are of $\deg 1$, so that we can talk about the Hilbert polynomial and $d(M)$. In particular $\chi_M^I(n) = l(M/I^{n+1}M)$ and $\deg \chi_M^I = d(M)$.
- Let $\delta(M)$ be the min value $x_1, \dots, x_n \in \mathfrak{m} = \text{Jacobson radical of } A$ such that $l(M/\sum x_i M) < \infty$. Still, let A be a semilocal Noetherian ring and M finite A -module. We have $\dim M = d(M) = \delta(M)$.
- Let A Noetherian. $\dim M = 0$ iff M is of finite length.
 - Recall $\dim(M) = \dim A/\text{ann}(M) =: \dim B$. If B is of $\dim 0$ then B is Artinian, so M is f.g. iff $l(M) < \infty$.
 - Conversely, suppose M is of finite length, then $\text{Supp } M$ is a finite set of maximal ideals since $A/\text{ann}(M)$ is Artinian, so $\dim M = 0$ since $V(\text{ann } M) = \text{Supp } M$.
- (Principle Ideal Theorem). Let A Noetherian and $I = (x_1, \dots, x_d)$, then if $\mathfrak{p}$ is a minimal prime ideal over I , then $\text{ht } \mathfrak{p} \leq d$. In particular, $A_{\mathfrak{p}}$ is finite dimensional.
- (Regularity). Let $(A, \mathfrak{m})$ Noetherian local. Recall that if $\mathfrak{m}$ is generated by less than r elements, then $\text{ht } \mathfrak{m} = \dim A < r$. Now, suppose $\dim A = r$, then A is regular if $\dim_{A/\mathfrak{m}} \mathfrak{m}/\mathfrak{m}^2 = \dim A$, in particular, iff $\mathfrak{m}$ is generated by r elements.
 - A $\dim 1$ regular local ring is just a DVR. In general, a regular local ring is an ID.
 - Let A $\dim d$ regular local, then $\text{gr}_{\mathfrak{m}} A \cong k[X_1, \dots, X_d]$.
 - If the graded ring $\text{gr}_{\mathfrak{m}} A$ is non-reduced, $\text{Spec } A$ is not smooth at $\mathfrak{m}$.

- Recall that $\dim B \otimes \kappa(\mathfrak{p}) = \sup_{\varphi^{-1}(P)=\mathfrak{p}} \dim B_P/PB_P$. Also, $\dim B_P/\mathfrak{p}B_P$ is the supremum of the length of chains of prime ideals $P_0 \subset \dots \subset P_n = P$ such that all of them lie over $\mathfrak{p}$.
- Suppose $\varphi : A \rightarrow B$ is flat between Noetherian rings, or more generally the going-down property holds, then $\text{ht } P = \text{ht } \mathfrak{p} + \dim B_P/\mathfrak{p}B_P$. In particular, if A, B Noetherian local, then $\dim B = \dim A + \dim B/\mathfrak{m}B$.
- Let $\varphi : A \rightarrow B$ between Noetherian rings. If $\mathfrak{p} \supset \mathfrak{q}$ in $\text{Spec } A$, and going-up (going-down) holds, then $\dim B \otimes \kappa(\mathfrak{p}) \geq \dim B \otimes \kappa(\mathfrak{q})$ ($\dim B \otimes \kappa(\mathfrak{p}) \leq \dim B \otimes \kappa(\mathfrak{q})$).

1.6 Associated Prime Ideals

- Minimal elements in $\text{Ass } M$ correspond to irreducible components of the support of M as a quasi-coherent sheaf; i.e., $\mathfrak{p} \in \text{Ass}(M)$ corresponds to $V(\mathfrak{p})$.
 - If we consider $\text{Ass}(M_S)$, this is exactly $\text{Ass}(M) \cap \text{Spec}(A_S)$: i.e., the support of M intersecting with the subset $\text{Spec } A_S$.
 - In particular, $\mathfrak{p} \in \text{Ass}(M)$ iff $\mathfrak{p}A_{\mathfrak{p}} \in \text{Ass}(M_{\mathfrak{p}})$.
 - As a result, minimal elements of $\text{Ass}(M)$ coincide with minimal elements of $\text{Supp}(M)$.
- We define primary submodules of M in the same fashion as for primary ideals: if a is an annihilator of M/N , then $a \in \sqrt{\text{ann}(M/N)}$.
 - Given finiteness conditions, $N \subset M$ is primary iff $\text{Ass}(M/N) = \{\mathfrak{p}\}$.
 - This is saying that a primary submodule is where $\text{Supp}(M/N)$ is an irreducible set (though may not be an irreducible component).
 - Such submodule N is called $\mathfrak{p}$ -primary.
 - This also explains why irreducible submodules are primary.
- Explicitly, let $\mathfrak{p}_i$ be the minimal primes over I . There are $\mathfrak{p}_i$ -primary ideals q_i such that $I = \bigcap q_i$.

1.7 Integral Extensions

- Suppose that $E \subset R \subset F$ and F/E is algebraic. Then R is itself a subfield.
 - For finite extension case we can consider $\times_a : R \rightarrow R$ for any $a \in R$. In general an explicit construction works.
- Let $A \rightarrow B$ be an integral inclusion. Then if B is a field, so is A . This implies that if $A \rightarrow B$ is integral, then $\text{Spec } \varphi$ sends closed points to closed points.
 - Being an integral extension is a local property: if $A \rightarrow B$ is integral, then so is $A_{\mathfrak{p}} \rightarrow B \otimes A_{\mathfrak{p}}$.
 - Also, if φ is injective, then $B \otimes A_{\mathfrak{p}} = \varphi(A \setminus \mathfrak{p})^{-1}B \neq 0$. In particular, if φ is injective and integral then $\text{Spec } \varphi$ is surjective.
- Finiteness (as a module) over a ring corresponds to integral extension.
- (Going Up/Going Down). If B/A is integral, then going-up holds. If furthermore B is an ID and A is integrally closed, then going-down holds.
 - Key for proving going-down: If A integrally closed in its field of fractions K . L a normal field extension of F , then if B is the integral closure of A in L , then all prime ideals in B lying over $\mathfrak{p}$ are conjugates over K .

- This creates, say given $P' \in \text{Spec } B$ over $\mathfrak{p}'$, a $P \in \text{Spec } B$ over $\mathfrak{p} \subset \mathfrak{p}'$, where we consider a larger extension ring that contains B and is the integral closure of A in some normal extension of $K = QF(A)$, and apply going-up theorem to the integral extension C/B .
- Flatness gives going-down theorem.
- If B/A is integral, then so is $B[x_1, \dots, x_n]/A[x_1, \dots, x_n]$.

1.8 Valuation Rings

- If $A \subset K$ is a subring, then $B = \bigcap R$, where $R \supset A$ valuation rings, is the integral closure of A .
- Let $A \subset K$ valuation ring. We have a canonical valuation: $v : K \rightarrow G \cup \{\infty\}$ with $G = \{xR, x \in K^\times\}$, where we define $v(0) = \infty$ and $xR \leq yR$ iff $xR \supset yR$. We then have $R_v = \{x \in K, v(x) \geq 0\} = R$. This is not unique but unique up to some order preserving isomorphism on the codomain.
- Characterization for DVR: dim 1 normal Noetherian local ring. For Dedekind domain: a field or a dim 1 Noetherian normal domain, or if every ideal is a finite product of prime ideals, or if all localizations at prime ideals are DVRs.
- One way for constructing Dedekind domain: let A dim 1 Noetherian ID, $K = QF(A)$, L/K algebraic, $B = \bar{A}$ in L , then B is Dedekind.
- Suppose A normal Noetherian ID, then $A = \bigcap_{\text{ht } \mathfrak{p}=1} A_{\mathfrak{p}}$.
- A valuation ring or a Dedekind domain is a Prüfer domain: a module over the ring is flat iff torsion-free.
- A Krull ring A satisfies $A = \bigcap_{\text{ht } \mathfrak{p}=1} A_{\mathfrak{p}}$, and that each $A_{\mathfrak{p}}$ DVR, and that every $x \in A$ lives in at most a finite number of $\mathfrak{p}$ with height 1.
 - Essentially, this is saying that $\text{Spec } A$ is reconstructed from codim 1 data, and that principal divisor makes sense.
 - A Dedekind domain is exactly a dim 1 Krull ring.
 - As noted earlier, a normal Noetherian ID is a Krull ring.
- Suppose $A \rightarrow B$ turns B to a finite A -module, then fibers of the spectrum map is finite. The fiber is given as $\text{Spec } B \times_{\text{Spec } A} \{x\}$, which corresponds to $\text{Spec } B \otimes_A k$ and this is a finite k -vector space.

1.9 Regular Sequences

- (Regular and Quasi-Regular Sequence). Let M A -module, $(a_1, \dots, a_n)$ is a M -sequence if $a_i x \neq 0$ for any $x \in M / \sum_{j < i} a_j M$ and that $M / \sum a_i M \neq 0$. Let $I = \sum a_i A$. The sequence is a M -quasi-regular sequence if for any $F \in M[X_1, \dots, X_n]$ of homogeneous degree ν , $F(a) \in I^{\nu+1}M$ implies all coefficients of F are in IM .
 - Note that $F(a)$ denotes an element in $I^\nu M$, so this is saying if $\sum x_i m_i \in I^{\nu+1}M$ where $x_i \in I^\nu M$, then $m_i \in IM$.
 - M -sequences are M -quasi-regular sequences.
 - A sequence is M -quasi-regular iff the map $\varphi : M/IM[x_1, \dots, x_n] \rightarrow \bigoplus I^\nu M / I^{\nu+1}M$ is an isomorphism, where f homogeneous of degree ν is sent to the class of $F(a)$ in $I^\nu M / I^{\nu+1}M$.

- Geometrical meaning of regular sequences: $f \in A$ is a non-zero divisor iff $f \notin \mathfrak{p}$ for every $\mathfrak{p} \in \text{Ass}(A)$. Correspondingly, $\dim X_j \cap V(f) < \dim X_j$ for each associated component X_j . Generalizing this, a sequence is regular if for each i , $\dim X_j \cap V(f_1) \cap \dots \cap V(f_i) < \dim X_j \cap V(f_1) \cap \dots \cap V(f_{i-1})$ for each associated component.
 - We later see that for A Noetherian local, $\dim A/\mathfrak{p} \geq \text{depth } M$ if $\mathfrak{p} \in \text{Ass}(M)$. Indeed, consider the associated component $V(\mathfrak{p})$. A regular sequence of length r in $\mathfrak{m}$ means that we can gradually reduce the dimension of $V(\mathfrak{p})$, each step by at least 1. This implies $\dim V(\mathfrak{p}) \geq \text{depth}(M)$.
 - Essentially, a system of parameter reduces the dimension by 1 each step, and a regular sequence reduces the dimension of each associated component by at least 1 at each step.
 - More generally, $\dim M = \sup \dim A/\mathfrak{p} \geq \inf \dim A/\mathfrak{p} \geq \text{depth } M$.
- (Koszul Complex). Let $x_1, \dots, x_n \in A$, we define $K_p = \bigoplus A e_{i_1, \dots, i_p}$ for $p \leq n$, and the boundary maps $d(e_{i_1, \dots, i_p}) = \sum (-1)^{r-1} x_{i_r} e_{i_1, \dots, \hat{i}_r, \dots, i_p}$ and the homology groups correspondingly. In particular, $H_0(\underline{x}, M) = M / \sum x_i M$ and $H_n(\underline{x}, M) \cong \{\xi, x_i \xi = 0\}$. In general $(x_1, \dots, x_n)$ annihilates all $H_p(\underline{x}, M)$. If $x_1, \dots, x_n$ is regular then $H_p(\underline{x}, M) = 0$ for all p .
- Let A Noetherian and M finite A -module. $IM \neq M$. $\text{Ext}_A^i(N, M) = 0$ for all finite A -module N with $\text{Supp}(N) \subset V(I)$ and $i < n$ iff there exists an M -sequence of length n in I .
- In fact, we have $n =: \text{depth}(I, M) = \inf\{i, \text{Ext}_A^i(A/I, M) \neq 0\}$. For Noetherian local ring A , we write $\text{depth } M = \text{depth}(\mathfrak{m}, M)$.
 - In particular, suppose A Noetherian local, M and N finite A -modules, then $\text{Ext}_A^i(N, M) = 0$ for $i < k - r$, where $k = \text{depth } M$ and $r = \dim N$.
- Let A Noetherian, M finite A -module with $M \neq IM$, where $I = (y_1, \dots, y_n)$, then $\text{depth}(I, M)$ is the number of successive zero terms from the left in $H_n(\underline{y}, M), H_{n-1}(\underline{y}, M), \dots, H_0(\underline{y}, M)$.

1.10 Cohen-Macaulay Rings

- Essentially this is some sort of equi-dimensional condition: in general $\text{depth } M \leq \dim M$, and M is a CM module if $\text{depth } M = \dim M$. This implies that M has no embedded associated primes. A CM local ring A satisfies $\text{ht } I + \dim A/I = \dim A$.
- Equivalently, A is a CM ring if all ideals I satisfy unmixed condition, i.e., $\text{Ass } A/I = \min A/I$, or if there is no embedded prime divisors of I .
- A ring is CM iff the regular sequences are the same as (part of) systems of parameter.
- Note that if A is a Noetherian local ring, then a maximal regular sequence in $\mathfrak{p}$ extends to a maximal regular sequence in $\mathfrak{m}$. In particular, $\text{depth } A/(x_1, \dots, x_r) = \text{depth } A - r$ if $(x_1, \dots, x_r)$ is a regular sequence.

2 Algebraic Geometry

2.1 Sheaves and Ringed Spaces

- Stalk-wise injectivity/surjectivity for presheaf maps does not imply section-wise. On the other hand, stalk-wise injectivity implies section-wise injectivity for sheaf maps.
 - In fact, we might just define surjective (pre)sheaf maps to be those surjective on stalks.
- $\mathcal{F}$ is a sheaf iff the sequence $0 \rightarrow \mathcal{F}(U) \rightarrow \prod \mathcal{F}(U_i) \rightarrow \prod \mathcal{F}(U_i \cap U_j)$ is exact.
- Two ways of sheafification:

- $\mathcal{F}^+(U) = \{f : U \rightarrow \bigsqcup \mathcal{F}_x \text{ locally compatible}\}.$
- $\mathcal{F}^+(V) = \text{colim}_{\mathcal{V}} \mathcal{F}_{\mathcal{V}}(V),$ where $F_{\mathcal{V}}$ is the kernel of the above sequence.
- Gluing sheaves: let $\mathcal{F}_i$ be a sheaf on U_i , where (U_i) open cover. Let (φ_{ij}) isomorphism between $\mathcal{F}_i|_{U_i \cap U_j}$ and $\mathcal{F}_j|_{U_i \cap U_j}$ satisfying the cocycle conditions. There exists a gluing $\mathcal{F}$ such that $\phi_i : \mathcal{F}|_{U_i} \cong \mathcal{F}_i$ isomorphisms with $\varphi_{ij} \circ \phi_i = \phi_j.$
- (Support of a sheaf): $\text{Supp } \mathcal{F} = \{x, \mathcal{F}_x \neq 0\}.$ This is in general not closed.
 - If $\text{Supp } \mathcal{F}$ is a finite number of closed points, then $\mathcal{F}$ is a skyscraper sheaf, with $\mathcal{F}(U) = \bigoplus_{x \in U} \mathcal{F}_x.$ (Consider $\mathcal{F}(U)$ as the locally compatible tuples in $\prod_{x \in U} \mathcal{F}_x.$)
- We can glue morphisms between ringed spaces in the sense that if $f_i : U_i \rightarrow Y$ morphisms that agree on intersections, we have unique $f : X \rightarrow Y.$
- (Sheaf of ideal): $\mathcal{J}$ is a sheaf of ideal if $\mathcal{J}(U) \subset \mathcal{O}_X(U)$ is an ideal for every $U.$ Let $V(\mathcal{J}) = \{x \in X, \mathcal{J}_x \neq \mathcal{O}_{X,x}\}$ with structure sheaf $j^{-1}(\mathcal{O}_X/\mathcal{J}).$
- A morphism $f : X \rightarrow Y$ between ringed spaces is closed/open immersion if $\mathcal{O}_{Y,f(x)} \rightarrow \mathcal{O}_{X,x}$ is surjective/isomorphism.
 - $j : V(\mathcal{J}) \rightarrow X$ is a closed immersion.
 - We can factor a closed immersion into an isomorphism with $V(\mathcal{J})$ followed by the canonical immersion.
- (Quotient of ringed spaces): let G acts on $X.$ We define $X/G =: Y$ by $\mathcal{O}_Y(U) = (\mathcal{O}_X(p^{-1}(U)))^G.$
 - For schemes quotient might not exists (and can be different from the quotient as ringed spaces).

2.2 Schemes

- Consider $f : \text{Spec } \mathbb{Z}[t] \rightarrow \text{Spec } \mathbb{Z}.$ $\text{Spec } \mathbb{Z}[t] = f^{-1}(\{0\}) \cup \bigcup f^{-1}(p\mathbb{Z}).$
 - If $\mathfrak{p} \in f^{-1}(\{0\}),$ then $\mathfrak{p} \cap \mathbb{Z} = \{0\}.$ In particular, we can consider the localization $S^{-1}\mathfrak{p}$ in $S^{-1}\mathbb{Z}[t] = \mathbb{Q}[t],$ where $S = \mathbb{Z} \setminus \{0\}.$ Therefore, $f^{-1}(\{0\})$ correspond to $\text{Spec } \mathbb{Q}[t].$
 - If $\mathfrak{p} \in f^{-1}(p\mathbb{Z}),$ then $(p) = \mathbb{Z} \cap \mathfrak{p}.$ We can then correspond $\mathfrak{p}$ with a prime ideal in $\mathbb{Z}/(p)[t] = \mathbb{F}_p[t].$ Therefore, $f^{-1}(\{p\})$ correspond to $\text{Spec } \mathbb{F}_p[t].$
 - We conclude that $\text{Spec } \mathbb{Z}[t] = \text{Spec } \mathbb{Q}[t] \cup \bigcup \text{Spec } \mathbb{F}_p[t],$ a family of affine lines.
- A scheme is a ringed space that is also locally affine schemes.
- Let $X_f = \{x \in X, f_x \in \mathcal{O}_{X,x}^\times\}.$ If X qcqs, then $\mathcal{O}_X(X)_f \rightarrow \mathcal{O}_X(X_f)$ is an isomorphism.
- A closed subscheme consists of a closed immersion $j : (Z, \mathcal{O}_Z) \rightarrow (X, \mathcal{O}_X).$ Affine locally, j is the same as $\text{Spec } A/J \rightarrow \text{Spec } A.$
 - For this one, we show qcqs of Z if X affine, and move to affine opens to conclude that $\ker j^\# = \ker i^\#$ where $i : \text{Spec } A/J \rightarrow \text{Spec } A$ (so $Z \cong \text{Spec } A/J$ is compatible with the immersions).
- Given Y affine $\text{Hom}(X, Y) \cong \text{Hom}(\mathcal{O}_Y(Y), \mathcal{O}_X(X)).$
 - If X is also affine, $f : X \rightarrow Y$ induces $\varphi : \mathcal{O}_Y(Y) \rightarrow \mathcal{O}_X(X),$ which further induces $f_\varphi : X \rightarrow Y,$ and we show that $f = f_\varphi$ by showing that $f^\# = f_\varphi^\#$ agree on stalks.
 - For general case, glue the morphisms $U_i \rightarrow Y$ on affine locals.
- A scheme over $\text{Spec } A$ is equivalent to a scheme with structure sheaf of A -algebras.

- Let X a scheme over k . A k -rational point of X corresponds to sections $\text{Spec } k \rightarrow X$. This is equivalent to picking $x \in X$ such that $\kappa(x) = k$.
 - For example, if $X = \text{Spec } k[T_1, \dots, T_n]/I$, then k -rational points correspond to maximal ideals with residue field k . This also corresponds to points in k^n that annihilate functions in I (if k algebraically closed).
 - Similarly, points in $\mathbb{P}^n$ corresponds to k -rational points in $\mathbb{P}_k^n$ via the identification sending $[\alpha_0 : \dots : \alpha_n] \rightarrow I = (\alpha_i T_j - \alpha_j T_i)$. Similarly, this corresponds to maximal ideals with residue field k .
 - Suppose $I = (P_1, \dots, P_m)$, then k -rational points of $\text{Proj } k[T_0, \dots, T_n]/I$ correspond to $Z_+(P_1, \dots, P_m) \subset \mathbb{P}^n$.
- Let B graded. $\text{Proj } B$ has basis $D_+(f)$, where f homogeneous, such that $D_+(f) \cong \text{Spec } B_{(f)} = \{\frac{a}{f^n}, \deg a = n \cdot \deg f\} \subset B_f$.
- For example, $\mathbb{P}_A^n = \text{Proj } A[x_1, \dots, x_n]$ with $B_{(T_i)} = A[\frac{T_1}{T_i}, \dots, \frac{T_n}{T_i}]$.
- Embedding $\text{Spec } k[\alpha_1, \dots, \alpha_n]$ as a dense open subscheme of a projective variety: consider $\text{Proj } k[\alpha_1, \dots, \alpha_n, X]$, with $\deg X = 1$ and $\deg \alpha_i = 0$.
- Recall that quasi-coherent sheaves of ideals correspond to closed subschemes of a scheme X . Now, let $f : X \rightarrow Y$ quasi-compact, let $\mathcal{J} = \ker f^\# : \mathcal{O}_Y \rightarrow f_* \mathcal{O}_X$, and let $Z = V(\mathcal{J})$, then Z is a scheme:
 - It suffices to show that $\mathcal{J}$ is quasi-coherent. Now,

$$\begin{aligned}
\mathcal{J}(D(g)) &= \ker \mathcal{O}_Y(D(g)) \rightarrow \mathcal{O}_X(f^{-1}(D(g))) \\
&= \ker A_g \rightarrow \mathcal{O}_X(\text{Spec } B_i \cap f^{-1}(D(g))) \\
&= \ker A_g \rightarrow \prod B_{i, \phi_i(g)} \rightarrow \prod B_{i, \phi_i(g) f_{ij}}
\end{aligned}$$

where φ_i induces $U_i \rightarrow U$. On the other hand, due to the *quasi-compactness* of f , we can commute finite product with localizations and conclude that $\mathcal{J}(U)_g$ equals the same kernel.

- In fact, Z is the scheme-theoretic closure of $f(X)$ in Y in the following sense: there exists $g : X \rightarrow Z$ that factors f , and for any other factorization, there exists a natural $Z \rightarrow Z'$ closed immersion that factors the factorization through this one.
- The categorical quotient in the category of schemes might not be geometrically meaningful: for example, let $\mathbb{G}_m$ acts on $\text{Spec } k[t]$ by $\lambda \cdot t = \lambda t$. Say k is algebraically closed. Then $\text{Spec } k[t]/\mathbb{G}_m = \text{Spec } k$ is a single point, while there are two orbits. It is not the orbit space in general.
- In general we might be collapsing several orbits into one point. See Mumford's work on geometric invariant theory for more information.

2.3 Reduced & Integral Schemes

- X is reduced if $\mathcal{O}_{X,x}$ is reduced for every $x \in X$. For any X , we define an associated reduced closed subscheme $X_{\text{red}} = (X, \mathcal{O}_X/\mathcal{N})$, where $\mathcal{N}(U) = \{s \in \mathcal{O}_X(U), s_x = 0, \forall x \in U\}$.
- Let X be an algebraic variety over a field k . For $f \in \mathcal{O}_X(X)$, we define $\tilde{f} : X^0 \rightarrow \bar{k}$ sending x to $f_x \in \kappa(x) \subset \bar{k}$. This defines $\mathcal{O}_X(X) \rightarrow F(X, k) = \text{Hom}(X^0, \bar{k})$ a group homomorphism.
 - Note that $\kappa(x)$ might not be a subfield of $\bar{k}$ if x is not closed. For example, if x is the generic point of $\text{Spec } k[x]$, then $\kappa(x) = k(T)$ is transcendental over k .

- In general, this gives $\mathcal{O}_X \rightarrow \mathcal{F}_X$, and the kernel is exactly the sheaf of ideals $\mathcal{N}$. In other words, this map is injective iff X is reduced.
- X is integral if it is reduced and irreducible. This implies that $\mathcal{O}_{X,x}$ is an integral domain for every x .
 - If X is integral, then $\mathcal{O}_X(U) \rightarrow \mathcal{O}_{X,x}$ and $\mathcal{O}_{X,x} \rightarrow \mathcal{O}_{X,\xi}$ are injective. In particular, $\mathcal{O}_X(U) = \bigcap \mathcal{O}_{X,x}$.
 - $K(X) = \mathcal{O}_{X,\xi}$ is called the ring of rational functions on X . f is regular at x if $f \in \mathcal{O}_{X,x}$.

2.4 Morphisms

- Fiber over y for $f : X \rightarrow Y$: $X_y = X \times_Y \text{Spec } \kappa(y)$. This is topologically homeomorphic to $f^{-1}(y)$. We can consider $X_y \rightarrow \text{Spec } \kappa(y)$ as a family of schemes over $\kappa(y)$ parametrized by y .
- Let X algebraic variety over k . For K/k algebraic, we denote X_K to be the base change via $\text{Spec } K \rightarrow \text{Spec } k$.
 - $\dim X_K = \dim X$.
 - If K/k separable and X reduced, so is X_K .
 - If K/k purely inseparable, $X_K \rightarrow X$ is a homeomorphism.
- (Separated Morphisms). X is separated over Y if $X \rightarrow X \times_Y X$ is a closed immersion, or if $\Delta(X)$ is closed. X is separated iff for every pair of open affines U, V , $U \cap V$ is affine and $\mathcal{O}_X(U) \otimes \mathcal{O}_X(V) \rightarrow \mathcal{O}_X(U \cap V)$ is surjective.
 - Essentially, because a closed immersion is affine and can be verified Zariski-locally.
 - The Valuative Criterion: Let $f : X \rightarrow Y$ quasi-separated. f is separated/universally closed/proper if for every $\text{Spec } V \rightarrow Y$ and $\text{Spec } K \rightarrow X$ Y -morphism where V valuation ring and $K = QF(V)$, the corresponding map $X_V(V) \rightarrow X_V(K)$ is injective/surjective/bijective.
- (Proper Morphisms). A map $f : X \rightarrow Y$ is universally closed if f is closed and so is any base change of f . $f : X \rightarrow Y$ is proper if it is of finite type, separated, and universally closed. This can also be verified locally.
- For affine schemes, proper implies finiteness.
 - By Artin-Tate & Noether Normalization: B is a f.g. A -module if it is a f.g. A -algebra and is integral over A .
 - Or more easily one can directly prove the above proposition by considering the maximal power x_i^d such that x_i^d not generated by $(x_i, \dots, x_i^{d-1})$ over A for each generator x_i .
 - More generally, if $X/\text{Spec } A$ is proper, then $\mathcal{O}_X(X)$ is integral over A .
- Let X proper over $\text{Spec } \mathcal{O}_K$, then $X_K(K) = X(\mathcal{O}_K)$.
- Any projective morphism is proper. Also, due to the Segre embedding, the family of projective morphisms satisfies the “good” properties. The same properties hold for quasi-projective morphisms (decomposed to an open immersion and a projective morphism).
- (Chow’s Lemma). Let Y Noetherian, any $X \rightarrow Y$ proper is a projective morphism after modifying outside of a dense open subset.
- In fact, separated morphisms can be detected on the universal reduced closed subscheme: $X \rightarrow Y$ is separated iff $X_{\text{red}} \rightarrow Y_{\text{red}}$ is. The analogous holds for proper morphisms, assume the map is of finite type to begin with.
- Finite and integral morphisms are affine local. Finite morphisms are of finite type, quasi-finite, and proper. Also, finite morphisms between affine schemes is projective.

2.5 Normal Schemes

- (Normal Scheme). X is normal if X is irreducible and $\mathcal{O}_{X,x}$ is normal for every x .
 - Let X quasi-compact, then it suffices to detect normality at closed points. (Note that for every x there exists $y \in \{x\}$ closed, and $\mathcal{O}_{X,x}$ is a localization of $\mathcal{O}_{X,y}$.)
 - Similarly, reducedness, dimension, or a sheaf $\mathcal{F} = 0$ are detectable at stalks of closed points (assume quasi-compactness).
 - X is normal implies that X is reduced. So X is integral.
- A Dedekind domain is a normal Noetherian ID of $\dim \leq 1$, and a Dedekind scheme is a normal locally Noetherian scheme of $\dim \leq 1$. (When $\dim = 0$, we only have fields.)
- Normality implies unique extension of regular functions when the codimension is large: $\mathcal{O}_X(X) \rightarrow \mathcal{O}_X(X \setminus F)$ is an isomorphism if F is of codimension ≥ 2 .
 - This essentially follows from $A = \bigcap A_{\mathfrak{p}}$ over all ht 1 prime ideals.
- Let X integral. The normalization of X is a morphism $X' \rightarrow X$ with X' normal, such that any $Y \rightarrow X$ dominant with Y normal factors through X' . For example, $\text{Spec } \bar{A} \rightarrow \text{Spec } A$ is a normalization morphism.
 - Let X integral. A morphism $f : Y \rightarrow X$ is a normalization iff Y normal, f birational and integral.

2.6 Regular Schemes

- The tangent space $T_{X,x}$ is defined as the dual of the $\kappa(x)$ -vector space $\mathfrak{m}_x/\mathfrak{m}_x^2 = \mathfrak{m}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x)$. We have $\dim T_{X,x} \geq \dim \mathcal{O}_{X,x}$.
 - For example, let $X = \text{Spec } k[x_1, \dots, x_n]$, then we define $D_x : k[x_1, \dots, x_n] \rightarrow (k^n)^\vee$, where $D_x P$ sends $(t_1, \dots, t_n)$ to $\sum \frac{\partial P}{\partial x_i}(x) t_i$. By considering the Taylor expansion, we see that for $\mathfrak{m} = (x_1 - \lambda_1, \dots, x_n - \lambda_n)$, we have $\mathfrak{m}/\mathfrak{m}^2 = (k^n)^\vee$ and that $T_{X,x} \cong k^n$.
 - Similarly, let $\mathfrak{m}$ be a maximal ideal in $k[T_1, \dots, T_n]$. $\varphi_x : \mathfrak{m}/\mathfrak{m}^2 \rightarrow (\kappa(x)^n)^\vee$ is an isomorphism iff $\kappa(x)/k$ is separable.
 - If $X = V(I) \subset \mathbb{A}^n$, then $T_{X,x} = \{(t_1, \dots, t_n) \in k^n, \sum \frac{\partial P}{\partial x_i}(x) t_i = 0 \text{ for every } P \in I\}$; i.e., the tangent space is the subspace of directions where the directional derivative along them for any $P \in I$ is zero. In particular, $T_{X,x} = (D_x I)^\perp$.
 - Recall that $(A, \mathfrak{m})$ is regular iff $\dim \mathfrak{m}/\mathfrak{m}^2 = \dim A$, or iff $\mathfrak{m}$ is generated by $\dim A$ elements. We call a point x as a regular point if $\mathcal{O}_{X,x}$ is regular, and a singular point otherwise.
 - In particular, a Noetherian connected scheme of $\dim 1$ is normal iff it is regular.
 - A system of parameters is similar to a coordinate system: if $(A, \mathfrak{m})$ Noetherian regular local of $\dim d$, then a system of parameters is a system of d generators for $\mathfrak{m}$.
 - For example, $k[x, y]/(x^2 - y^3)$ is not regular at $(0, 0)$. In fact, if $(A, \mathfrak{m})$ is a regular Noetherian local ring, then A/fA is regular iff $f \notin \mathfrak{m}^2$. Here, $x^2 - y^3 \in (x, y)^2$ so it fails to be regular.
 - More generally, A/I is regular iff I is generated by r elements of a system of parameters of A , where $r + \dim A/I = \dim A$. The particular case above says that f is in a system of parameter iff $f \notin \mathfrak{m}^2$.
- Regular Noetherian local rings are normal and regular at localizations.
- Let $X = V(I)$ with $I = (f_1, \dots, f_r)$. Note that $\dim T_{X,x} = n - \dim D_x I$, where $\dim D_x I$ is the rank of the Jacobian by the definition. So, X is regular at x iff the rank of $J_x = (\frac{\partial F_i}{\partial x_j}(x))$ is $n - \dim \mathcal{O}_{X,x}$.

- Let X normal. $\text{Reg } X$ is open and $\text{codim}(X_{\text{Sing}}, X) \geq 2$.
- Let X algebraic variety over k and regular at $x \in X(k)$, then $\hat{\mathcal{O}}_{X,x} \cong k[[x_1, \dots, x_d]]$ where $d = \dim \mathcal{O}_{X,x}$, where the completion is $\mathfrak{m}_x$ -adic.
- Every rational point is closed, every closed point can be viewed as a geometric point. Smooth points are always regular.

2.7 Flatness and Smoothness

- Flatness is the property that assures “dimensions are right”. In general, for a map between locally Noetherian schemes, $\dim \mathcal{O}_{X_y, x} \geq \dim \mathcal{O}_{X, x} - \dim \mathcal{O}_{Y, y}$, and equality holds if f is flat.
 - Essentially, if $X = \text{Spec } B$ and $Y = \text{Spec } A$ with A reduced Noetherian local, $t \in A$ not nilpotent nor unit, we have $\dim A/tA = \dim A - 1$ and $\dim B/tB \geq \dim B - 1$, where equality holds if flatness is given. Now the result follows by induction.
 - For flat morphisms between irreducible algebraic varieties, the dimension of the fibers is constant, $\dim X_y = \dim X - \dim Y$.
 - Blowup fails to be flat at the points that are blowed up.
- A morphism of finite type between locally Noetherian schemes is unramified at x if $\mathcal{O}_{Y, y} \rightarrow \mathcal{O}_{X, x}$ satisfies $\mathfrak{m}_y \mathcal{O}_{X, x} = \mathfrak{m}_x$, and if $\kappa(y) \rightarrow \kappa(x)$ is separable. f is étale at x if it is unramified and flat at x .
 - For example, $\text{Spec } L \rightarrow \text{Spec } K$ is étale if L/K is separable.
 - f being unramified is equivalent to X_y being finite, reduced, and $\kappa(x)/\kappa(y)$ separable.
 - $\text{Spec } k[T]/(P) \rightarrow \text{Spec } k$ is étale at x if x (representing $Q \mid P$) is simple in P and separable.
- Let $f : X \rightarrow Y$ étale, then $\dim \mathcal{O}_{X, x} = \dim \mathcal{O}_{Y, y}$ with X_y finite, and $T_{X, x} \cong T_{Y, y} \otimes_{\kappa(y)} \kappa(x)$.
- X over $\text{Spec } k$ is smooth at x if the points lying over x in $X_{\bar{k}}$ are all regular. If X is an algebraic variety with x closed, then X is regular at x if it is smooth at x , and the converse holds if $\kappa(x)/k$ is separable.
 - In general this would still be true, as J_x and $J_{x'}$, and $\dim \mathcal{O}_{X, x}$ and $\dim \mathcal{O}_{X_{\bar{k}}, x'}$ are independent of the choice of x' .
- Let X and Y locally Noetherian and $f : X \rightarrow Y$ of finite type. f is smooth at $x \in X$ if it is flat at x and $X_y \rightarrow \text{Spec } \kappa(y)$ is smooth. Note that it suffices to check smoothness at closed points of X_y over $y \in Y$ closed. f is smooth of relative dimension n if the non-empty fibers are equi-dimensional of $\dim n$, and an étale morphism is smooth of relative $\dim 0$.
- Let $\mathfrak{n}_x$ be the maximal ideal of $\mathcal{O}_{X_y, x}$, we have an exact sequence of $\kappa(x)$ vector spaces:

$$0 \longrightarrow T_{X_y, x} \longrightarrow T_{X, x} \longrightarrow T_{Y, y} \otimes \kappa(x) \longrightarrow 0$$

In particular, étale morphisms induce isomorphisms on the tangent spaces.

2.8 Zariski's Main Theorem

- (Zariski's Main Theorem): Let Y normal locally Noetherian integral, $f : X \rightarrow Y$ separated quasi-finite birational morphism of finite type, then f is an open immersion.
 - As for the motivation, the initial version is that: if Y is normal, $f : X \rightarrow Y$ birational, and $y \in Y$ with isolated point $x \in X_y$, then the birational morphism is an isomorphism in a neighborhood of x .
 - We can “compactify” quasi-finite morphisms to finite morphisms. Now, the theorem is saying that after compactification, we assume f is finite; now since Y is normal, f must be an open immersion (locally, it is finite and therefore an open immersion, and we use the fact that a locally open immersion is an open immersion for f separated and X, Y irreducible).
- The theorem implies that the set of points where f is étale is open if f is locally of finite type.
- Also, if Y Dedekind with generic point η , then if $f : X \rightarrow Y$ dominant of finite type, with X irreducible, then X_η is empty or of dimension equals $\dim X_\eta$.

2.9 Coherent Sheaves

- A quasi-coherent sheaf $\mathcal{F}$ is an $\mathcal{O}_X$ -module such that for each $x \in X$, there exists a neighborhood $U \ni x$ and an exact sequence of $\mathcal{O}_X$ -modules:

$$\mathcal{O}_X^{(J)}|_U \rightarrow \mathcal{O}_X^{(I)}|_U \rightarrow \mathcal{F}|_U \rightarrow 0.$$

In some sense quasi-coherent sheaves correspond to modules with a presentation. Recall that there exists a surjective $\mathcal{O}_X^{(I)} \rightarrow \mathcal{F}$ iff $\mathcal{F}$ is generated by I (i.e., $I = \{s\} \subset \mathcal{F}(X)$ such that $\{s_x\}$ generates $\mathcal{F}_x$ for each x).

- For a quasi-coherent sheaf, if X is Noetherian or separated & quasi-compact, then for $f \in \mathcal{O}_X(X)$, $\mathcal{F}(X)_f \cong \mathcal{F}(X_f)$. In fact, $\mathcal{F}$ is quasi-coherent iff for every open affine U , $\mathcal{F}(U)^\sim \cong \mathcal{F}|_U$.
- A coherent sheaf is an $\mathcal{O}_X$ -module that is finitely generated (locally, we have surjective $\mathcal{O}_X^n|_U \rightarrow \mathcal{F}|_U$), and if the kernel for every homomorphism $\mathcal{O}_X^n|_U \rightarrow \mathcal{F}|_U$ is finitely generated. The notion of coherent sheaves correspond to finitely presented modules; if X locally Noetherian, then $\mathcal{F}$ is coherent iff it is finitely generated, iff for every U open affine, $\mathcal{F}(U)$ is f.g. over $\mathcal{O}_X(U)$.
- Given an $\mathcal{O}_Y$ -module $\mathcal{G}$, we can pull it back via f by $f^*\mathcal{G} = f^{-1}\mathcal{G} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X$ to get a $\mathcal{O}_X$ -module.
- We define quasi-coherent sheaves on projective schemes via the following: let B be a graded ring, we define $B(n)$ to be the graded ring with $B(n)_d = B_{n+d}$ and $\mathcal{O}_X(n) = (B(n))^\sim$, with $\mathcal{O}_X(n)(D_+(f)) = f^n B_{(f)}$, where $M_{(f)} = \{\frac{a}{f^n}, \deg a = n \deg f\}$.
 - We can also define the twist of a quasi-coherent sheaf $\mathcal{F}$ by $\mathcal{F}(n) = \mathcal{F} \otimes \mathcal{O}_X(n)$.
 - We can easily see that $\mathcal{O}_X(n) \otimes \mathcal{O}_X(m) = \mathcal{O}_X(n+m)$. Also, $\mathcal{O}_X(n)(X) = B_n$.
 - Let $\mathcal{F}$ f.g. quasi-coherent sheaf over a projective scheme X , there exists $n_0 \geq 0$ such that $\mathcal{F}(n)$ is f.g. by global sections if $n \geq n_0$.
- If $X = \text{Proj } B$, then any quasi-coherent sheaf $\mathcal{F}$ satisfies $\mathcal{F} \cong (\bigoplus \mathcal{F}(n)(X))^\sim$.
- Let $i : X \rightarrow \mathbb{P}_A^d$ be an immersion, the sheaf $\mathcal{O}_X(1) = i^*\mathcal{O}_{\mathbb{P}_A^d}(1)$ is called a very ample sheaf.

- Let $Y = \text{Proj } A[T_0, \dots, T_d]$, X a scheme over A . Morphisms $X \rightarrow Y$ over $\text{Spec } A$ correspond to invertible sheaves $\mathcal{L}$ over X generated by $d+1$ global sections, via $f \rightarrow f^* \mathcal{O}_Y(1)$. Intuitively, we can correspond $X_{s_i} = \{x \in X, \mathcal{L}_x = (s_i)_x \mathcal{O}_{X,x}\}$, and sent X_{s_i} to $Y_i = D_+(T_i)$.
 - This sort of says that projective spaces can be viewed as the classifying spaces for line bundles.
- An invertible sheaf $\mathcal{L}$ over a quasi-compact scheme X is ample if for every $\mathcal{F}$ quasi-coherent, there exists some n_0 such that $n \geq n_0$ implies that $\mathcal{F} \otimes \mathcal{L}^{\otimes n}$ is generated by global sections.
- Let $f : X \rightarrow \text{Spec } A$, if X is Noetherian or f is separated, then there exists $m \geq 1$ such that $\mathcal{L}^{\otimes m}$ is very ample.
- $\mathcal{L}$ is ample if there exists $s_1, \dots, s_r \in \mathcal{L}(X)$ such that X_{s_i} is affine for every i and that $X = \bigcup X_{s_i}$.

3 Homotopical Algebra

3.1 Simplicial Sets

- The realization functor is left adjoint to the singular functor. We have natural maps $X \rightarrow \text{Sing } |X|$ and $|\text{Sing } Y| \rightarrow Y$. Both of these are weak equivalences, where a map of simplicial sets is a weak equivalence if it is after applying realization.
- We define the mapping space $\underline{\mathcal{S}}(X, Y)$ to be the simplicial set with $\underline{\mathcal{S}}(X, Y)_n = \text{Hom}(X \times \Delta^n, Y)$. A map $f : X \rightarrow Y$ is a simplicial homotopy equivalence if there exists $g : Y \rightarrow X$ such that $f \circ g$ and $g \circ f$ are both homotopic to the identity in the sense of 1-cells in the mapping space.
- We denote the category of simplicial Abelian groups by $\mathcal{A}$. It is also simplicial in the sense that $\underline{\mathcal{A}}(M, N)_n = \mathcal{A}(M \otimes \mathbb{Z}[\Delta^n], N)$, where $\mathbb{Z}[\Delta^n]$ is the free simplicial abelian group associated to the simplicial set.
- We define homotopy groups of simplicial sets/Abelian groups via $\pi_*(|X|)$, i.e., the homotopy groups of the realization.
- A simplicial Abelian group M gives rise to the Moore complex C_* , where $M_n \rightarrow M_{n-1}$ via the alternating sum $d_0 - d_1 + d_2 - \dots \pm d_n$. This satisfies that $H_*(C_*(M)) = \pi_*(M)$. We can also define $H_*(X) = H_*(C_*\mathbb{Z}[X])$ for X a simplicial set in general.
- We can also define homology via the normalized chain complex, where $C_n^{\text{norm}}(M) = \bigcap_{i=0}^{n-1} \ker d_i : M_n \rightarrow M_{n-1}$ and the boundary map via d_n . C^{norm} gives an equivalence of categories between $\mathcal{A}$ and $\mathbf{Ch}^{\geq 0}$. The normalized complex is homotopy equivalent to the Moore complex (and hence gives the same homology groups).
- We define the suspension of $X \in \mathcal{S}$ via $S^1 \wedge X$. $S^n := S^1 \wedge S^{n-1}$. For a pointed set X , we define $\tilde{\mathbb{Z}}[X] = \mathbb{Z}[X]/\mathbb{Z}[*]$. For a simplicial set X , its reduced homology is given as the homotopy groups of $\tilde{\mathbb{Z}}[X]$. Certain properties of $\tilde{\mathbb{Z}}$ guarantees the wedge axiom, Künneth theorem, and excision.
 - In particular, $\tilde{\mathbb{Z}}[S^n]$ has the n -th homotopy group $\mathbb{Z}$ and anything else 0, and hence plays the role of the Eilenberg-MacLane spaces.
- Adjunction holds in the sense that $\underline{\mathcal{S}}_*(X \wedge Y, Z) \cong \underline{\mathcal{S}}_*(Y, \underline{\mathcal{S}}_*(X, Z))$.
- We define $\Omega X = \underline{\mathcal{S}}_*(S^1, \text{Sing } |X|) \cong \text{Sing } \Omega|X|$. Note that although $X \rightarrow \text{Sing } |X|$ is a weak equivalence, $\underline{\mathcal{S}}_*(A, X) \rightarrow \underline{\mathcal{S}}_*(A, \text{Sing } |X|)$ is not guaranteed to be one.
- Cohomology group of $X \in \mathcal{S}$ is given by $\tilde{H}^n(X) = \pi_0 \underline{\mathcal{S}}_*(X, \tilde{\mathbb{Z}}[S^n])$.

3.2 Spectra

- A spectrum is a sequence of simplicial sets E^* with maps $S^1 \wedge E^k \rightarrow E^{k+1}$. If the adjoint maps $E^k \rightarrow \Omega E^{k+1}$ are equivalences, we call the spectrum a Ω -spectrum.
- For example, $\mathbf{S} = (S^k)$ and $H\mathbb{Z} = (\tilde{\mathbb{Z}}[S^k])$.
- There is a pair of adjoint functors $R : \mathbf{Spt} \rightarrow \mathcal{S}_*$ via $E \rightarrow E^0$ and $\Sigma^\infty : \mathcal{S}_* \rightarrow \mathbf{Spt}$ via $\Sigma^\infty X \rightarrow (S^* \wedge X)$.
- Let E be a spectrum. The stable homotopy groups of E are defined as $\pi_q E = \text{colim } \pi_{q+k} E^k$ (and $\pi_{-n} E = \text{colim } \pi_k(\Omega^n E^k)$). A map of spectra is a stable equivalence if it induces isomorphisms on stable homotopy groups. Inverting all stable equivalence gives the stable homotopy category **HoSpt**.
- A spectrum gives a (co)homology theory by $E_n(X) = \pi_n(E \wedge X)$ and $E^n(X) = \pi_{-n}(E^X)$. The (co)homology group given by $H\mathbb{Z}$ is the same as the reduced (co)homology of X .

3.3 Model Structure

- One thing good about $\mathcal{S}$ is that every object behaves like a CW complex: if $K \subset L$, then there exists a functorial factorization $K = K(-1) \subset K(0) \subset \dots \subset \bigcup K(n) = L$ where each $K(i-1) \subset K(i)$ is given by attaching cells:

$$\begin{array}{ccc} \bigsqcup \partial \Delta^i & \longrightarrow & \bigsqcup \Delta^i \\ \downarrow & & \downarrow \\ K(i-1) & \longrightarrow & K(i) \end{array}$$

- Trivial fibrations are maps that have the right lifting property with respect to $\partial \Delta^n \subset \Delta^n$, and fibrations are maps that have the right lifting property with respect to $\Lambda_i^n \subset \Delta^n$.
- Kan complexes are the fibrant objects. $\text{Sing } Y$ is a Kan complex, and so is the nerve of a groupoid $\mathcal{G}$.
- Small object argument is used for many proofs. Essentially a map $\bigcup X_n \rightarrow Y$ comes from some map $X_n \rightarrow Y$.
- If $f : X \rightarrow Y$ is a (trivial) fibration and $A \subset B$, then so is $\underline{\mathcal{S}}(B, X) \rightarrow \underline{\mathcal{S}}(A, X) \times_{\underline{\mathcal{S}}(A, Y)} \underline{\mathcal{S}}(B, Y)$. The proof uses some tricks about adjunction. A weak equivalence between Kan complexes is a homotopy equivalence.
- For a Kan complex $X \in \mathcal{S}_*$, we define $\pi_q X = \pi_0 \underline{\mathcal{S}}_*(S^q, X)$. We have $\pi_* X \cong \pi_* |X|$.
- We have a fibrant replacement functor Ex^∞ , i.e., $X \subset \text{Ex}^\infty X$ is a weak equivalence and $\text{Ex}^\infty X$ is fibrant. We can then define $\pi_q X = \pi_0 \underline{\mathcal{S}}_*(S^q, \text{Ex}^\infty X)$.
- (Proper Model Categories & Homotopy Pullback) $\mathcal{M}$ is left proper if in

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow i & & \downarrow \\ C & \xrightarrow{g} & D \end{array}$$

where i a cofibration and f a weak equivalence, g is a weak equivalence. Dually we define right proper model categories. All the classical model category structures are proper. A commuting square in $\mathcal{M}$ is a homotopy pullback square

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & D \end{array}$$

if for functorial factorization $B \rightarrow X \rightarrow D$ and $C \rightarrow Y \rightarrow D$ into trivial cofibrations followed by fibrations, $A \rightarrow X \times_D Y$ is a weak equivalence. $X \times_D Y$ is called the homotopy pullback.

- If $\mathcal{M}$ is proper then the homotopy pullbacks are homotopy invariant. If the diagram is a pullback and $B \rightarrow D$ a fibration, the square is said to be homotopy cartesian.
- In **Spt**, we define a fibration (weak equivalence) to be a pointwise fibration (weak equivalence), and a cofibration to be those with the correct left lifting property. This gives a model structure.
- The stable structure on the other hand, define the weak equivalences to be the stable equivalences and fibrations to be those with the correct right lifting property. This is a cofibrantly generated model structure.

- Consider the category $\mathcal{S}^{\mathcal{C}}$. A natural transformation is a pointwise weak equivalence (fibration) if $X(c) \rightarrow Y(c)$ is a weak equivalence (fibration) for each $c \in \mathcal{C}$. This gives a model category structure on $\mathcal{S}^{\mathcal{C}}$ called the projective structure.
 - A cellular inclusion if a composite $X_0 \rightarrow \dots \rightarrow \operatorname{colim} X_n$ where each $X_i \rightarrow X_{i+1}$ is the pushout along a disjoint union of maps $\partial\Delta^n \times \mathcal{C}(c, -) \rightarrow \Delta^n \times \mathcal{C}(c, -) \rightarrow$.
 - A cofibration is exactly a retract of a cellular inclusion.

3.4 Motivic Spaces

- The category of motivic spaces contains the pointed simplicial presheaves on $\mathcal{S}m/\mathcal{S}$. We define $\mathcal{M}_{\mathcal{S}} = \mathcal{S}_*^{(\mathcal{S}m/\mathcal{S})^{op}}$. $\mathcal{M}_{\mathcal{S}}$ is given a model structure, the projective structure. It also has smash product $X \wedge Y(U) = X(U) \wedge Y(U)$ and an internal morphism object $\underline{\mathcal{M}}_{\mathcal{S}}$ right adjoint to the smash.
 - $X \in \mathcal{M}_{\mathcal{S}}$ is locally fibrant if it is a Nisnevich sheaf up to homotopy. For $X \in \mathcal{M}_{\mathcal{S}}$ we define KX to be the functorial factorization $* \rightarrow KX \rightarrow X$ into a cofibration followed by a trivial fibration (in the projective structure). $X \rightarrow Y \in \mathcal{M}_{\mathcal{S}}$ is a local equivalence if for all locally fibrant Z the map $\underline{\mathcal{M}}_{\mathcal{S}}(KY, Z) \rightarrow \underline{\mathcal{M}}_{\mathcal{S}}(KX, Z)$ is a pointwise equivalence. Local cofibrations are the cofibrations in the projective structures, and local fibrations are the maps with the correct right lifting property.
 - This gives a model structure on $\mathcal{M}_{\mathcal{S}}$ with local weak equivalences detected pointwisely on locally fibrant objects.
 - $X \in \mathcal{M}_{\mathcal{S}}$ is $\mathbb{A}^1$ -fibrant if it is locally fibrant s.t. $X(\mathbb{A}^1 \times_{\mathcal{S}} -) \rightarrow X(-)$ is a pointwise weak equivalence. An $\mathbb{A}^1$ -equivalence is defined analogous to local equivalence. $\mathbb{A}^1$ -cofibrations are the local cofibrations and $\mathbb{A}^1$ -fibrations are those with the correct right lifting property.
 - This gives a model structure on $\mathcal{M}_{\mathcal{S}}$; its homotopy category is equivalent to the unstable homotopy category.
 - We want motivic homology theories to be stable under smashing with the Tate object $T = S^1 \wedge \mathbb{G}_m$ and to commute with filtered colimits. We define $f\mathcal{M}_{\mathcal{S}} \subset \mathcal{M}_{\mathcal{S}}$ to be the category of finite spaces (sort of like CW complexes).
 - A functor $X : f\mathcal{M}_{\mathcal{S}} \rightarrow \mathcal{M}_{\mathcal{S}}$ is continuous if X induces maps $\underline{\mathcal{M}}_{\mathcal{S}}(v, w) \rightarrow \underline{\mathcal{M}}_{\mathcal{S}}(Xv, Xw)$. Such functors form a category $\mathcal{F}_{\mathcal{S}}$. Note that continuous functors give map $S^1 \wedge XS^n \rightarrow XS^{n+1}$ so (XS^n) give a spectrum.
 - We have a smash product in $\mathcal{F}_{\mathcal{S}}$, where $X \wedge Y \rightarrow Z$ is in correspondence with maps $Xu \wedge Yv \rightarrow Z(u \wedge v)$ that are natural in u and v .
 - The motivic sphere spectrum is the inclusion $\mathbf{S} : f\mathcal{M}_{\mathcal{S}} \rightarrow \mathcal{M}_{\mathcal{S}}$, and $\mathbf{S} \wedge X = X$. We can talk about $\mathbf{S}$ -algebras.
 - We have a pointwise structure on $\mathcal{F}_{\mathcal{S}}$ via the $\mathbb{A}^1$ structure on $\mathcal{M}_{\mathcal{S}}$. We define KX to be the functorial factorization $* \rightarrow FX \rightarrow X$ into a cofibration followed by a trivial fibration in the pointwise structure.
 - $X \in \mathcal{F}_{\mathcal{S}}$ is ht-fibrant if it is pointwise fibrant such that for $\mathbb{A}^1$ -equivalences $\phi : v \rightarrow w$ in $f\mathcal{M}_{\mathcal{S}}$, $Xv \rightarrow Xw$ is an $\mathbb{A}^1$ -equivalence. An ht-equivalence is defined analogous to a local or an $\mathbb{A}^1$ -equivalence in $\mathcal{M}_{\mathcal{S}}$. ht-cofibrations are the pointwise cofibrations and ht-fibrations are those with the correct right lifting structure.
 - ht-structure is a model category structure on $\mathcal{F}_{\mathcal{S}}$ that lifts to model structures on algebras and modules in $\mathcal{F}_{\mathcal{S}}$.

3.5 Motivic Spectra

- A T-spectrum (or motivic spectrum) is a sequence $(E^n) \subset \mathcal{M}_{\mathcal{S}}$ with $T \wedge E^n \rightarrow E^{n+1}$. These form a category $\mathbf{Spt}_{\mathcal{S}}$ that comes with a model structure whose homotopy category is equivalent to the motivic stable homotopy category.
- Given X a spectrum in $\mathbf{Spt}$, we can define $(QX)^n = \text{colim} \text{Sing } \Omega^i |X^{n+i}|$, this is a Ω -spectrum with a canonical map $X \rightarrow QX$. Now in $\mathbf{Spt}_{\mathcal{S}}$, we declare a map $X \rightarrow Y$ to be a stable equivalence if $(QX)^n \rightarrow (QY)^n$ is an $\mathbb{A}^1$ -equivalence for each n . Similarly we define stable fibrations and hence a stable structure on $\mathbf{Spt}_{\mathcal{S}}$.
- Let $T^n = T^1 \wedge T^{n-1}$ and $T^0 = S^0$. For $X \in \mathcal{F}_{\mathcal{S}}$, we define $evX \in \mathbf{Spt}_{\mathcal{S}}$ to be the spectrum $(X(T^n))$. This gives a functor $\mathcal{F}_{\mathcal{S}} \rightarrow \mathbf{Spt}_{\mathcal{S}}$ that preserves limits and colimits. This forces the stable structure on $\mathcal{F}_{\mathcal{S}}$ (after functorial replacement by homotopy functors (i.e., those sending weak equivalences to equivalences, or the fibrant objects in the ht-structure) via the ht-structure).
- There is a left adjoint $F : \mathbf{Spt}_{\mathcal{S}} \rightarrow \mathcal{F}_{\mathcal{S}}$ and this is a Quillen equivalence after putting the stable structure on $\mathcal{F}_{\mathcal{S}}$. The stable model structure induces model category structure on algebras and modules over $\mathcal{F}_{\mathcal{S}}$.

4 Geometric Invariant Theory & Moduli

- One goal was to construct good quotient: for example, $\mathbb{C}^\times$ acts on $\mathbb{C}$, but the fiber of the map $\mathbb{C} \rightarrow \mathbb{C}/\mathbb{C}^\times$ are not closed. We can form categorical quotient for affine schemes, by letting $\text{Spec } A/G = \text{Spec } A^G$, but it is not always the “good” geometric quotient.
- (Geometric Quotient). Q is a geometric quotient if there exists $\pi : X \rightarrow Q$ such that π surjective with fibers G -orbits, $\pi^{-1}(U)$ open iff U open, and that $\Gamma(U, \mathcal{O}_U) = \Gamma(\pi^{-1}(U), \mathcal{O}_{\pi^{-1}(U)})^G$.
- Good categorical quotients exist for affine varieties, given furthermore that G is reductive. However, it is not always desired.
 - For example, $\mathbb{C}^\times$ acts on $\text{Spec } \mathbb{C}[x] = \mathbb{A}^1$ by sending $\lambda \cdot f(x)$ to $f(\lambda x)$. Now, $\mathbb{A}^1/\mathbb{C}^\times = \text{Spec } \mathbb{C}$ consists of only one point, while there are two orbits.
 - Let $\mathbb{C}^\times$ acts on $\mathbb{A}^2$ by sending $t \cdot (x, y)$ to (tx, ty) . Then $(\mathbb{A}^2)^{\mathbb{C}^\times} = \mathbb{C}$, and $\mathbb{C}^2/\mathbb{C}^\times = \text{Spec } \mathbb{C}$ has only one point, while the orbits are $\{(0, 0)\}$, $\{\mathbb{C}^\times \times 0\}$, $\{0 \times \mathbb{C}^\times\}$, and (complement of 0 in) $\text{Spec } \mathbb{C}[xy^{-1}] = \mathbb{A}^1$. In short, the dimension is not right. Note that the closure of the orbits contain $(0, 0)$.
 - On the other hand, let $\mathbb{C}^\times$ acts on $\mathbb{A}^2$ by sending $t \cdot (x, y)$ to $(tx, t^{-1}y)$, then the orbits are $\{(0, 0)\}$, $\{\mathbb{C}^\times \times 0\}$, $\{0 \times \mathbb{C}^\times\}$, and (complement of 0 in) $\text{Spec } \mathbb{C}[xy] = \mathbb{A}^1$. The orbits in the last one are closed.
 - In fact, points in $X//G$ corresponds to closed orbits.
- Assume G reductive. A point x is stable if G_x is finite and $G \cdot x$ is closed. The set of stable points X^s is open, and $X^s \rightarrow X^s//G$ is a geometric quotient. In addition, if $X^s \neq \emptyset$, then $\dim X - \dim G = \dim X//G$.
- One thing not so good is that $U//G \rightarrow X//G$ is not always an open embedding even if $U \rightarrow X$ is. For example, consider $U = \mathbb{A}^2 \setminus \{(0, 0)\}$, then $U//G = \mathbb{P}^1$ while $X//G$ is a point. (We can glue things up to form $U//G$).
- For $X \subset \mathbb{P}^m$ projective variety, one can show that X^{ss} is covered by G -invariant affines where $x \in X^{ss}$ iff $x \in X_f$ for some G -invariant f . Note that in general we can't just glue things up, because it is not necessarily true that X can be covered by G -invariant open affines.

- We then form $X//G = X^{ss}//G$, where the latter comes by gluing. $X \rightarrow X^{ss}//G$ is only rational and defined on semi-stable points.
 - This is a categorical quotient and almost geometric.
 - $X//G$ is a projective variety.
 - In general $X^s/G \subset X//G$ and $\dim X//G = \dim X - \dim G$.
- Let $V \subset \mathbb{P}^n$ be a smooth hypersurface of degree d , we can view it as a point in $\mathbb{P}^{N_d-1}$ where $N_d = \binom{n+d}{d}$. $[V]$ is semi-stable but not stable w.r.t. the $\mathbb{P}GL(n+1)$ action.
 - A quadric over $\mathbb{C}$ in $\mathbb{P}^m$ is classified by the rank. In other words, $Q \simeq_{\mathbb{P}GL(m+1)} V(x_0^2 + \dots x_k^2)$ for some unique k . There are hence $m+1$ orbits of the action of $\mathbb{P}GL(m+1)$.
 - One can show, with a 1-parameter subgroup argument, that the closure of the orbit $G \cdot (x_0^2 + \dots + x_k^2)$ contains $G \cdot (x_0^2 + \dots + x_{k-1}^2)$. Also, $G \cdot x_0^2$ is not semistable.
 - In fact, $\mathbb{P}^{N_d}//\mathbb{P}GL(m+1)$ consists of one point, and there is a unique polystable orbit: the orbit of smooth quadrics. This orbit is open in $X = \mathbb{P}^{N_d}$ but closed in X^{ss} .

5 Equivariant Homotopy Theory

We fix G a finite group or a compact Lie group (or perhaps more generally, a compact topological group).

5.1 The Unstable Theory

- A G -space X is defined via a G -action and expected compatibility axioms. G -equivariant maps are also defined as expected. We write the category of G -spaces as $G\mathbf{Top}$, and there are two choices for what the morphisms are: we can use the G -equivariant maps between two G -spaces, or we can use the G -space of all maps between the underlying spaces with G acting via conjugations, which allows $G\mathbf{Top}$ to be enriched over itself. We denote the former as $\text{Map}^G(X, Y)$ and the latter as $G\mathbf{Top}(X, Y)$. Note that $(G\mathbf{Top}(X, Y))^G = \text{Map}^G(X, Y)$.
- Let X a G -set, X^H is the fixed points under a subgroup action, and G_x is the isotropy group, the elements in G that fixes x . Note that $X^H \simeq \text{Map}^G(G/H, X)$.
- Let $G \rightarrow H$ a group map. We naturally have a functor $H\mathbf{Top} \rightarrow G\mathbf{Top}$. This map has left and right adjoints, which can be understood as the Kan extensions along $BG \rightarrow BH$, where BG and BH are the one-point groupoid with automorphism set G and H .
- Let V be a finite-dimensional real representation of G (i.e., an inner product space on which G acts in a way compatible with the inner product). The one-point compactification of V is denoted S^V , and this is exactly $D(V)/S(V)$. S^V is called a representation sphere.
- A G -homotopy is a map $X \times I \rightarrow Y$ between G -spaces where G acts trivially on I .
- We define G -CW complexes by gluing cells, where the cells are $G/H \times D^n$ and $G/H \times S^n$ with trivial actions where H is any subgroup of G .
 - Note that $[G/H \times S^n, X] \simeq \pi_n(X^H) =: \pi_n^H(X)$. We define a weak-equivalence to be a map such that $\pi_n^H(X) \rightarrow \pi_n^H(Y)$ is an isomorphism for all H .
 - The spirit is that $G/H \times D^n$ represents a cell on which H acts trivially. For example, consider S^1 acts on $\mathbb{R}^2$ by rotation. Let $V = \mathbb{R}^2$ with this action. S^V is a G -CW complex with two fixed points $S^1/S^1 \times *$ and one 1-cell $S^1 \times I$ attached to the endpoints (rotate the geodesic between two endpoints via the S^1 -action).

- Let $\theta : \{\text{conjugacy classes of subgroups of } G\} \rightarrow \{x \in \mathbb{Z}, x \geq -1\}$. We say a map is θ -connected if f^H is $\theta(H)$ -connected for every H , and a G -CW complex is θ -dimensional if all cells of orbit type G/H have dimension at most $\theta(H)$. The key point is the equivariant Whitehead theorem: a weak equivalence between G -CW complexes is a G -homotopy equivalence.
 - In fact we can define a model structure on $G\mathbf{Top}$, where fibrations or weak equivalences are those maps f such that f^H is a fibration or a weak equivalence for every subgroup H .

5.2 The Orbit Category and Elmendorf's Theorem

- The orbit category $\mathcal{O}_G$ is the full subcategory of $G\mathbf{Top}$ with objects G/H . The morphisms $G/H \rightarrow G/K$ are identified with $(G/K)^H$, and are specified by what they do to eH . It is equivalent to the category of finite transitive G -sets and G -maps (this comes from the orbit-stabilizer relation).
- Given a G -space X , we can define a functor $X^{(-)} : \mathcal{O}_G^{\text{op}} \rightarrow \mathbf{Top}$. This assignment is a functor $\psi : G\mathbf{Top} \rightarrow \text{Fun}(\mathcal{O}_G^{\text{op}}, \mathbf{Top})$. $\text{Fun}(\mathcal{O}_G^{\text{op}}, \mathbf{Top})$ has a projective model structure with pointwise weak equivalences and fibrations.
- (Elmendorf). ψ is the right adjoint in a Quillen equivalence, the left adjoint θ is evaluation at G/e .

5.3 The Stable Theory

- For motivation, we consider the $\mathcal{S}$ -category, in which the objects are topological spaces and morphisms are $\text{Map}(X, Y) = \text{colim Map}(\Sigma^n X, \Sigma^n Y)$ where the latter is in $\mathbf{Top}$.
 - Atiyah duality basically says $T\nu$ and M_+ are dual in the $\mathcal{S}$ -category, where an embedding $f : M \rightarrow \mathbb{R}^n$ is fixed.
 - More specifically, the PT construction gives a map $S^n \simeq \mathbb{R}_+^n \rightarrow \mathbb{R}^n/\mathbb{R}^n - M \simeq T\nu \rightarrow T\nu \wedge M_+$, which gives the coevaluation. Consider the embedding $\iota : M \rightarrow \nu \times M$ in which the normal bundle is trivial, this leads to (by the PT construction again), a map $(\nu \times M)_+ \simeq T\nu \wedge M_+ \rightarrow T\nu_\iota \simeq \Sigma^n M_+ \rightarrow S^n$ where the last map is by projecting M to a point, which is the evaluation.
- (Equivariant Whitney's Theorem). A G -manifold is a manifold M with smooth G -action. Let M compact G -manifold, then there exists a G -equivariant embedding $M \rightarrow V$ where V is some finite-dimensional real G -representation.
 - This leads to the equivariant version of everything, PT construction, Atiyah duality, etc. However, we first need to define the right $\mathcal{S}$ -category equivariantly, where $\text{Map}(X, Y) := \text{colim}_V \text{Map}(\Sigma^V X, \Sigma^V Y)$ where V takes place over all finite-dimensional real representations V .
- (Orbit Category). Earlier we defined the orbit category $\mathcal{O}_G$ in which the objects are G/H for H some closed subgroup, and morphisms G -equivariant maps. For the equivariant $\mathcal{S}$ -category, we need more morphisms in the orbit category called the transfer maps.
 - Suppose M embeds in some G -representation V , then the PT construction gives a map $S^V \rightarrow T\nu \rightarrow T(\nu \oplus \tau) \rightarrow S^V \wedge M_+$. Suppose $K \subset H$, we get maps $S^V \rightarrow S^V \wedge (H/K)_+$, and inducing this gives maps $(G/H)_+ \wedge S^V \rightarrow (G/K)_+ \wedge S^V$, which are the extra maps we need in $\mathcal{O}_G$.
 - Now we have two functors from $\mathcal{O}_G$ to the equivariant $\mathcal{S}$ -category. One of them is covariant and sends a map to its representative in the colimit, while the other one is contravariant via these transfer maps.

- (Burnside Category and Mackey Functor). The Burnside category B_G is the full subcategory of the $\mathcal{S}$ -category spanned by orbits G/H 's. The maps in B_G is a compositions of two maps in the image of the two functors depicted above. It is enriched in $\mathbf{Top}$. The algebraic Burnside category is the homotopy category $\pi_0 B_G$ and is enriched in $\mathbf{Ab}$. A Mackey functor is a $\mathbf{Ab}$ -enriched functor $\pi_0 B_G \rightarrow \mathbf{Ab}$.
- We first can define the category of spectra via the $\mathcal{S}$ -category: the goal is to construct a complete/cocomplete symmetric monoidal category $\mathbf{Sp}$ which contains the $\mathcal{S}$ -category as a full subcategory and admits a symmetric monoidal functor $\Sigma^\infty : \mathbf{Top} \rightarrow \mathbf{Sp}$ which is the left adjoint to $\Omega^\infty : \mathbf{Sp} \rightarrow \mathbf{Top}$. In some sense $\mathbf{Sp}$ is the smallest such category.
 - The definition is based on a more general construction. Let $\mathcal{D}$ small and symmetric monoidal and enriched in $\mathbf{Top}$. The category of $\mathcal{D}$ -spaces is $\text{Fun}(\mathcal{D}, \mathbf{Top}_*)$ of enriched functors, and it is symmetric monoidal under Day convolution: $(F \wedge G)(z) = \text{colim}_{x \otimes y = z} F(x) \wedge G(y) = \int^{x, y \in \mathcal{D}} F(x) \wedge G(y) \wedge \mathcal{D}(x \otimes y, z)$.
 - For $d \in \mathcal{D}$, there is an evaluation $\text{Ev}_d : \text{Fun}(\mathcal{D}, \mathbf{Top}_*) \rightarrow \mathbf{Top}_*$ by sending X to $X(d)$. It is adjoint to a functor F_d , and the unit in $\text{Fun}(\mathcal{D}, \mathbf{Top}_*)$ is $F_0 S^0$. Let R commutative monoid object in $\text{Fun}(\mathcal{D}, \mathbf{Top}_*)$, we can talk about $\mathbf{Mod}_R$ the category of R -modules. It is also symmetric monoidal.
 - Let $\mathcal{D} = \mathbb{N}$ with only the identity maps and $[m] \otimes [n] = [m + n]$. The category of $S_{\mathbb{N}}$ -modules is called prespectra, where $S_{\mathbb{N}}$ sends $[n]$ to S^n . Note that $S_{\mathbb{N}}$ is not a commutative monoid which causes problem. In particular, the category is only monoidal not symmetric monoidal.
 - * An Ω -prespectrum is a prespectrum such that $X_n \simeq \Omega^m X_{m+n}$ via the structure maps.
 - * $\pi_q(X) := \text{colim } \pi_{n+q} X(n)$. A map is a π_* -isomorphism if it induces isomorphisms on homotopy groups.
 - This motivates the definition of orthogonal spectra. Let $\mathcal{J}$ be the category of finite-dimensional real inner product spaces V with morphisms linear isometric isomorphisms. $S_{\mathcal{J}}$ sending V to S^V gives a commutative monoid, and the category of $S_{\mathcal{J}}$ -modules is called the category of orthogonal spectra. In this section a spectrum is just an orthogonal spectrum.
 - * An Ω -spectrum is an orthogonal spectrum X such that $X(U) \simeq \Omega^V X(U \oplus V)$ via the structure maps.
 - * A map of orthogonal spectra $f : X \rightarrow Y$ is a π_* -isomorphism if Uf is a π_* -isomorphism of prespectra, where $U : \mathbf{Sp}^{\mathcal{J}} \rightarrow \mathbf{Sp}^{\mathbb{N}}$ is the pullback.
- (Bousfield Localization). Let $\mathcal{C}$ model category and S a set of maps in $\mathcal{C}$. Bousfield localization produces a new model structure on $\mathcal{C}$ such that morphisms in S are weak equivalences. Localization is given by fibrant replacement. In fact, weak equivalences are S -local equivalences, the cofibrations are exactly those in $\mathcal{C}$, and the fibrants are the S -local objects.
- Now consider the model structure on $\mathcal{D}$ -spaces, or prespectra, or orthogonal spectra. We can first define a level-wise structure, where weak equivalences and fibrations are detected pointwise in $\mathbf{Top}_*$. We can apply Bousfield localization to the stable equivalences: $f : X \rightarrow Y$ such that $[Y, Z] \simeq [X, Z]$ for all Ω -spectrum Z . This gives the stable model structure. The stable equivalences are the same as π_* -isomorphisms in $\mathbf{Sp}^{\mathcal{J}}$ or $\mathbf{Sp}^{\mathbb{N}}$.
 - This implies that the stable model structure is stable: $X \rightarrow \Omega \Sigma X$ is a π_* -isomorphism.
- $P \dashv U$ is a Quillen equivalence and gives $\text{Ho}(\mathbf{Sp}^{\mathbb{N}}) \simeq \text{Ho}(\mathbf{Sp}^{\mathcal{J}})$. The homotopy category is called the stable category, and it has a triangulated structure by suspension. There is also a Quillen adjunction $\Sigma^\infty \dashv \Omega^\infty$ between $\mathbf{Top}_*$ and $\mathbf{Sp}^{\mathcal{J}}$.

- (Monads and Algebras). Let M a monad on $\mathcal{C}$, then the category $\mathcal{C}[M]$ of algebras over M is the category whose objects are pairs $X \in \mathcal{C}$ and structure maps $m : MX \rightarrow X$ satisfying associativity and unitality.
 - Some examples: $\mathbb{T}X := \bigvee X^{\wedge n}$ gives a monad over $\mathbf{Sp}^{\mathcal{J}}$ whose algebras are associative monoids. $\mathbb{P}X := \bigvee X^{\wedge n} / \Sigma_n$ gives a monad over $\mathbf{Sp}^{\mathcal{J}}$ whose algebras are commutative monoids (i.e., commutative ring spectra).
 - There is also an adjunction $\mathcal{C} \dashv U$ between $\mathcal{C}$ and $\mathcal{C}[M]$. Suppose $\mathcal{C}$ is a model category, we can show that $\mathcal{C}[M]$ is complete. We want: 1. $\mathcal{C}[M]$ being cocomplete and 2. a model structure on $\mathcal{C}[M]$ from the model structure on $\mathcal{C}$.
 - * $\mathcal{C}[M]$ is cocomplete if M preserves reflexive coequalizers. $\mathbb{T}$ and $\mathbb{P}$ do under mild hypotheses.
 - * Under mild hypotheses $\mathcal{C}[M]$ inherits a model structure from $\mathcal{C}$. $\mathbb{T}$ satisfies the hypotheses while $\mathbb{P}$ does not.
- (Universe). Recall that a representation (G acting on a finite dimensional real inner product space V) is irreducible if there is no non-trivial invariant subspace. A universe is a countably infinite-dimensional real inner product space with a G -action through isometries. There is a collection R of irreducible representations such that U contains countably many copies of each representation in R , and R always contains all trivial representations. U is called a complete universe if R contains all irreducible representations.
 - The motivation: we want to know which orbits G/H are dualizable after suspension. It connects closely to the existence of G -embeddings $G/H \rightarrow V$ for some representation V that is contained in U . Now we know G/H is embedded in $\mathbb{R}[G/H]$ whose basis elements are e_{gH} . It therefore burns down to if each component of the irreducible decomposition of $\mathbb{R}[G/H]$ is contained in U (or in R).
 - Note that such an embedding exists if we have the stabilizer $G_p = H$ for some $p \in V$, forgetting the linear structure. Therefore, if G acts trivially on U , then there is no such embedding besides the trivial one, and this is why the homotopy groups are exactly the nonequivalent homotopy groups.
- (G -prespectra). A G -prespectrum X is the data of a based G -space $X(V)$ for each finite-dimensional subspace $V \subset U$ and G -maps $S^W \wedge X(V) \rightarrow X(V \oplus W)$ satisfying associativity. It is an Ω -prespectrum if for all orthogonal pairs V and W , $X(V) \simeq \Omega^W X(V \oplus W)$ via the structure maps.
 - Note that here $\Omega^W X(V \oplus W) := \text{Map}(S^W, X(V \oplus W))$ because we want Ω^W to be adjoint to Σ^W , the smash product with S^W .
 - We define the homotopy groups by:
 - * $\pi_q^H(X) = \text{colim}_V \pi_q^H \Omega^V X(V)$.
 - * $\pi_{-q}^H(X) = \text{colim}_{V \supset \mathbb{R}^q} \pi_0^H \Omega^{V - \mathbb{R}^q} X(V)$.
 - We can put a model structure on the category of G -prespectra where the weak equivalences are the π_* -isomorphisms. The equivariant stable homotopy category structured by U is the homotopy category of G -prespectra.
- (Orthogonal G -spectra). In the data of a G -prespectra, we only have structure maps for inclusions $V \rightarrow V \oplus W$. In some sense we want to encode the structure of all G -equivariant isometries. Let $I_G(V, V')$ to be the collection of such maps between V and V' , both of which are finite-dimensional subspaces of U .
 - We have a bundle $I_G(V, V') \times V'$ over $I_G(V, V')$. The subbundle $E(V, V')$ contains pairs (f, x) where $x \notin f(V)$. $\tilde{I}_G(V, V')$ denotes the Thom space of this bundle. This in some sense is topologizing $S^{V' \ominus f(V)}$ where f is varying.

- Let $\tilde{I}_G$ be the category whose objects are finite-dimensional subspaces of U and morphisms are $\tilde{I}_G(V, V')$. An orthogonal G -spectrum is a $\mathbf{Top}_*$ -enriched functor $\tilde{I}_G \rightarrow G\mathbf{Top}_*$.
- More explicitly, an orthogonal G -spectrum X contains the following data: based G -space $X(V)$ for each finite-dimensional $V \subset U$, and structure maps $\tilde{I}_G(V, V') \wedge X(V') \rightarrow X(V)$.
- We denote the category of orthogonal G -spectra to be $\mathbf{Sp}^G$. It is symmetric monoidal under the Day convolution, and a map in $\mathbf{Sp}^G$ is a π_* -isomorphism if the map is after applying the forgetful functor to the category of G -prespectra.
- The unit in $\mathbf{Sp}^G$ is $\mathbb{S}$, where $\mathbb{S}(V) = S^V$.
- (The Naive Case). We can be more specific for the naive case, where $U = \mathbb{R}^\infty$. A naive orthogonal G -spectrum E contains the following data: a based G -space E_n for each n and maps $\tilde{I}_G(\mathbb{R}^n, \mathbb{R}^m) \wedge E_n \rightarrow E_m$. Note that $\tilde{I}_G(\mathbb{R}^n, \mathbb{R}^m)$ is just the space of isometries $\mathbb{R}^n \rightarrow \mathbb{R}^m$. In particular, when $n = m$, this gives an $O(n)$ action on E_n ; when $n = 0$, this gives $S^m \wedge E_0 \rightarrow E_m$; in general this is $O(m)_+ \wedge_{O(m-n)} S^{m-n}$, and therefore gives $O(m)_+ \wedge_{O(m-n)} S^{m-n} \wedge E_n \rightarrow E_m$.
 - In other words, we can ask the structure maps to be $S^{m-n} \wedge E_n \rightarrow E_m$ that is G -equivariant and is $O(m-n) \times O(n)$ -equivariant.
- The evaluation at V has an adjoint $F_V : G\mathbf{Top}_* \rightarrow \mathbf{Sp}^G$. We can put a stable model structure on $\mathbf{Sp}^G$ where the weak equivalences are π_* -isomorphisms, fibrations are level-wise fibrations, and cofibrations are generated by $F_V((G/H \times S^{n-1})_+ \rightarrow (G/H \times D^n)_+)$.
- For $H \subset G$ a subgroup we have restriction $i_H : \mathbf{Sp}^G \rightarrow \mathbf{Sp}^H$. This has a left adjoint $G_+ \wedge_H -$ and a right adjoint $F_H(G_+, -)$, which are also left and right Quillen functors.
- $\mathbf{Sp}^G$ is closed symmetric monoidal, and is tensored and cotensored over $G\mathbf{Top}$ via $A_+ \wedge -$ level-wise and $F(A_+, -)$ level-wise. We now define the notion of fixed points, each of which is a functor $\mathbf{Sp}^G \rightarrow \mathbf{Sp}^{W_G H}$ for a closed subgroup H , where $W_G H = N_G H / H$ and $N_G H = \{g, gHg^{-1} = H\}$:
 - (Categorical). $X^H := F(G/H_+, X)$. This is analogous to $(-)^H$ for $G\mathbf{Top}$.
 - (Homotopy). $X^{hH} := F(EH_+, X)^H$, where EH is a H -space. This means levelwise taking the set of all maps and make it a H -space via H -conjugation.
 - (Geometric). The unique functor ϕ^H such that:
 - * if X and Y both cofibrant, then $\phi^H(X) \wedge \phi^H(Y) \simeq \phi^H(X \wedge Y)$,
 - * preserves homotopy colimits,
 - * $\phi^H \Sigma^\infty X \simeq \Sigma^\infty X^H$.
 - A map $X \rightarrow Y$ in $\mathbf{Sp}^G$ is a weak equivalence if it is after applying $(-)^H$ or ϕ^H for every subgroup H .
- Note that a finite G -set is a coproduct of orbits G/H . This motivates another definition for the Burnside category. Let $\text{Span}(\mathcal{C})$ be the category whose objects are those in $\mathcal{C}$ and morphisms are equivalent classes of diagrams

$$\begin{array}{ccc} & Z & \\ \swarrow & & \searrow \\ X & & Y. \end{array}$$

Compositions are defined by pulling back. $\text{Span}^+(\mathcal{C})$ is the preadditive completion of $\text{Span}(\mathcal{C})$, where each Hom set is the Grothendieck group of the Hom set in $\text{Span}(\mathcal{C})$. We now define $B_G = \text{Span}^+(G\mathbf{Set}^{\text{fin}})$. This is equivalent to $\pi_0 \tilde{B}_G$, where $\tilde{B}_G$ is the full subcategory of $\mathbf{Sp}^G$ spanned by finite G -sets. We now define a Mackey functor to be an enriched functor $B_G \rightarrow \mathbf{Ab}$.

- Note that a Mackey functor F gives rise to functors $F_* : \mathcal{O}_G \rightarrow \mathbf{Ab}$ and $F^* : \mathcal{O}_G^{\text{op}} \rightarrow \mathbf{Ab}$ via the inclusion $\mathcal{O}_G^{\text{op}} \rightarrow B_G$ or $\mathcal{O}_G \rightarrow B_G$. For example, the homotopy group π_*^H lifts to a Mackey functor. In fact, a Mackey functor is determined by its restrictions to $\mathcal{O}_G$. F^* and F_* always satisfy the Chevalley condition.
- (Cohomology). Let M be a Mackey functor. There is associated an equivariant Eilenberg-Mac Lane spectrum $HM \in \mathbf{Sp}^G$ whose homotopy groups are determined by M . we define

$$H^n(X; M) := [X, \Sigma^n HM].$$

More generally we can define an $RO(G)$ -graded cohomology theory by

$$H^V(X; M) := [X, \Sigma^V HM].$$

- The category of Mackey functors is Abelian.

5.4 Brown Representability for $\mathbf{Sp}^G$

- (Representation Ring). $RO(G)$ is the Grothendieck group of (isomorphism classes of) finite-dimensional real G -representations. For example, if $G = C_2$, then $RO(G) = \mathbb{Z}\{\mathbf{1}\} \oplus \mathbb{Z}\{\sigma\}$ where σ is the sign representation. For a representation V , we can define its character $\chi_V : G \rightarrow \mathbb{R}$ where $\chi_V(g) = (g \cdot -)$. $RO(G)$ can be thought as the ring of functions $G \rightarrow \mathbb{R}$ that are characters.
- (Real Equivariant Bott Periodicity) In particular take $G = C_2$ and $\mathbb{R}^{p,q}$ to be the C_2 -representation with p -copies of trivial representations and q copies of sign representations. Let $KR^0(X)$ to be the Grothendieck group of real equivariant vector bundles on X . We have $KR^0(S^{1,1} \wedge X) \simeq KR^0(X)$, and we can define $KR^{p,q}(X) := KR^0(S^{p+n, q+n} \wedge X)$ for some large n , and obtain a $RO(C_2)$ -graded cohomology theory.
- (Actual Definition for $RO(G)$ -graded Cohomology). One issue for defining the $RO(G)$ -graded cohomology theory is that you cannot really work with isomorphism classes of representations. Instead, let U be a universe and $RO(G; U)$ the category of finite-dimensional representations embedded equivariantly in U , with morphisms G -equivariant isometric isomorphisms. We define two maps to be homotopic if their one-point compactification are stably homotopic. We can therefore talk about $\text{Ho}(RO(G; U))$.
- Explicitly, an $RO(G)$ -graded cohomology theory assigns an Abelian group $E^V(X)$ for each pair $(V, X) \in \text{Ho}(RO(G; U)) \times \text{Ho}(G\mathbf{Top})^{\text{op}}$. There should be isomorphisms $\sigma_W : E^V(X) \rightarrow E^{V \oplus W}(\Sigma^W X)$, such that for each V , E^V satisfies the wedge and cofiber axioms, and that for an isometric isomorphism $\alpha : W \rightarrow W'$, we have a commutative diagram:

$$\begin{array}{ccc} E^V(X) & \xrightarrow{\sigma_W} & E^{V \oplus W}(\Sigma^W X) \\ \sigma_{W'} \downarrow & & \downarrow (\text{id} \oplus \alpha, \text{id}) \\ E^{V \oplus W'}(\Sigma^{W'} X) & \xrightarrow{(\text{id} \oplus \text{id}, \alpha)} & E^{V \oplus W'}(\Sigma^W X) \end{array}$$

- (Brown Representability). For $E \in \mathbf{Sp}^G$, we can define $E_G^V(X) := [S^V \wedge X, E]_G$ which gives an $RO(G)$ -graded cohomology theory. Conversely, any $RO(G)$ -graded cohomology theory is represented by a G -spectrum.

6 Higher Topos Theory

6.1 Introduction

- (The homotopy type of) A topological space is encoding essentially the same data as an ∞ -groupoid does. Therefore, we can define an ∞ -category to be a category enriched over **Top**.
- On the other hand, we can define an ∞ -category via inner horn lifting property. $\text{Sing } X$ and $N\mathcal{C}$ are ∞ -categories, and the former is a Kan complex.
 - We do not require the uniqueness of the lift, because we only need the lift to be unique up to higher homotopy. Consider the fundamental groupoid of a space for example.
- We can view the homotopy category of a topological category as a category enriched over $\mathcal{H}$, the homotopy category of CW complexes, and two topological category should be seen as weak equivalent if the induced functor on $\mathcal{H}$ -enriched categories is an equivalence.
 - However, note that $h\mathcal{C}$ alone does not determine $\mathcal{C}$ up to equivalence; we need an equivalence $h\mathcal{C} \rightarrow h\mathcal{D}$ to say $\mathcal{C} \simeq \mathcal{D}$.
- To connect the two notions, we use simplicial categories. Note that one can go between a simplicial category and a topological category via Sing and $|-|$. One can also talk about the $\mathcal{H}$ -enriched category $h\mathcal{C}$ for a simplicial category $\mathcal{C}$.
 - We define the simplicial nerve of a simplicial category $N\mathcal{C}$ via $\mathbf{sSet}(\Delta^n, N\mathcal{C}) = \mathbf{sCat}(\mathbb{C}[\Delta^n], \mathcal{C})$ where $\mathbb{C}[-]$ sends an ordered set to a simplicial category. We can extend $\mathbb{C}$ to $\mathbb{C} : \mathbf{sSet} \rightarrow \mathbf{sCat}$, and it is left adjoint to N .
 - Let $\mathcal{C}$ simplicial category. If $\text{Map}(X, Y)$ is a Kan complex for every pair (X, Y) , then $N\mathcal{C}$ is an ∞ -category. In particular, if $\mathcal{C}$ is a topological category, then $N\mathcal{C} := N\text{Sing } \mathcal{C}$ is an ∞ -category.
 - In fact one can show that N and $\mathbb{C}[-]$ are homotopy inverse to each other:

$$|\text{Map}_{\mathbb{C}[N\mathcal{C}]}(X, Y)| \rightarrow \text{Map}_{\mathcal{C}}(X, Y)$$
 is a weak homotopy equivalence of topological spaces, where $X, Y \in \mathcal{C}$ a topological category.
- We define $\mathcal{C}^{\text{op}}$ by inverting the mapping spaces.
- Let $\mathcal{C}$ ∞ -category, we can define $\text{Hom}_{\mathcal{C}}^R(x, y)$ with n -simplices being the collection of $z : \Delta^{n+1} \rightarrow \mathcal{C}$ such that $z|_{\Delta^{\{n+1\}}} = y$ and $z|_{\Delta^{\{0,1,\dots,n\}}} = x$. We can similarly define $\text{Hom}_{\mathcal{C}}^L(x, y)$ and this is homotopy equivalent to $\text{Hom}_{\mathcal{C}}^R(x, y)$.
 - $\text{Hom}_{\mathcal{C}}^R(x, y)$ is a Kan complex and represents $\text{Map}_{\mathcal{C}}(x, y) \in \mathcal{H}$. $\mathcal{C}$ being an ∞ -category allows compositions of maps.
- $h : \mathbf{sSet} \rightarrow \mathbf{Cat}$ is the left adjoint to N .
 - For $\mathcal{C}$ an ∞ -category, we can define $\pi(\mathcal{C})$ that is naturally equivalent to $h\mathcal{C}$, with a more concrete construction: objects are objects in $\mathcal{C}$ and morphisms are homotopy classes of edges $\Delta^1 \rightarrow \mathcal{C}$ that goes between x and $y \in \mathcal{C}$ (two edges being homotopic is an equivalence relation on edges joining x and y , and compositions are well-defined in $\pi(\mathcal{C})$).
 - A morphism $f : X \rightarrow Y$ in an ∞ -category $\mathcal{C}$ is an equivalence if it is an isomorphism in $h\mathcal{C}$.
- The inclusion of Kan complexes into ∞ -categories has a right adjoint: $\mathcal{C} \rightarrow \mathcal{C}'$ where $\mathcal{C}'$ is the largest Kan complex contained in $\mathcal{C}$ (every edge in $\mathcal{C}'$ is an equivalence).

- Working on $h\mathcal{C}$ is somewhat similar to working on $\mathcal{C}$, but one distinction is that the homotopy between two maps in $\mathcal{C}$ should be a datum. Commutative diagrams in $h\mathcal{C}$ corresponds to homotopy commutative diagrams in $\mathcal{C}$ (for example, a 1-functor $\mathcal{C} \rightarrow \mathcal{H}$ does not always lift to $\mathcal{C} \rightarrow \mathbf{Top}$).
 - This motivates the definition of a homotopy coherent diagram in an ∞ -category $\mathcal{C}$, namely it is a functor $\mathcal{J} \rightarrow h\mathcal{C}$ with additional data that allows us to lift the functor to $\mathcal{J} \rightarrow \mathcal{C}$. We can also define a homotopy coherent diagram to be a map of simplicial sets $N\mathcal{J} \rightarrow \mathcal{C}$.
- The general slogan: if two objects in $\mathcal{C}$ are equivalent, then the ∞ -categorical notions should be the same for them.
 - Let $\mathcal{C}$ be an ∞ -category, then $\mathrm{Fun}(K, \mathcal{C}) := \mathbf{sSet}(K, \mathcal{C})$. This in particular gives the notion of the ∞ -category of functors between ∞ -categories.
 - One can define the join of ∞ -categories such that $N(\mathcal{C} \star \mathcal{C}') = N\mathcal{C} \star N\mathcal{C}'$. In particular, the join of ∞ -categories is an ∞ -category.
 - * Let $K \in \mathbf{sSet}$, the left/right cones $K^{\triangleleft}$ and $K^{\triangleright}$ are defined as $\Delta^0 \star K$ and $K \star \Delta^0$. Both have a distinguished vertex.
 - $F : \mathcal{C} \rightarrow \mathcal{D}$ is essentially surjective if hF is, and is fully faithful if $\mathrm{Map}_{h\mathcal{C}}(X, Y) \rightarrow \mathrm{Map}_{h\mathcal{D}}(FX, FY)$ is an isomorphism in $\mathcal{H}$.
 - A subcategory of $\mathcal{C}$ is a pullback after choosing a subcategory $h\mathcal{C}' \subset h\mathcal{C}$:

$$\begin{array}{ccc} \mathcal{C}' & \longrightarrow & \mathcal{C} \\ \downarrow & & \downarrow \\ Nh\mathcal{C}' & \longrightarrow & Nh\mathcal{C} \end{array}$$

- Let $\mathcal{C}$ ∞ -category, defining the initial/final objects to be the ones in $\mathcal{C}$ as a 1-category would be too strict. Instead, we say $Y \in \mathcal{C}$ is final if $\mathrm{Hom}_{\mathcal{C}}^R(X, Y)$ is contractible.
- The above notions allow us to define colimit/limit for $p : K \rightarrow \mathcal{C}$ a map between simplicial sets as the initial/final object in diagrams $K \rightarrow \mathcal{C}$ under/below p . On the other hand one can also identify such under/overcategories with the category of maps $K^{\triangleleft}$ or $K^{\triangleright} \rightarrow \mathcal{C}$.
- In particular, we define $N\mathbf{Kan} = \mathfrak{S}$ to be the ∞ -category of spaces, where $\mathbf{Kan} \subset \mathbf{sSet}$ is simplicial.

6.2 Fibrations of Simplicial Sets

- $X \rightarrow S$ is a Kan (trivial) fibration if it has RLP for all $\Lambda_k^n \rightarrow \Delta^n$ ($\partial\Delta^n \rightarrow \Delta^n$). $A \rightarrow B$ is anodyne (cofibration) if it has LLP for all Kan (trivial) fibration.
 - We can define left/right/inner fibration/anodyne by restricting k (e.g., an ∞ -category is characterized by $\mathcal{C} \rightarrow *$ inner fibration).
 - The small object argument shows that any map $X \rightarrow Z$ can be factorized into a cofibration/anodyne followed by trivial/Kan fibration (perhaps with left/right/inner).
- (Cofibered in Groupoids over $\mathcal{D}$). Let $F : \mathcal{D} \rightarrow \mathbf{Gpd}$ a functor. We can construct $\mathcal{C}_F$ with a functor $\mathcal{C}_F \rightarrow \mathcal{D}$. However, not all $\mathcal{C} \rightarrow \mathcal{D}$ arise from a functor to $\mathbf{Gpd}$, so we need the notion of $\mathcal{C}$ being cofibered in groupoids over $\mathcal{D}$, and informally functors $F : \mathcal{D} \rightarrow \mathbf{Gpd}$ and categories cofibered in groupoids over $\mathcal{D}$ are equivalent.
 - In particular, $F : \mathcal{C} \rightarrow \mathcal{D}$ is cofibered in groupoids over $\mathcal{D}$ iff NF is a left fibration of simplicial sets.

- Precisely, we can construct a Quillen equivalence $\mathbf{sSet}^{\mathbb{C}[S]} \rightarrow \mathbf{sSet}_{/S}$ where fibrants are the left fibrations over S . This suitable model structure on $\mathbf{sSet}_{/S}$ is called the covariant model structure.
- Kan complexes are those $X \rightarrow *$ with RLP w.r.t. left fibration, so for any left fibration $p : X \rightarrow S$, the fiber $X_s = X \times_S \{s\}$ is a Kan complex (an ∞ -groupoid).
 - In particular, we can define $f_l : X_s \rightarrow X_{s'}$ given an edge $s \rightarrow s'$ with higher coherence data. So a left fibration over S is more or less like a functor $S \rightarrow \mathfrak{S}$.
 - In fact, $p : X \rightarrow S$ is a Kan fibration iff $X_s \rightarrow X_{s'}$ is an isomorphism in $\mathcal{H}$. This gives a construction $X_{s'} \rightarrow X_s$.
- A functor $\mathcal{C} \rightarrow \mathfrak{S}$ is a “cosheaf of spaces” on $\mathcal{C}$. Examples of such include the functors sending $D \rightarrow \text{Map}_{\mathcal{C}}(C, D)$, and such functors arise from a left fibration $f : \mathcal{C}_{\mathcal{C}} \rightarrow \mathcal{C}$. In fact, $\mathcal{C}_p \rightarrow \mathcal{C}$ is a left fibration for $p : K \rightarrow \mathcal{C}$ a diagram.
- (Straightening & Unstraightening). Let S simplicial set and $\mathcal{C}$ simplicial category. $\phi : \mathbb{C}[S] \rightarrow \mathcal{C}^{\text{op}}$ a simplicial functor. With the correct model structure, we have a Quillen adjunction:

$$\mathbf{sSet}_{/S} \begin{array}{c} \xrightarrow{\text{St}_{\phi}} \\ \xleftarrow{\text{Un}_{\phi}} \end{array} \mathbf{sSet}^{\mathcal{C}}$$

and this is a Quillen equivalence if ϕ is an equivalence of simplicial categories. When ϕ is the identity, we call the maps St_S and Un_S .

- The general philosophy: we can define straightening $\text{St} : \mathbf{sSet}_{/S} \rightarrow \mathcal{M}^{\mathbb{C}[S]}$, where the target has a matching model structure as on $\mathbf{sSet}_{/S}$ via the Quillen inverse Un . For an arbitrary $\mathcal{M}$, we cannot expect to always being able to pull the model structure back.
- Let $\mathcal{C}$ a category with small colimit, then a cosimplicial object $C^* \in \mathcal{C}$ determines a functor $\mathbf{sSet} \rightarrow \mathcal{C}$ via the coend construction. We denote $FS = |S|_{C^*}$. F has a right adjoint Sing_{C^*} .
 - In particular, suppose $X = \{x\} \in \mathbf{sSet}$. We can identify St_X with the geometric realization $| - |_{Q^*}$ for some Q^* cosimplicial in $\mathbf{sSet}$.
 - One can show that $| - |_{Q^*}$ and Sing_{Q^*} give a Quillen equivalence for $\mathbf{sSet}$ and itself, with the correct model structure.
- (Joyal Model Structure). We now have a model structure on $\mathbf{sSet}$ where cofibrations are inclusions, fibrants are ∞ -categories, and weak equivalences are those that are equivalence after applying $\mathbb{C}[-]$ and move to $\mathbf{sCat}$. Also, $(\mathbb{C}, N)$ gives a Quillen equivalence between $\mathbf{sSet}$ (with this model structure) and $\mathbf{sCat}$.
- $X \rightarrow S$ being an inner fibration implies that each fiber is an ∞ -category. However the fibers do not really functorially depend on S .
- A correspondence between two categories is a functor $M : \mathcal{C}^{\text{op}} \times \mathcal{C}' \rightarrow \mathbf{Set}$. We can form $\mathcal{C} \star^{\mathcal{M}} \mathcal{C}'$. Given $F : \mathcal{C}' \rightarrow \mathcal{C}$, we can define $M_F(c, c') = \mathcal{C}(c, Fc')$.
 - Given an inner fibration $X \rightarrow S$ and $f : s \rightarrow s'$ in S an edge, we can define a correspondence between X_s and $X_{s'}$.
 - We might want to view an inner fibration as a functor from S to the ∞ -category $\infty\mathbf{Cat}$ where morphisms in the latter are correspondences. However, this does not work because correspondences do not compose nicely. The correct notion would be Cartesian fibrations.
- (Minimal Kan Complexes). We can classify Kan complexes via minimal Kan complexes; an equivalence between the latter is an isomorphism. In fact, a more general notion, minimal ∞ -categories, is possible.

- (n -Truncated). We can define n -categories essentially by saying two maps $\Delta^m \rightarrow \mathcal{C}$ are the same if they agree on the boundary (where $m \geq n$).
 - This leads to the notion of k -truncated: explicitly, a Kan complex K is k -truncated if $\pi_i(X, x) \simeq *$ for $k < i$ and $x \in X$. All the mapping spaces of $\mathcal{C}$ are $(k-1)$ -truncated iff the minimal model of $\mathcal{C}$ is an k -category.
- We want to know whether a correspondence M arises from such a functor. This requires the notion of Cartesian morphisms, and motivates the following definition:
 - Let $p : X \rightarrow S$ inner fibration (RLP for Λ_k^n where $0 < k < n$). $f : x \rightarrow y$ an edge in X is called p -Cartesian if $X_{/f} \rightarrow X_{/y} \times_{S_{/py}} S_{/pf}$ is a trivial Kan fibration.
 - Let $p : N(\mathcal{M}) \rightarrow \Delta^1$ a map, then $f : x \rightarrow y$ in $\mathcal{M}$ which maps isomorphically onto Δ^1 is p -Cartesian iff it is Cartesian in the classical sense.
 - In particular, let $p : \mathcal{C} \rightarrow \mathcal{D}$ an inner fibration between ∞ -categories and $f : c \rightarrow c'$ morphism in $\mathcal{C}$. f is an equivalence iff it is p -Cartesian and pf is an equivalence.
- Let $p : X \rightarrow S$ between simplicial sets. p is a Cartesian fibration if it is an inner fibration, and that for every edge $f : x \rightarrow y$ in S and $\tilde{y} \in X$ lifting y , there exists $\tilde{f} : \tilde{x} \rightarrow \tilde{y}$ p -Cartesian that lifts f .
 - Now let $p : X \rightarrow S$ an inner fibration, we can associate an ∞ -category X_s to each $s \in S$, and a correspondence $X_e : X \times_S \Delta^1 \rightarrow \Delta^1$ where $(X_e)_0 = X_s$ and $(X_e)_1 = X_{s'}$.
 - Being a Cartesian fibration is essentially saying that the correspondences come from functors $X_{s'} \rightarrow X_s$. Or in other word a Cartesian fibration gives a functor $S^{\text{op}} \rightarrow \infty\mathbf{Cat}$. However, it should be noted that equivalence between Cartesian fibrations do not give equivalence between functors. This motivates the definition of marked simplicial sets.
 - An inner fibration is a right fibration iff it is Cartesian and that every fiber is a Kan complex, or if every edge in X is p -Cartesian.
- We can always talk about locally p -Cartesian or locally Cartesian fibration by pulling back to $X \times_S \Delta^1 \rightarrow \Delta^1$.
 - A locally Cartesian fibration is a Cartesian fibration iff every locally p -Cartesian edge in X is p -Cartesian.
- Let $p : \mathcal{C} \rightarrow \mathcal{D}$ (co)Cartesian fibration. $q : K \rightarrow \mathcal{C}$ diagram. Then $\mathcal{C}_{/q} \rightarrow \mathcal{D}_{/pq}$ is a (co)Cartesian fibration, and an edge in $\mathcal{C}_{/q}$ is p' -(co)Cartesian iff the image of f in $\mathcal{C}$ is.
- Let $p : \mathcal{C} \rightarrow \mathcal{D}$ an inner fibration of ∞ -categories, $\bar{e} : pX \rightarrow pY$ a morphism in $\mathcal{D}$, $e : X' \rightarrow Y$ locally p -Cartesian lifting $\bar{e}$. Then we have a fiber sequence in $\mathcal{H} : \text{Map}_{\mathcal{C}_{pX}}(X, X') \rightarrow \text{Map}_{\mathcal{C}}(X, Y) \rightarrow \text{Map}_{\mathcal{D}}(pX, pY)$.
- Let $p : \mathcal{C} \rightarrow \mathcal{D}$ inner fibration of ∞ -cats. Let $f : Y \rightarrow Z$ in $\mathcal{C}$. f is p -Cartesian iff for every $X \in \mathcal{C}$, the following diagram is homotopy Cartesian:

$$\begin{array}{ccc}
 \text{Map}(X, Y) & \longrightarrow & \text{Map}(X, Z) \\
 \downarrow & & \downarrow \\
 \text{Map}(pX, pY) & \longrightarrow & \text{Map}(pX, pZ)
 \end{array}$$

6.3 The ∞ -Category of ∞ -Categories

- We define $\infty\mathbf{Cat}$ to be the ∞ -category $N(\infty\mathbf{Cat}^\Delta)$ where $\infty\mathbf{Cat}^\Delta$ is the simplicial category with mapping spaces defined by the taking all invertible functors between two ∞ -categories.

- We want to do fibrant replacements: let $p : X \rightarrow S$, we can factor it through $q \circ \phi : X \rightarrow Y \rightarrow S$ where q is a Cartesian fibration. We want to know what edges in X has image q -Cartesian in Y , and this requires the notion of marked simplicial sets.
 - Motivation: with right fibrations we can induce functors from the category of right fibrations over S to $\text{Fun}(S^{\text{op}}, \mathbf{Space})$. However, this does not work when we go to Cartesian fibrations, because two Cartesian fibrations can be isomorphic over S with completely different collections of Cartesian edges. We therefore need to not only remember the fibration but also the collection of Cartesian edges.
 - In essence, a marked simplicial set $(X, \mathcal{E})$ consists of $X \in \mathbf{sSet}$ and $\mathcal{E}$ a family of edges in X that contains all degenerate edges. We denote the category $\mathbf{sSet}^+$. We want to endow each $\mathbf{sSet}^+/_S$ a model structure.
 - We define a notion called marked anodyne maps in $\mathbf{sSet}^+$ so that $p : X \rightarrow S$ has RLP w.r.t. all marked anodyne maps iff p is an inner fibration, edge e marked iff $p(e)$ marked and e p -Cartesian, and that every marked edge in S can be lifted to a marked edge in X .
 - Two default marked simplicial sets: $S^\flat$ where marked edges are the degenerates, and $S^\sharp$ where every edge is marked.
- $X^\sharp$ is used to denote, given $p : X \rightarrow S$, the marked simplicial set $(X, \mathcal{E})$ where $\mathcal{E}$ contains all p -Cartesian edges. Now, $Y \rightarrow S^\sharp$ has RLP w.r.t. marked anodyne maps iff $Y \rightarrow S$ is Cartesian and $Y = Y^\flat$.
- We can now define a model structure on $\mathbf{sSet}^+/_S$ where cofibrations are those that are cofibrations in $\mathbf{sSet}$ and weak equivalences are the Cartesian equivalences.
 - Now, X is fibrant iff $X \simeq Y^\sharp$, where $Y \rightarrow S$ is a Cartesian fibration.
- (Fibered Categories). Let $\chi : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Cat}$ be a functor. The Grothendieck construction assigns a new category $\tilde{\mathcal{C}}$ where objects are $(C, \eta), C \in \mathcal{C}, \eta \in \chi(C)$, and morphisms are $(f, \alpha), f : C \rightarrow C', \alpha : \eta \rightarrow \chi(f)(\eta')$. There is an equivalence between functors $\mathcal{C}^{\text{op}} \rightarrow \mathbf{Cat}$ and categories fibered over $\mathcal{C}$ (given $\phi : C \rightarrow C'$ and F' over C' , we can lift ϕ to $\mathcal{F}$).
 - We try to mimic the approach by replacing $\mathcal{C}$ by a simplicial set S , $\mathbf{Cat}$ by $\infty\mathbf{Cat}$, and fibered categories by Cartesian fibrations. Note that the category of simplicial functors $\mathbb{C}[S]^{\text{op}} \rightarrow \mathbf{sSet}^+$ equipped with the projective model structure has the underlying ∞ -category equivalent to $\text{Fun}(S^{\text{op}}, \infty\mathbf{Cat})$.
 - (Straightening-Unstraightening). Let $\phi : \mathbb{C}[S] \rightarrow \mathcal{C}^{\text{op}}$ simplicial functor, we have an adjunction:

$$\mathbf{sSet}^+/_S \xrightleftharpoons[\text{Un}_\phi^+]{\text{St}_\phi^+} (\mathbf{sSet}^+)^{\mathcal{C}}$$
 and this is a Quillen adjunction where on the LHS we have the Cartesian model structure and on the RHS we have the projective model structure. This is a Quillen equivalence if ϕ is an equivalence, which in particular gives our description above.
 - Recall that given a simplicial model category, we can first restrict to the fibrant-cofibrant objects (and the subcategory is Kan-enriched), and then apply the coherent nerve functor. This gives an underlying ∞ -category.
- Let f be a map between Cartesian fibrations over S as marked simplicial sets. f is a categorical equivalence iff fiberwise f induces a categorical equivalence, or iff $X^\sharp \rightarrow Y^\sharp$ is a Cartesian equivalence in $\mathbf{sSet}^+/_S$.
- $p : X \rightarrow S$ is a categorical fibration if it is a Cartesian fibration, and the converse is true if S is Kan.

- (Universal Fibration). We have a natural inclusion $\infty\mathbf{Cat}^\Delta \rightarrow \mathbf{sSet}^+$ which can be seen as a fibrant object $\mathcal{F} \in (\mathbf{sSet}^+)^{\infty\mathbf{Cat}^\Delta}$. Applying $\mathrm{Un}_{\infty\mathbf{Cat}^{\mathrm{op}}}^+$, we identify it with a Cartesian fibration $q : Z \rightarrow \infty\mathbf{Cat}^{\mathrm{op}}$ as an object in $\mathbf{sSet}_{/\infty\mathbf{Cat}^{\mathrm{op}}}^+$, and this is called the universal Cartesian fibration. For every $\mathcal{C} \in \infty\mathbf{Cat}$, its fiber of $q : Z \times_{\infty\mathbf{Cat}^{\mathrm{op}}} \{\mathcal{C}\}$ is canonically equivalent to $\mathcal{C}$.
 - In fact, every Cartesian fibration between small simplicial sets $X \rightarrow S$ is classified by a map $\phi : S \rightarrow \infty\mathbf{Cat}^{\mathrm{op}}$.
- (Limit of a Diagram). Let $p : K \rightarrow \infty\mathbf{Cat}^{\mathrm{op}}$ a diagram classified by a Cartesian fibration $p : X \rightarrow K$, then $\lim p \simeq \mathrm{Map}_K^b(K^\sharp, X^\sharp)$ in $h\infty\mathbf{Cat}$. Similarly, let $p : K \rightarrow \mathbf{Space}$ classified by $X \rightarrow K$ left fibration, then there is a natural isomorphism $\lim p \simeq \mathrm{Map}_K(K, X)$ in $\mathcal{H}$.
- (Colimit of a Diagram). Let $p : K^{\mathrm{op}} \rightarrow \infty\mathbf{Cat}$ a diagram classifying a Cartesian fibration $X \rightarrow K$, then $\mathrm{colim} p \simeq X^\sharp$ in $h\infty\mathbf{Cat}$. Similarly, let $p : X \rightarrow \mathbf{Space}$ classifying a left fibration $X \rightarrow K$, then $\mathrm{colim} p \simeq X$ in $\mathcal{H}$.

6.4 Limits and Colimits

- (Cofinal Maps). We use cofinal maps to compute colimit after replacing the diagram with a smaller one. The explicit definition: $p : S \rightarrow T$ is cofinal if for any $X \rightarrow T$ right fibration, $\mathrm{Map}_T(T, X) \rightarrow \mathrm{Map}_T(S, X)$ is a homotopy equivalence.
 - Let $K' \rightarrow K$ cofinal and $K \rightarrow \mathcal{C}$, where $\mathcal{C}$ an ∞ -category. There is a homotopy equivalences of Kan complexes between fibers of $\mathcal{C}_{p/} \rightarrow \mathcal{C}$ and $\mathcal{C}_{p \circ v/} \rightarrow \mathcal{C}$.
 - In particular, if $K^\triangleright \rightarrow \mathcal{C}$ is the colimit of $K \rightarrow \mathcal{C}$, then the restriction to $K'^\triangleright$ is the colimit of $K' \rightarrow \mathcal{C}$. The converse tells that $K' \rightarrow K$ is cofinal.
 - Cofinal maps are weak equivalences. $i : A \rightarrow B$ is cofinal iff it is a contravariant equivalence in $\mathbf{sSet}_{/B}$. In particular, if B Kan, then i is cofinal iff i is a weak homotopy equivalence.
- (Smooth Maps). Let $p : X \rightarrow Y$ between simplicial sets. In general p induces a Quillen adjunction $(p_!, p^*)$ between the contravariant model categories $\mathbf{sSet}_{/X}$ and $\mathbf{sSet}_{/Y}$. p^* has a right adjoint p_* , but (p^*, p_*) is not necessarily a Quillen adjunction.
 - p is a smooth map if for every p' pulled back from p , (p'^*, p'_*) is a Quillen adjunction. This happens iff p'^* preserves contravariant equivalences.
 - Now, suppose we have a pullback diagram

$$\begin{array}{ccc} X' & \xrightarrow{f'} & X \\ \downarrow & & \downarrow p \\ S' & \xrightarrow{f} & S \end{array}$$

with f cofinal and p smooth, then f' is cofinal.

- Let $f : \mathcal{C} \rightarrow \mathcal{D}$ where $\mathcal{D}$ an ∞ -category. f is cofinal iff for every $D \in \mathcal{D}$, $\mathcal{C} \times_{\mathcal{D}} \mathcal{D}_{D/}$ is weakly contractible. This in particular leads to Quillen's Theorem A in his work *Higher K-Theory*.
- For every $p : X \rightarrow \mathcal{C}$, we can find $\mathcal{C}^{p/}$ by the adjoint functor theorem, such that $\mathrm{Hom}_{\mathbf{sSet}}(Y, \mathcal{C}^{p/}) \simeq \mathrm{Hom}_{\mathbf{sSet}_{/X}}(X \diamond Y, \mathcal{C})$, where $\diamond$ is a new operation such that $X \diamond Y \rightarrow X \star Y$ is a categorical equivalence. Now, we can form an equivalence $\mathcal{C}_{p/} \rightarrow \mathcal{C}^{p/}$, which sends initial object from the former to the latter. Therefore we can replace $\mathcal{C}_{p/}$ by $\mathcal{C}^{p/}$ in the definition of colimit.
- (Parametrized Colimit). The idea is that $\mathrm{colim} p$ should depend functorially on p . We define a notion $X \diamond_S Y$ for X, Y over S , that is compatible with base change: $(X_T \diamond_T Y_T) \simeq (X \diamond_S Y)_T$.

- Let $K \in \mathbf{sSet}_{/S}$. The functor sending $X \in \mathbf{sSet}_{/S}$ to $X \diamond_S S \in (\mathbf{sSet}_{/S})_{K/}$ has a right adjoint, where a map $p_S : K \rightarrow Y$ over S is sent to $Y^{p_S/}$.
- Let $q : Y \rightarrow S$ a Cartesian fibration and $p_S : K \rightarrow Y$ over S . Suppose that $p_s : K_s \rightarrow Y_s$ has a colimit in Y_s for each $s \in S$, and that $q \circ p_S$ is a coCartesian fibration, then there exists p'_S such that

$$\begin{array}{ccc} K & \xrightarrow{p_S} & Y \\ \downarrow & \nearrow p'_S & \downarrow q \\ K \diamond_S S & \longrightarrow & S \end{array}$$

commutes, and that $p'_s : K_s \diamond \{s\} \rightarrow Y_s$ is a colimit of p_s . The collection of such $p'_s \in \text{Fun}_S(K \diamond_S S, Y)$ is a contractible Kan complex.

- (Decomposition of Diagrams). We want to analyze $\text{colim } p$ after restricting to some family of simplicial subsets K_I of K . We can also work in more general settings where each K_I just maps to K .
 - Now, fix $F : \mathcal{J} \rightarrow \mathbf{sSet}_{/K}$. We construct a simplicial set K_F such that there is a map $K_F \rightarrow \Delta^1$ where the fiber over 0 is K and over 1 is $N(\mathcal{J})$.
 - Suppose that for each I the map $p_I : K_I \rightarrow \mathcal{C}$ has a colimit q_I . There exists $\pi'_I : K_I^\triangleright \rightarrow K_F$ that commutes with π_I which sends the cone point to the vertex in $N(\mathcal{J})$ corresponds to I , then there exists $q : K_F \rightarrow \mathcal{C}$ such that $q \circ \pi'_I = q_I$ and $q|_K = p$. Moreover, $\mathcal{C}_{q/} \rightarrow \mathcal{C}_{p/}$ is a trivial fibration.
 - Now suppose $N(\mathcal{J}) \subset K_F$ is right anodyne, then $\mathcal{C}_{q/} \rightarrow \mathcal{C}_{p/}$ and $\mathcal{C}_{q/} \rightarrow \mathcal{C}_{q|_{N(\mathcal{J})}}$ are trivial fibrations, so in particular colimits of p can be identified with colimits of $q|_{N(\mathcal{J})}$.
- Suppose $\mathcal{C}$ and $\mathcal{J}$ are fibrant simplicial categories (the mapping spaces are Kan complexes). $C \in \mathcal{C}$ being a homotopy colimit of $F : \mathcal{J} \rightarrow \mathcal{C}$ (together with a collection of compatible maps $FI \rightarrow C$) in $\mathcal{C}$ iff after applying N and extending NF by maps $FI \rightarrow C$ to a map $N(\mathcal{J})^\triangleright \rightarrow N(\mathcal{C})$, we get a colimit diagram in $N(\mathcal{C})$.
- (Classical Kan Extension). Let $i : \mathcal{J} \rightarrow \mathcal{J}'$ be a functor between 1-categories. We have a natural functor $i^* : \mathcal{C}^{\mathcal{J}'} \rightarrow \mathcal{C}^{\mathcal{J}}$. The left adjoint to this functor is called the left Kan extension along i , and it exists if $\mathcal{C}$ has sufficiently many colimits.
- (Relative Colimit). Let $f : \mathcal{C} \rightarrow \mathcal{D}$ be an inner fibration and $\bar{p} : K^\triangleright \rightarrow \mathcal{C}$ a diagram. Let $p = \bar{p}|_K$. $\bar{p}$ is a f -colimit of p if $\mathcal{C}_{\bar{p}/} \rightarrow \mathcal{C}_{p/} \times_{\mathcal{D}_{f\bar{p}/}} \mathcal{D}_{f\bar{p}/}$ is a trivial fibration (or equivalently, a categorical equivalence). In particular, when $\mathcal{D} = *$, this agrees with colimit, and when an edge $e : \Delta^1 \rightarrow \mathcal{C}$ is a f -colimit iff it is f -coCartesian.
 - Similar to colimit, relative colimits can be computed after precomposition with a cofinal map.
 - Let $q : X \rightarrow S$ locally coCartesian fibration. $\bar{p} : K^\triangleright \rightarrow X_s$ is a q -colimit (after composition with $X_s \rightarrow X$) iff for every edge $e : s \rightarrow s'$ in S , $e_! \circ \bar{p}$ is a colimit in $X_{s'}$. In particular, relative colimit can be detected fiberwise.
- (Kan Extensions along Inclusions). Let $\mathcal{C}^0 \subset \mathcal{C}$ be a full subcategory, and for a diagram $p : K \rightarrow \mathcal{C}$, we denote $\mathcal{C}_{/p}^0 := \mathcal{C}_{/p} \times_{\mathcal{C}} \mathcal{C}^0$. In the following diagram where p an inner fibration,

$$\begin{array}{ccc} \mathcal{C}^0 & \xrightarrow{F_0} & \mathcal{D} \\ \downarrow & \nearrow F & \downarrow p \\ \mathcal{C} & \longrightarrow & \mathcal{D}' \end{array}$$

we say F is a p -left Kan extension of F_0 if for every $C \in \mathcal{C}$ the extension

$$\begin{array}{ccc} \mathcal{C}_{/C}^0 & \xrightarrow{F_C} & \mathcal{D} \\ \downarrow & \nearrow \bar{F}_C & \downarrow p \\ (\mathcal{C}_{/C}^0)^\triangleright & \longrightarrow & \mathcal{D}' \end{array}$$

gives $F(C)$ as a p -colimit of F_C . In particular, if $C \in \mathcal{C}^0$ this is automatic. In the case $\mathcal{C} \rightarrow \mathcal{C}^\triangleright$, the map $\bar{F}$ is a left p -Kan extension iff it is a p -colimit of F .

- A p -left Kan extension exists iff for every $C \in \mathcal{C}$, $\mathcal{C}_{/C}^0 \rightarrow \mathcal{C}^0 \rightarrow \mathcal{D}$ admits a p -colimit.
- In particular, if every fiber of p admits small colimits and $\mathcal{D}_E \rightarrow \mathcal{D}_{E'}$ preserves small colimits, then p -left Kan extension exists.
- Now suppose p is a categorical fibration, and every $F_0 \in \text{Map}_{\mathcal{D}'}(\mathcal{C}', \mathcal{D})$ admits a p -left Kan extension, then the left Kan extension functor $i_!$ is left adjoint to the precomposition i^* .
- (Kan Extensions along General Functors). We focus on the absolute case. Note that in general, one do not expect the unit $f \rightarrow \delta^* \delta_! f$ to be an equivalence. We take this unit as part of the data. Let $\delta : K \rightarrow K'$ and $f : K \rightarrow \mathcal{D}$ a diagram. A left Kan extension of f along δ is a functor $f' : K' \rightarrow \mathcal{D}$ and $f \rightarrow f' \circ \delta$ in $\text{Fun}(K, \mathcal{D})$.
 - Under this definition, $\delta_!$ is left adjoint to δ^* . Note that $\delta_! \delta^* F \rightarrow F$ does not need to be an equivalence as well. However, if δ is a fully faithful inclusion, then $f \rightarrow \delta^* \delta_! f$ is an equivalence.
- (Homotopy Coproduct). Let $\{X_\alpha\}$ be objects in a fibrant simplicial category $\mathcal{C}$. We can view the coproduct over X_α as the colimit for a functor from a discrete category A to $\mathcal{C}$. X is a homotopy coproduct if $\text{Map}(X, Y) \simeq \prod \text{Map}(X_\alpha, Y)$ as Kan complexes. We can also pass to the homotopy category $h\mathcal{C}$ and ask for an equivalence in $\mathcal{H}$.
 - In particular, suppose $p_\alpha : K_\alpha \rightarrow \mathcal{C}$ diagrams in an ∞ -category $\mathcal{C}$ with colimits X_α . Let $K = \bigsqcup K_\alpha$, then the colimit of $p = \bigsqcup p_\alpha$ is $\bigsqcup X_\alpha$ (and the former exists iff the latter does).
- (Pushouts). A square in an ∞ -category $\mathcal{C}$ is a diagram $\Delta^1 \times \Delta^1 \rightarrow \mathcal{C}$. Note that $(\Lambda_0^2)^\triangleright \simeq \Delta^1 \times \Delta^1 \simeq (\Lambda_2^2)^\triangleleft$. The square is a pushout or coCartesian (a pullback or Cartesian) if the corresponding diagram is a colimit (limit). Let $\mathcal{C}$ be a fibrant simplicial category, then a diagram is a homotopy pushout iff the square after applying Map is a pullback in **Kan**.
 - As in the classical case, given

$$\begin{array}{ccccc} X & \longrightarrow & Y & \longrightarrow & Z \\ \downarrow & & \downarrow & & \downarrow \\ X' & \longrightarrow & Y' & \longrightarrow & Z' \end{array}$$

where the left square is a pushout, then the right square is a pushout iff the outer square is a pushout.

- We can “decompose” colimits into a pushout of less complicated colimits. More specifically, let $p : K' = K \sqcup_L L' \rightarrow \mathcal{C}$, where $L \rightarrow L'$ is a monomorphism and colimits exist after restriction to L , L' , and K (call them Z , X , and Y), then $\text{colim } p = X \sqcup_Z Y$.
- As in the classical case, an ∞ -category admits all finite colimits iff it admits all pushouts and has an initial object. An ∞ -functor preserves all finite colimits iff it preserves pushouts and an initial object.

- Similarly, an ∞ -category admits κ -small colimits if it admits κ -small coproducts and pushouts. An ∞ -functor preserves all κ -small colimits iff it preserves pushouts and κ -small coproducts. (We can replace coproducts by coequalizers in the first sentence.)
- An ordinary category can be seen as tensored over **Set** if it admits coproducts. In particular, $S \otimes X := \bigsqcup_S X$, and this satisfies $\mathbf{Set}(S \otimes X, Y) = \mathbf{Set}(S, \mathbf{Set}(X, Y))$. Analogously, an ∞ -category is tensored over **Space** if it admits small colimits.
- Let $\mathcal{D}$ closed monoidal and $\mathcal{C}$ enriched over $\mathcal{D}$. A $\mathcal{D}$ -enriched functor $G : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$ is representable if there exists $C \in \mathcal{C}$ and $\eta : 1_{\mathcal{D}} \rightarrow GC$ such that $\text{Map}(X, C) \simeq \text{Map}(X, C) \otimes 1_{\mathcal{D}} \rightarrow \text{Map}(X, C) \otimes GC \rightarrow GX$ is an isomorphism in $\mathcal{D}$.
 - Now, let $F : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Space}$ an ∞ -functor. F is representable if $hF : h\mathcal{C} \rightarrow \mathcal{H}$ is representable. Here, $\eta : 1 \rightarrow FC$ can be regarded as $\eta \in \pi_0(FC)$.
 - In particular, let $f : \tilde{\mathcal{C}} \rightarrow \mathcal{C}$ be a right fibration, then it is classified by some $F : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Space}$. Now, F is representable by (C, η) iff $\tilde{\mathcal{C}}$ is final, where $C = f\tilde{\mathcal{C}}$ and $\eta \in \pi_0 FC \simeq \pi_0(\tilde{\mathcal{C}} \times_{\mathcal{C}} C)$ contains $\tilde{\mathcal{C}}$. We say f is representable if $\tilde{\mathcal{C}}$ has a final object.
 - Let $p : K \rightarrow \mathcal{C}$ be a diagram with constant value X . We denote $X \otimes K$ to be the colimit of p if such colimit exists. This is well-defined up to unique weak equivalence and depends on the weak homotopy type of K .
- (Retracts and Idempotents). Let $\mathcal{C}$ be a category. If Y is a retract of X , i.e., there exists $\text{id} = i \circ r : Y \rightarrow X \rightarrow Y$, then we have in particular $f := r \circ i : X \rightarrow X$ idempotent, and Y can be viewed as a equalizer $(\text{id}_X, r \circ i)$. We have an injective map from the isomorphism classes of retracts of X to the set of idempotent maps $X \rightarrow X$. This is in particular a bijection (and we say $\mathcal{C}$ is idempotent complete) if $\mathcal{C}$ admits equalizers.
 - Now, in ∞ -categorical setting, we say Y is a retract of X if it is the case in $h\mathcal{C}$, or if there exists a 2-simplex between $i \circ r$ and id . The analogous map still exists, but we cannot recover Y as a finite limit involving X .
- We can manually define simplicial sets $\text{Ret}, \text{Idem}, \text{Idem}^+$, where the latter two are ∞ -categories, with natural maps $\text{Ret} \rightarrow \text{Idem}^+$ and $\text{Idem} \rightarrow \text{Idem}^+$. We call an idempotent (weak retraction diagram, strong retraction diagram) in $\mathcal{C}$ to be a functor from Idem ($\text{Ret}, \text{Idem}^+$) to $\mathcal{C}$.
 - $\text{Fun}(\text{Idem}^+, \mathcal{C}) \rightarrow \text{Fun}(\text{Ret}, \mathcal{C})$ is a trivial fibration, so every weak retraction diagram can be extended to a strong retraction diagram.
 - An idempotent is effective if it can be extended to a strong retraction diagram, iff the functor admits a limit/colimit. $\mathcal{C}$ is said to be idempotent complete if every idempotent is effective.
 - A strong retraction diagram can be seen as the Kan extension of the restriction to an idempotent. If $\mathcal{C}$ is idempotent complete, then $\text{Fun}(\text{Idem}^+, \mathcal{C}) \rightarrow \text{Fun}(\text{Idem}, \mathcal{C})$ is a trivial fibration. In particular, every idempotent can be extended to a strong retraction diagram iff this map is a trivial fibration.

6.5 Presentable and Accessible ∞ -Categories

- (Yoneda Lemma). The classical Yoneda lemma says that $\mathcal{C} \rightarrow \text{Hom}(\mathcal{C}^{\text{op}}, \mathbf{Set})$ is fully faithful. In ∞ -categorical setting, we replace the target from presheaves of sets to presheaves of spaces. For a simplicial set S , we denote $\text{Fun}(S^{\text{op}}, \mathbf{Space})$ to be $\mathcal{P}(S)$, referred to as the ∞ -category of presheaves on S . We in fact have a map $S \rightarrow \mathcal{P}(S)$ which is also fully faithful. In addition, the map preserves all small limits.

- There are other models for $\mathcal{P}(S)$ that are canonically equivalent to each other. For example, $N(s\mathcal{P}'(S))$ where $s\mathcal{P}'(S)$ is the full subcategory of $\mathbf{sSet}_{/S}$ spanned by right fibrations over S , or $N(s\mathcal{P}''(S))$ where $s\mathcal{P}''(S)$ is the simplicial subcategory of simplicial presheaves over $\mathbb{C}[S]$ spanned by fibrant-cofibrant objects.
- $\mathcal{P}(S)$ admits all small limits and colimits.
- (Idempotent Completion). We say f exhibits $\mathcal{D}$ as an idempotent completion for $\mathcal{C}$ if $f : \mathcal{C} \rightarrow \mathcal{D}$ is fully faithful, $\mathcal{D}$ is idempotent complete, and that every object in $\mathcal{D}$ is a retract of an object in $f\mathcal{C}$.
 - We can define an idempotent completion by considering the subcategory of $\mathcal{P}(\mathcal{C})$ spanned by retracts. In particular, it is unique in the sense that if $\mathcal{E}$ is idempotent complete, then $\mathrm{Fun}(\mathcal{D}, \mathcal{E}) \rightarrow \mathrm{Fun}(\mathcal{C}, \mathcal{E})$ is an equivalence of ∞ -categories.
- We can informally view $\mathcal{P}(S)$ as the limit of the diagram $S^{\mathrm{op}} \rightarrow \infty\mathbf{Cat}$ where everything is sent to $\mathbf{Space}$. In addition we have $\mathrm{Fun}(\mathcal{C}, \mathcal{P}(S)) \simeq \mathrm{Fun}(\mathcal{C} \times S^{\mathrm{op}}, \mathbf{Space})$.
- Let $\mathrm{Fun}^L(\mathcal{D}, \mathcal{C})$ be the functors that are left adjoints. When $\mathcal{D} = \mathcal{P}(S)$, such functors are exactly those which preserve colimits. (Suppose $\mathcal{C}$ admits small colimits) Every functor $S \rightarrow \mathcal{C}$ extends to a colimit-preserving functor $\mathcal{P}(S) \rightarrow \mathcal{C}$. In other words, $\mathrm{Fun}^L(\mathcal{P}(S), \mathcal{C}) \rightarrow \mathrm{Fun}(S^{\mathrm{op}}, \mathcal{C})$ is an equivalence.
 - Colimits-preserving functors $f : \mathcal{P}(S) \rightarrow \mathcal{C}$ are exactly those that are left Kan extensions of their restrictions to the essential image of the Yoneda embedding. In particular, $F \simeq \mathrm{colim}_{(j(s) \rightarrow F) \in \mathcal{P}(S)_{jF}^{\mathrm{rep}}} j(s)$ where j is the Yoneda embedding.
 - Say $\mathcal{C}' \subset \mathcal{C}$ is stable under colimits if for any small diagram $K \rightarrow \mathcal{C}'$ which has a colimit $K^{\triangleright} \rightarrow \mathcal{C}$, the colimit factors through $\mathcal{C}'$. A collection of objects A generates $\mathcal{C}$ under colimits if any full subcategory $\mathcal{C}' \subset \mathcal{C}$ that is stable under colimits and contains A is equal to $\mathcal{C}$. In particular, $\mathcal{P}(S)$ is generated by S under small colimits via the Yoneda embedding.
- Let $f : S \rightarrow \mathcal{C}$ be a diagram, we want to know when $\mathcal{C} \simeq \mathcal{P}(S)$. If $\mathcal{C}$ admits small colimits, we can extend this map to $\mathcal{P}(S) \rightarrow \mathcal{C}$, and the extension is an equivalence iff f is fully faithful, generates $\mathcal{C}$ under colimits, and that the essential image of f consists of complete compact objects of $\mathcal{C}$.
- Let $\mathcal{M} \rightarrow \Delta^1$ be a Cartesian fibration, this can be viewed as a functor $(\Delta^1)^{\mathrm{op}} \rightarrow \infty\mathbf{Cat}$. We in particular get a functor $\mathcal{D} \rightarrow \mathcal{C}$ between ∞ -categories. There is a bijection between functors $\mathcal{D} \rightarrow \mathcal{C}$ and equivalence classes of Cartesian fibrations $\mathcal{M} \rightarrow \Delta^1$ with equivalences $\mathcal{C} \rightarrow \mathcal{M}_{\{0\}}$ and $\mathcal{D} \rightarrow \mathcal{M}_{\{1\}}$.
- (Adjunction). An adjunction between $\mathcal{C}$ and $\mathcal{D}$ is a map $p : \mathcal{M} \rightarrow \Delta^1$ that is both a Cartesian and a coCartesian fibration, with equivalences $\mathcal{C} \rightarrow \mathcal{M}_{\{0\}}$ and $\mathcal{D} \rightarrow \mathcal{M}_{\{1\}}$. Let $f : \mathcal{C} \rightarrow \mathcal{D}$ and $g : \mathcal{D} \rightarrow \mathcal{C}$ to be functors associated to p , we say f is left adjoint to g and vice versa.
 - According to the above, if f admits a right adjoint, then it is unique up to equivalences.
 - Adjunction is preserved by compositions. Adjoints can be determined fiberwise in the sense that $X \rightarrow S$ is a coCartesian fibration iff all the maps $X_{s'} \rightarrow X_s$ have left adjoints.
 - Adjunction can also be characterized by the existence of a unit in $\mathrm{Fun}(\mathcal{C}, \mathcal{C})$.
 - If f is a left adjoint then so is hf . Conversely, if hf admits a right adjoint then so does f .
- Left adjoints preserve colimits, and right adjoints preserve limits.
- Let $\mathcal{C}, \mathcal{D}$ fibrant simplicial categories with $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{C}$ adjoints. Then the induced functors NF and NG are adjoints.

- In particular, let $\mathbf{A}$ and $\mathbf{A}'$ be simplicial model categories with F, G a pair of a simplicial Quillen adjunction. Let $\mathcal{M}$ be the correspondence between (F, G) , and $\mathcal{M}^\circ$ be the full subcategory of fibrant-cofibrant objects. Then $N\mathcal{M}^\circ$ gives an adjunction between $N\mathbf{A}^\circ$ and $N\mathbf{A}'^\circ$. We say the induced adjunction between $N\mathbf{A}^\circ$ and $N\mathbf{A}'^\circ$ to be the left and right derived functors of the Quillen adjunction.
- Assume $\mathcal{C}$ admits pullbacks and $C \in \mathcal{C}$, let (F, G) be an adjunction between $\mathcal{C}$ and $\mathcal{D}$. Then the induced functor $\mathcal{C}^{C/} \rightarrow \mathcal{D}^{FC/}$ admits a right adjoint $\mathcal{D}^{FC/} \rightarrow \mathcal{C}^{GFC/} \rightarrow \mathcal{C}^{C/}$ where the second map is induced by the unit $C \rightarrow GFC$.
- Note that we have a colimit-preserving map $\mathcal{P}(f) : \mathcal{P}(\mathcal{C}) \rightarrow \mathcal{P}(\mathcal{C}')$ given a map $f : \mathcal{C} \rightarrow \mathcal{C}'$, which corresponds to the composition of f with the Yoneda embedding: $\mathcal{C}' \rightarrow \mathcal{P}(\mathcal{C}')$. $\mathcal{P}(f)$ is the left adjoint to the natural map $\mathcal{P}(\mathcal{C}') \rightarrow \mathcal{P}(\mathcal{C})$.
- Suppose we want to localize $\mathbf{Ab}$ by maps with kernel and cokernel p -torsion. We can take the full subcategory of uniquely p -divisible groups, and this gives a localization of $\mathbf{Ab}$ with the size controlled. The localization is given by $A \rightarrow A \otimes \mathbb{Z}[1/p]$, and it is left adjoint to the inclusion. We can do the similar for $\mathbf{Sp}$ via the Bousfield localization by tensoring with $\mathbb{S}[1/p]$. We want to do this for ∞ -categories in general.
- (Localization). A functor $f : \mathcal{C} \rightarrow \mathcal{D}$ is a localization if it admits a fully faithful right adjoint g . We identify $\mathcal{D}$ with its image in $\mathcal{C}$ via g , and let $L = g \circ f : \mathcal{C} \rightarrow \mathcal{C}$.
 - L is a localization iff L is left adjoint to the inclusion $LC \rightarrow \mathcal{C}$, iff there exists a natural transformation $\mathcal{C} \times \Delta^1 \rightarrow \mathcal{C}$ from id to L such that $LC \rightarrow LLC$ is an equivalence for each $C \in \mathcal{C}$.
 - Localization satisfies the universal property in the sense that $\text{Fun}(LC, \mathcal{D}) \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D})$ is fully faithful, and the essential image contains those functors $\mathcal{C} \rightarrow \mathcal{D}$ where any map f in $\mathcal{C}$ such that Lf is an equivalence is sent to an equivalence in $\mathcal{D}$.
- (Orthogonal Maps and Factorization System). A map of set can be (uniquely) factorized as the composition of a surjective map followed by an injective map. We want to do the same for ∞ -categories. In particular, say $f \perp g$ if for every commutative diagram in $\mathcal{C}$

$$\begin{array}{ccc} A & \longrightarrow & X \\ f \downarrow & & \downarrow g \\ B & \longrightarrow & Y \end{array}$$

$\text{Map}_{\mathcal{C}_{A/Y}}(B, X)$ is contractible. Or in other words, the space of lifts $B \rightarrow X$ is contractible.

- Now, (S_L, S_R) is called a factorization system where S_L and S_R are collections of morphisms in $\mathcal{C}$ if every map in S_L is left orthogonal to every map in S_R , they are closed under retracts, and any map in $\mathcal{C}$ can be factored to a composition of a map in S_L and a map in S_R up to a 2-homotopy. In fact, $S_L = {}^\perp S_R$ and $S_R = S_L^\perp$.
- In fact, such factorizations are canonical, in the sense that $\text{Fun}'(\Delta^2, \mathcal{C}) \rightarrow \text{Fun}(\Lambda_0^2, \mathcal{C})$ is a trivial Kan fibration, where the former represents the full subcategory of $\text{Fun}(\Delta^2, \mathcal{C})$ where the upper two edges are sent to maps in S_L and S_R .
- Filtered topological or fibrant simplicial categories are defined analogous to how filtered categories are defined.
- An ∞ -category $\mathcal{C}$ is κ -filtered if for every κ -small simplicial set K and a map $K \rightarrow \mathcal{C}$, there exists an extension $K^\triangleright \rightarrow \mathcal{C}$. An ω -filtered ∞ -category is said to be filtered. In particular, a topological category is filtered iff its nerve is.
 - In other words, every κ -small diagram in a κ -small category admits an upper bound.

- If $\mathcal{C}$ is κ -filtered then there exists a κ -filtered poset A with $NA \rightarrow \mathcal{C}$ cofinal. $\mathcal{C}$ is κ -filtered iff $\mathcal{C} \rightarrow \text{Fun}(K, \mathcal{C})$ is cofinal for every κ -small simplicial set K .
- (Right Exact Functors). For Abelian categories admitting finite colimits, a functor is right exact iff it preserves finite colimits. For a functor between ∞ -categories, F is κ -right exact if for any right fibration $\mathcal{B}' \rightarrow \mathcal{B}$ where $\mathcal{B}'$ is κ -filtered, $\mathcal{A} \times_{\mathcal{B}} \mathcal{B}'$ is κ -filtered. In particular, if $\mathcal{A}$ admits κ -small colimits, then a functor is κ -right exact iff it preserves κ -small colimits.
- An ∞ -category is κ -closed if every diagram $p : K \rightarrow \mathcal{C}$ where K is κ -small admits a colimit.
 - This gives a way for defining the “colimit functor”. More generally, let $\mathcal{C}$ be an ∞ -category and K a simplicial set, such that every map $K \rightarrow \mathcal{C}$ has a colimit $K^{\triangleright} \rightarrow \mathcal{C}$, then we can define $\mathcal{D} \subset \text{Fun}(K^{\triangleright}, \mathcal{C})$ the subcategory spanned by colimit diagrams. Now, we can define $\text{colim}_K : \text{Fun}(K, \mathcal{C}) \rightarrow \mathcal{D} \subset \text{Fun}(K^{\triangleright}, \mathcal{C}) \rightarrow \mathcal{C}$, where the first map comes from the section of the trivial fibration $\mathcal{D} \rightarrow \text{Fun}(K, \mathcal{C})$ and the last map comes from evaluating at the cone point. This gives a colimit in $\mathcal{C}$ for each diagram.
 - $\mathcal{J}$ is κ -filtered iff the colimit functor $\text{colim}_{\mathcal{J}} : \text{Fun}(\mathcal{J}, \mathbf{Space}) \rightarrow \mathbf{Space}$ preserves κ -small colimits.
- (Compact Objects). For ordinary categories, an object C is compact if $\text{Hom}(C, -)$ commutes with filtered colimits. Analogously, in an ∞ -category $\mathcal{C}$, an object C is κ -compact if the corepresentable functor $j_C : \mathcal{C} \rightarrow \widehat{\mathbf{Space}}$ is κ -continuous (i.e., preserves κ -filtered colimits) where the latter denotes the category of large spaces. A left fibration $\tilde{\mathcal{C}} \rightarrow \mathcal{C}$ is κ -compact if it is classified by a κ -continuous $\mathcal{C} \rightarrow \widehat{\mathbf{Space}}$.
- (Ind-Object). We call an Ind-object of a category $\mathcal{C}$ to be a diagram $f : \mathcal{J} \rightarrow \mathcal{C}$ where $\mathcal{J}$ is a filtered category. Or we can identify f with $\text{colim } X_i$. This gives a category $\text{Ind}(\mathcal{C})$ which contains $\mathcal{C}$ as a full subcategory. Moreover, any map $\mathcal{C} \rightarrow \mathcal{D}$ that preserves filtered colimits can be extended to a map $\text{Ind}(\mathcal{C}) \rightarrow \mathcal{D}$ whose restriction to $\mathcal{C}$ is isomorphic to the map we begin with, and this extension is unique up to a unique isomorphism. We want to do the same for ∞ -categories.
 - Let $\mathcal{C}$ be the category of finitely presented groups, then $\text{Ind}(\mathcal{C})$ is the category of groups, in the sense that every group is a filtered colimit of finitely presented groups.
 - Note that $\mathcal{P}(\mathcal{C})$ is an ∞ -category freely generated by $\mathcal{C}$ under colimits. We want an analogous construction with filtered colimits. A hope might be to find $\text{Ind}(\mathcal{C})$ inside $\mathcal{P}(\mathcal{C})$: $\text{Ind}_{\kappa}(\mathcal{C})$ is the full subcategory of $\mathcal{P}(\mathcal{C})$ where the functors are classified by right fibrations $\tilde{\mathcal{C}} \rightarrow \mathcal{C}$ where $\tilde{\mathcal{C}}$ is κ -filtered. It is stable under κ -filtered colimits.
 - Suppose $\mathcal{C}$ admits κ -small colimits, then $F \in \text{Ind}_{\kappa}(\mathcal{C})$ iff F preserves κ -small limits. In addition, $j(C)$ is κ -compact in $\text{Ind}_{\kappa}(\mathcal{C})$.
 - In fact, suppose $\mathcal{D}$ admits κ -filtered colimits, there is an equivalence $\text{Map}_{\kappa}(\text{Ind}_{\kappa}(\mathcal{C}), \mathcal{D}) \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D})$ where the LHS is the ∞ -category of all κ -continuous functors. Similar to what we did for $\mathcal{P}(S)$, we can express every $F \in \text{Ind}_{\kappa}(\mathcal{C})$ as a κ -filtered colimit (which is essentially via the definition, because $\tilde{\mathcal{C}} \simeq \mathcal{C}_{/F}^{\text{rep}}$).
- This is actually an universal construction, in the sense that we can replace filtered by any collection of diagrams:
 - Let $\mathcal{K}$ be a collection of simplicial sets. An ∞ -category admits $\mathcal{K}$ -indexed colimits if it admits K -indexed colimits for each $K \in \mathcal{K}$. $\text{Fun}_{\mathcal{K}}(\mathcal{C}, \mathcal{D})$ is the full subcategory of $\text{Fun}(\mathcal{C}, \mathcal{D})$ spanned by functors which preserve $\mathcal{K}$ -indexed colimits.
 - Let $\mathcal{R}$ be a collection of diagrams $K_{\alpha}^{\triangleright} \rightarrow \mathcal{C}$. $\text{Fun}_{\mathcal{R}}(\mathcal{C}, \mathcal{D})$ denotes the full subcategory of $\text{Fun}(\mathcal{C}, \mathcal{D})$ spanned by functors that send diagrams in $\mathcal{R}$ to colimit diagrams in $\mathcal{D}$.
 - There exists an ∞ -category $\mathcal{P}_{\mathcal{R}}^{\mathcal{K}}(\mathcal{C})$ and $j : \mathcal{C} \rightarrow \mathcal{P}_{\mathcal{R}}^{\mathcal{K}}(\mathcal{C})$ such that:

- * $\mathcal{P}_{\mathcal{R}}^{\mathcal{K}}(\mathcal{C})$ admits $\mathcal{K}$ -indexed colimits.
- * If $\mathcal{D}$ admits $\mathcal{K}$ -indexed colimits, then there is an equivalence $\mathrm{Fun}_{\mathcal{K}}(\mathcal{P}_{\mathcal{R}}^{\mathcal{K}}(\mathcal{C}), \mathcal{D}) \rightarrow \mathrm{Fun}_{\mathcal{R}}(\mathcal{C}, \mathcal{D})$.
- * If every diagram in $\mathcal{R}$ is a colimit, then j is fully faithful.
- (Essentially Small). An ∞ -category $\mathcal{C}$ is essentially κ -small if the collection of equivalence classes of objects in $\mathcal{C}$ is κ -small, and for every $f : C \rightarrow D$, $\pi_i(\mathrm{Hom}^R(C, D), f)$ is κ -small.
 - Let $\mathcal{C} \rightarrow \mathcal{D}$ be a functor where the latter is essentially κ -small, to conclude that $\mathcal{C}$ is essentially κ -small, it suffices to have $\mathcal{C}_D$ essentially κ -small for each $D \in \mathcal{D}$.
 - An Kan complex is essentially κ -small if $\pi_i(X, x)$ is κ -small for each i and x .
 - Being essentially κ -small is equivalent to being κ -compact in $\infty\mathbf{Cat}$ or **Space**.
- (Locally Small). Relax the conditions a bit, we say an ∞ -category is locally small if $\mathrm{Map}(X, Y)$ is essentially small for each pair of objects, or if the full subcategory spanned by any small collection of objects is essentially small.
- (Accessible ∞ -Category). The motivation: for an ordinary category $\mathcal{C}$, a functor $F : \mathcal{C}^{\mathrm{op}} \rightarrow \mathbf{Set}$ leads to a fibered category $\tilde{\mathcal{C}} \rightarrow \mathcal{C}$. This fibered category is equivalent to $\mathcal{C}_{/C}$, where C is the colimit of the map $\tilde{\mathcal{C}} \rightarrow \mathcal{C}$ if $\mathcal{C}$ admits colimits. Here set-theoretical issues arise: if we just assume $\mathcal{C}$ admits small colimits then $\tilde{\mathcal{C}}$ is not necessarily small, while if we assume $\mathcal{C}$ small it is not reasonable to expect $\mathcal{C}$ admit small colimits. An accessible category is not small but controlled by some small subcategory $\mathcal{C}^0$, in the sense that $\mathcal{C}' \times_{\mathcal{C}} \mathcal{C}^0$ is small so the colimit is expected to exist. For the ∞ -categorical setting, we define an κ -accessible ∞ -category to be equivalent to $\mathrm{Ind}_{\kappa}(\mathcal{C}^0)$ for some small ∞ -category $\mathcal{C}^0$. We say $\mathcal{C}$ is accessible if it is κ -accessible for some κ .
 - $\mathcal{C}$ is κ -accessible if it is locally small and admits κ -filtered colimits, and the full subcategory $\mathcal{C}^{\kappa}$ of κ -compact objects is essentially small and generates $\mathcal{C}$ under small, κ -filtered colimits.
 - **Space** and $\mathcal{P}(\mathcal{C})$ are accessible for $\mathcal{C}$ small.
- (Accessible Functor). Let $\mathcal{C}$ be accessible. A functor $F : \mathcal{C} \rightarrow \mathcal{C}'$ is accessible if it is κ -continuous for some regular cardinal κ .
- If $\mathcal{C}$ is κ -accessible, then it is κ' -accessible for any $\kappa' \gg \kappa$ but generally not for $\kappa' > \kappa$.
- The category of κ -accessible ∞ -categories is equivalent to the category spanned by idempotent complete ∞ -categories, and idempotent completion denotes the former as a localization of the category of ∞ -categories. In fact, we know that a small ∞ -category is an accessible iff it is idempotent complete.
- Let $\mathcal{C}$ accessible and K small simplicial set, then $\mathrm{Fun}(K, \mathcal{C})$ is also accessible. If $p : K \rightarrow \mathcal{C}$ is a diagram, then $\mathcal{C}_{p/}$ is accessible (and same for $\mathcal{C}_{/p}$). The class of accessible categories is stable under fiber products.
- (Presentable ∞ -Category). An ∞ -category is presentable if it is accessible and admits small colimits.
 - An ∞ -category is presentable iff it is an accessible localization (i.e., an localization such that L is accessible) of $\mathcal{P}(\mathcal{D})$ for some $\mathcal{D}$ small. In fact an localization L of an accessible category $\mathcal{C}$ is accessible iff the essential image LC is accessible.
 - If $\mathcal{C}$ is κ -accessible, then $\mathcal{C}$ is presentable iff the full subcategory $\mathcal{C}^{\kappa}$ admits κ -small colimits. Roughly speaking this is because small colimits can be written as κ -filtered colimits and κ -small colimits of κ -compact objects. Similarly, assume $\mathcal{C}$ and $\mathcal{D}$ accessible, and the former κ -accessible, $f : \mathcal{C} \rightarrow \mathcal{D}$ preserves small colimits iff it is κ -continuous and $f|_{\mathcal{C}^{\kappa}}$ preserves κ -filtered colimits.

- $F \in \mathcal{P}(\mathcal{C})$ is representable if it is in the essential image of the Yoneda embedding. Now if F is representable, it preserves small limits. If $\mathcal{C}$ is presentable, the converse is true. So a presentable ∞ -category admits all small limits.
 - F is corepresentable iff F is accessible and preserves small limits.
 - (Adjoint Functor Theorem). A functor between presentable ∞ -categories has a right (left) adjoint iff it preserves small colimits (limits and is in addition accessible).
- We define $\mathbf{Pr}^{\mathbf{L}}, \mathbf{Pr}^{\mathbf{R}} \subset \infty\mathbf{Cat}$ to be the subcategories of presentable ∞ -categories, where functors in the former are those that preserve small colimits, and the functors in the latter are those that are accessible and preserve small limits.
- A presentable fibration is a morphism that is both Cartesian and coCartesian with the fibers presentable ∞ -categories. The equivalence classes of presentable fibrations $X \rightarrow S$ can be identified with $[S, \mathbf{Pr}^{\mathbf{L}}]$ or with $[S^{\mathrm{op}}, \mathbf{Pr}^{\mathbf{R}}]$, and in particular $\mathbf{Pr}^{\mathbf{L}}$ and $\mathbf{Pr}^{\mathbf{R}^{\mathrm{op}}}$ are equivalent in the homotopy category of $\infty\mathbf{Cat}$.
- Presentable categories are stable under many basic constructions, including:
 - products,
 - $\mathrm{Fun}(K, -)$'s for K small simplicial set,
 - $\mathrm{Fun}^L(\mathcal{C}, \mathcal{D})$ (which can be viewed as the internal hom-set in $\mathbf{Pr}^{\mathbf{L}}$),
 - $\mathcal{C}_{/p}$ for p small diagram,
 - fiber products with morphisms in $\mathbf{Pr}^{\mathbf{L}}$.
- $\mathbf{Pr}^{\mathbf{L}}$ and $\mathbf{Pr}^{\mathbf{R}}$ admit small limits and the inclusion functors preserve all small limits. Essentially, limits and colimits (first dualize to $\mathbf{Pr}^{\mathbf{R}}$ and then compute as limits) can be computed in $\infty\mathbf{Cat}$.
- A localization is determined up to equivalence by the collection of morphisms S that are mapped to equivalences. Such a collection arises from a accessible localization iff it is strongly saturated and of small generation. Moreover, given any collection S , we can find the smallest strongly saturated collection that contains S . This allows us to define $S^{-1}\mathcal{C}$ for any collection of morphisms S .
 - For any S , the full subcategory $S^{-1}\mathcal{C}$ of S -local objects is a localization of $\mathcal{C}$ and every localization arises that way. $S^{-1}\mathcal{C}$ and $T^{-1}\mathcal{C}$ coincide iff they generate the same strongly saturated collection of morphisms. (Suppose $\mathcal{C}$ accessible) The localization is accessible iff $S^{-1}\mathcal{C}$ is, or iff there exists a small subset of S that generates the same strongly saturated class of maps.
 - As expected, let $\mathcal{C}$ presentable, then $\mathrm{Fun}^L(S^{-1}\mathcal{C}, \mathcal{D}) \rightarrow \mathrm{Fun}(\mathcal{C}, \mathcal{D})$ is fully faithful, with the essential image being those that send S to equivalences.
 - Note that $L\mathcal{C}$ is accessible iff it is presentable.
- A similar phenomena holds for factorization systems: Let $\mathcal{C}$ presentable, S a saturated collection of maps that is also of small generation, then $(S, S^{\perp})$ gives a factorization system. Note that the converse need not hold.
- (Truncated Objects). Recall that in classical topology, we say a space X is k -truncated if all its homotopy groups of dimension higher than k vanish. A nice space X admits an essentially unique Postnikov tower: $X \rightarrow \dots \rightarrow \tau_{\leq n}X \rightarrow \dots \rightarrow \tau_{\leq -1}X$, where $\tau_{\leq k}X$ is k -truncated and is universal. $\tau_{\leq 0} = \pi_1 X$ and $\tau_{\leq 1}X = B\pi_1 X$. X can be recovered by taking homotopy limit of the tower. We can do the same for an object in an ∞ -category.
 - A Kan complex X is k -truncated if $\mathrm{Map}(S, X)$ is for every small simplicial set S . $C \in \mathcal{C}$ is k -truncated if for every $D \in \mathcal{C}$, $\mathrm{Map}(D, C)$ is.

- A morphism is k -truncated if $\text{Map}(E, C) \rightarrow \text{Map}(E, D)$ is for every E .
- Let $\mathcal{C}$ presentable, $\tau_{\leq k}\mathcal{C} \subset \mathcal{C}$ admits an accessible left adjoint, which is denoted as $\tau_{\leq k}$.
- **Space** is generated by compact projective objects under colimits. Having enough compact projective objects allows us to define projective resolution, which gives rise to non-abelian homological (or homotopical) algebra.
- An ∞ -category is κ -compactly generated if it is presentable and κ -accessible; in particular, iff there exists some small ∞ -category $\mathcal{D}$ that admits κ -small colimits, and that $\mathcal{C} \simeq \text{Ind}(\mathcal{D})$.
 - A localization L is κ -continuous iff $L\mathcal{C}$ is stable under κ -filtered colimits. If such, then $L\mathcal{C}$ is κ -compactly generated, and L carries κ -compact objects to κ -compact objects.

7 Characteristic Classes

7.1 Vector Bundles

- A n -dim. v.b. is trivial iff there are n linearly independent sections. For example, S^1 and S^3 are parallelizable.
 - We denote the tautological bundle over $\mathbb{P}^n$ to be γ_n^1 .
 - * $E(\gamma_n^1)$ is exactly the Möbius band so $\mathbb{P}^1$ is not parallelizable.
- A v.b. is Euclidean if there exists continuous $\mu : E \rightarrow \mathbb{R}$ such that the restriction to each fiber is positive definite and quadratic.
 - An Euclidean metric on TM would be the same as a Riemannian metric.
- (Whitney Sum): Let ξ_1, ξ_2 be two v.b.'s over B . We define $\xi_1 \oplus \xi_2$ via $\Delta^*(\xi_1, \xi_2)$ where Δ is the diagonal map. Note that each fiber is exactly the direct sum of the two summands' fibers.
 - $\xi_x \simeq (\xi_1)_x \oplus (\xi_2)_x$ implies $\xi \simeq \xi_1 \oplus \xi_2$.
 - Let $\xi \subset \eta$ be two v.b.'s over X with Euclidean metric on η . We can define $\xi^\perp$ fiberwise, and $\eta \simeq \xi \oplus \xi^\perp$.
 - In particular, let $M \subset N$ Riemannian, then $TM \oplus \nu \simeq f^*TN$.
 - $\bigwedge^k V$ is defined fiberwise, and is of rank C_n^k .
 - In fact, given a functor $\mathbf{Vect}_{\mathbb{R}} \times \dots \times \mathbf{Vect}_{\mathbb{R}} \rightarrow \mathbf{Vect}_{\mathbb{R}}$, we can always define a functor from k vector bundles to a vector bundle fiberwise, and the topology on the total space is canonical.

7.2 Stiefel-Whitney Class

- We can define the Stiefel-Whitney classes $(w_i(\xi)) \in H^i(B; \mathbb{Z}/2\mathbb{Z})$ for a vector bundle $\xi \rightarrow B$ such that
 - w_i is natural. $w_k(\xi) = 0$ if $k > \text{Rank}(\xi)$.
 - $w_0(\xi) = 1 \in H^0(B; \mathbb{Z}/2\mathbb{Z})$.
 - $w_k(\xi \oplus \eta) = \sum w_i(\xi) \cup w_j(\eta)$.
 - $w_1(\gamma_1^1) \neq 0$.

In particular, these indicate:

- $w_i(\epsilon) = 0$ if ϵ trivial and $i > 0$.

- If ξ has k linearly independent sections, then $w_n(\xi) = \dots = w_{n-k+1}(\xi) = 0$.
- Let $H\mathbb{P}(B; \mathbb{Z}/2\mathbb{Z})$ be the ring with all formal infinite series $a = a_1 + \dots$, then $w(\xi)$ has an inverse, denoted $\bar{w}(\xi)$ and $w(\xi) = \bar{w}(\eta)$ if $\xi \oplus \eta$ is trivial. In particular, $w(\nu) = \bar{w}(TM)$ for $M \rightarrow N$ embedding.
- $w(S^n) = \bar{w}(\nu_{S^n}) = 1$.
- $H^i(\mathbb{RP}^n; \mathbb{Z}/2\mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}$ with generator $a_i = a_1^i$ for $0 \leq i \leq n$ and $w(\gamma_n^1) = 1 + a$. $w(\gamma^\perp) = 1 + a + \dots + a^n$.
- $T\mathbb{RP}^n \simeq \text{Hom}(\gamma_n^1, \gamma^\perp)$. In particular, $T\mathbb{RP}^n \oplus \epsilon^1 = \text{Hom}(\gamma_n^1, \gamma^\perp) \oplus \text{Hom}(\gamma_n^1, \gamma_n^1) = (\gamma_n^1)^{n+1}$. Therefore, $w(\mathbb{RP}^n) = (1 + a)^{n+1}$.
 - It is easy to show that $w(T\mathbb{RP}^n) = 1$ iff $n + 1 = 2^k$; this shows that most projective spaces are not parallelizable.
 - This also shows that the bound for the Whitney immersion theorem is optimal.
- Let $[M] \in H_n(M; \mathbb{Z}/2\mathbb{Z})$ be a fundamental class, we can define $H^n(M; \mathbb{Z}/2\mathbb{Z}) \rightarrow \mathbb{Z}/2\mathbb{Z}$ via evaluating at $[M]$. The Stiefel-Whitney numbers are defined to be $\langle w_1(TM)^{r_1} \dots w_n(TM)^{r_n}, [M] \rangle$ for each $(r_1, \dots, r_n)$ that makes the dimension right.
 - In fact one can show that M can be realized as the boundary of some smooth compact manifold iff all Whitney numbers vanish.
- $w_1(\xi) = 0$ iff ξ is orientable: we can classify ξ as a map $f \in [X, BO(n)]$, and to lift f to $[X, BSO(n)]$, it suffices to show that its image in $[X, B\mathbb{Z}/2\mathbb{Z}]$ is zero via the SES of groups $SO(n) \rightarrow O(n) \rightarrow \mathbb{Z}/2\mathbb{Z}$. Now, via some diagram chasing we can show that it reduces to $w_1(\xi) = 0$.
- $\text{Spin}(n) \rightarrow SO(n)$ is the universal cover. We now have a fiber sequence $B\text{Spin}(n) \rightarrow BSO(n) \rightarrow K(\mathbb{Z}/2\mathbb{Z}, 2)$ and to lift $f \in [X, BSO(n)]$ to $[X, B\text{Spin}(n)]$, we need exactly the composition in $[X, K(\mathbb{Z}/2\mathbb{Z}, 2)]$ to be zero, or if $w_1(\xi) = w_2(\xi) = 0$, so this happens iff ξ has a spin structure.
- Let M^1 or $M^k \subset \mathbb{R}^{k+1}$, we can define natural maps $M^1 \rightarrow S^k$ or $M^k \rightarrow S^k$. Generalizing these gives $M^k \subset \mathbb{R}^{n+k}$ a map $M^k \rightarrow \text{Gr}_k(\mathbb{R}^{n+k}) \simeq \text{Gr}_n(\mathbb{R}^{n+k})$. This leads to a map of v.b.'s $TM \rightarrow \gamma^n(\mathbb{R}^{n+k})$. In fact, we can do this for every rank n ξ over M assuming compactness conditions. Taking colimits gives the infinite Grassmanian Gr_n , which classifies any rank n bundles up to homotopy.
- We now attempt to compute the $\mathbb{Z}/2\mathbb{Z}$ cohomology ring of Gr_n .
 - (Cell Structure). Let a Schubert symbol $\sigma = (\sigma_1, \dots, \sigma_n)$ be a sequence of integers $1 \leq \sigma_1 < \dots < \sigma_n \leq m$ and $e(\sigma) \subset \text{Gr}_n(\mathbb{R}^m)$ be the collection of $V \in \text{Gr}_n(\mathbb{R}^m)$ such that $\dim X \cap \mathbb{R}^{\sigma_i} = \dim X \cap \mathbb{R}^{\sigma_{i-1}} + 1 = i$. In other words, the sequence of integers $(\dim X \cap \mathbb{R}^j)_{1 \leq j \leq m}$ jumps exactly at each σ_i .
 - * Note that each $X \in e(\sigma)$ contains a unique orthonormal basis $(x_1, \dots, x_n) \in \mathbb{H}^{\sigma_1} \times \dots \times \mathbb{H}^{\sigma_n}$. It can then be shown that $e(\sigma)$ is an open cell of dimension $\sigma_1 - 1 + \sigma_2 - 2 + \dots + \sigma_n - n$.
 - * It can then be shown that the $\binom{m}{n}$ cells $e(\sigma)$ gives a cell structure on $\text{Gr}_n(\mathbb{R}^m)$, and letting $m \rightarrow \infty$ gives a cell structure on $\text{Gr}_n(\mathbb{R}^\infty)$.
 - * We can also show that the number of r -cells in $\text{Gr}_n(\mathbb{R}^m)$ is the number of partitions of r into at most n integers each no greater than $m - n$.
- We now claim that $H^*(\text{Gr}_n; \mathbb{Z}/2\mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}[w_1, \dots, w_n]$ where each $w_i = w_i(\gamma^n)$.

- To show that it contains the polynomial algebra, we need to show the w_i 's are algebraically independent. It suffices to show that after pulling back to some rank n bundle, for example the n -fold product of γ_n^1 on $\mathbb{RP}^\infty$. It's easy to see that $w(\prod \mathbb{RP}^n) = (1 + a_1) \dots (1 + a_n)$ so that each w_k is the symmetric function of degree k , and they are algebraically independent.
- Now we can show the two algebra agree by counting the dimensions: the number of r -cells is bounded, so is the rank of the r -th cohomology group according to cellular cohomology.
- (Thom Isomorphism). There exists $u \in H^n(E, E_0; \mathbb{Z}/2\mathbb{Z})$ such that u restricts to the unique non-zero class in each fiber (F, F_0) . Cup product with u gives an isomorphism $H^i(E) \simeq H^{i+n}(E, E_0)$. Now the Thom isomorphism is the composition of isomorphisms $H^i(B) \rightarrow H^i(E) \rightarrow H^{i+n}(E, E_0)$.
- (Definition of the Stiefel-Whitney Class). We assume there are Steenrod operations $\text{Sq}^i : H^n(X, Y) \rightarrow H^{n+i}(X, Y)$ that satisfies various axioms. We define $w_i(\xi) = \phi^{-1} \text{Sq}^i(u)$ where ϕ is the Thom isomorphism.

7.3 Orientation and Euler Class

- For an oriented vector space V we can define a generator $\mu_V \in H_n(V, V_0; \mathbb{Z})$ and hence a generator $u_V \in H^n(V, V_0; \mathbb{Z})$. Therefore, for an oriented bundle, we can find a generator $u_F \in H^n(F, F_0; \mathbb{Z})$ for each fiber F , and locally the choice of these u_F 's should glue to a cohomology class u .
 - In fact we can show that there uniquely exists globally $u \in H^n(E, E_0; \mathbb{Z})$ that restricts to the generator of each fiber. Cup product with u gives an isomorphism $H^k(E; \mathbb{Z}) \simeq H^{n+k}(E, E_0; \mathbb{Z})$.
 - The proof idea: we show that $H^n(B \times \mathbb{R}^n, B \times \mathbb{R}_0^n) \simeq H^0(B)$, and the class $1 \times e^n$ corresponding to $1 \in H^0(B)$ satisfies the condition. Now, we can glue two classes together via Mayer-Vietoris.
- (Euler Class). We define $e(\xi) \in H^n(B; \mathbb{Z})$ to be the class that corresponds to $u|_E \in H^n(E; \mathbb{Z})$ via the isomorphism $H^n(B; \mathbb{Z}) \simeq H^n(E; \mathbb{Z})$ (given by the deformation retract).
 - $e(\xi)$ is natural w.r.t. orientation-preserving bundle maps and changes sign if the orientation is reversed.
 - We can see that $\phi(e(\xi)) = u \cup u$ where ϕ is the Thom isomorphism. In particular, we know that $2e(\xi) = 0$ if $\text{Rank}(\xi)$ is odd.
 - Since $u \cup u = \text{Sq}^n(u)$, the image of $e(\xi)$ under the map $H^n(B; \mathbb{Z}) \rightarrow H^n(B; \mathbb{Z}/2\mathbb{Z})$ is $w_n(\xi)$.
 - $e(\xi \oplus \eta) = e(\xi) \cup e(\eta)$ and $e(\xi \times \eta) = e(\xi) \times e(\eta)$.
 - If ξ has a nowhere zero section, then $e(\xi) = 0$.
- A complex vector space has a canonical real orientation, because any complex basis $(a_j + ib_j)$ gives a real basis $(a_1, b_1, a_2, \dots, a_n, b_n)$ and any complex linear automorphism respects this real orientation. In particular, any complex n -bundle can be viewed as a real $2n$ -bundle and is automatically real orientable.
- Let $M^n \subset A^{n+k}$.
 - (Tubular Neighborhood). There exists an open neighborhood N_ϵ that is diffeomorphic to the total space of the normal bundle over M in A .
 - This gives $H^*(A, A - M) \simeq H^*(N_\epsilon, N_\epsilon - M) \simeq H^*(E, E_0)$.

- Consider the composition $H^k(A, A - M) \rightarrow H^k(A) \rightarrow H^k(M)$. The image of the fundamental class u' in $\mathbb{Z}/2\mathbb{Z}$ coefficient (or $\mathbb{Z}$ coefficient if ν_M is orientable) is $w_k(\nu^k)$ (or $e(\nu_k)$ if orientable).
- The image of u' in $H^k(A)$ is called the dual cohomology class of M in the sense of Poincaré duality. For instance, if $M^n \subset \mathbb{R}^{n+k}$, then $w_k(\nu_M)$ or $e(\nu_M) = 0$ as the dual cohomology class in $H^k(\mathbb{R}^{n+k})$ is zero.
- Note that M is orientable iff TM is orientable: $H^n(M, M - x) \simeq H^n(T_x M, T_x M - 0)$ via the results above. We can now push the generator that defines the orientation from one to the other by local arguments.
- (Slant Product). Consider $H^*(X) \otimes H^*(Y) \otimes H_*(Y) \rightarrow H^*(X)$. Now replace $H^*(X) \otimes H^*(Y)$ by $H^*(X \times Y)$ gives $H^*(X \times Y) \otimes H_*(Y) \rightarrow H^*(X)$, where the image of $p \otimes \beta$ is denoted as p/β .
 - Let the image of $u' \in H^n(M \times M, M \times M - M)$ to be $u'' \in H^n(M \times M)$. $u''/\mu = 1 \in H^0(M)$.
- Assume M orientable (or coefficient $\mathbb{Z}/2\mathbb{Z}$). We can define dual basis $(b_j^\#)$ for each basis (b_j) such that $\langle b_i \cup b_j^\#, \mu \rangle = \delta_{ij}$. This gives an identification $H^k(M) \simeq H^{n-k}(M)$. (Note that such an isomorphism is not canonical when M is not orientable.) Computation gives that $u'' = \sum (-1)^{\dim b_i} b_i \times b_i^\#$. This tells $e(TM) = \sum (-1)^{\dim b_i} b_i \cup b_i^\#$, which tells that $\langle e(TM), \mu \rangle = \sum (-1)^{\dim b_i} = \chi(M)$. (Or replace everything by $w_n(TM)$ and modulo 2. Note that $TM \simeq \nu_M$ where the latter is its normal bundle in $M \times M$).
- (Wu Class). Consider the homomorphism sending $x \in H^{n-k}(M)$ to $\langle \text{Sq}^k(x), \mu \rangle \in \mathbb{Z}/2\mathbb{Z}$. This is represented by $v_k \in H^k(M)$. $v = \sum v_i$ is the total Wu class, which satisfies $\langle v \cup x, \mu \rangle = \langle \text{Sq}(x), \mu \rangle$ where $\text{Sq} = \sum \text{Sq}^i$. Now $w_k = \sum_{i+j=k} \text{Sq}^i(v_j)$.

7.4 Local Systems and Cohomology

- A local system on X is a locally constant sheaf. If X path-connected, then equivalently, a local system with constant stalk L corresponds to a group homomorphism $\rho : \pi_1(X, x) \rightarrow \text{Aut}(L)$.
 - On one hand, let γ be a loop at x . It is easy to show that local systems over $[0, 1]$ is constant, so in particular is $\gamma^* \mathcal{L}$. This gives an isomorphism $(\gamma^* \mathcal{L})_0 \simeq (\gamma^* \mathcal{L})_1$ via parallel transport, i.e., an automorphism of L . On the other hand, let ρ be such a homomorphism. We consider the universal covering $\tilde{X}$ over X and define the local system via $\mathcal{L}(U) = \{f : \tilde{U} \rightarrow L, f(\gamma y) = \rho(\gamma)f(y)\}$ where $y \in \tilde{U}$ and $\gamma \in \pi_1(X)$.
 - We can define $H^j(X; \mathcal{L})$ via sheaf cohomology.

7.5 Obstructions

- Consider the fiber bundle $V_k(\xi)$ with fibers $V_k(F)$. There exists a cross-section over the $(n - k)$ -skeleton of B , and a cross-section over the $(n - k + 1)$ -skeleton exists iff certain class $\mathfrak{o}_j(\xi) \in H^j(B; \{\pi_{j-1} V_{n-j+1}(F)\})$ where $\{\pi_{j-1} V_{n-j+1}(F)\}$ is the local coefficient system where over each x there is $\pi_{j-1} V_{n-j+1}(F_x)$ where $j = n - k + 1$. The reduction of $\mathfrak{o}_j(\xi)$ to $\mathbb{Z}/2\mathbb{Z}$ -coefficient gives $w_j(\xi)$.
 - In fact the obstructions $\mathfrak{o}_j(\xi)$ are completely determined by the Stiefel-Whitney classes.
 - If ξ oriented, then $\mathfrak{o}_n(\xi) = e(\xi)$.
- (Gysin). Suppose ξ oriented. There is a LES:

$$\dots \longrightarrow H^k(B) \xrightarrow{\cup e} H^{n+k}(B) \longrightarrow H^{n+k}(E_0) \longrightarrow H^{k+1}(B) \longrightarrow \dots$$

This comes from the LES of cohomology group via the sequence $E_0 \rightarrow E \rightarrow (E, E_0)$ and the identification $H^*(B) \simeq H^*(E)$ and $H^{*+n}(E, E_0) \simeq H^*(B)$. The analogous holds for unoriented ξ .

- In particular consider the 2-fold cover $\tilde{B}$ and the line bundle over B with total space $\tilde{B} \times \mathbb{R}/((x, t) \sim (x', -t))$. We get a LES

$$\dots \longrightarrow H^k(B) \xrightarrow{\cup w_1} H^{k+1}(B) \longrightarrow H^{k+1}(\tilde{B}) \longrightarrow H^{k+1}(B) \longrightarrow \dots$$

- The 2-fold cover $\tilde{\text{Gr}}_n$ and the universal bundle $\tilde{\gamma}_n$ classifies oriented n -bundles. $H^*(\tilde{\text{Gr}}_n; \mathbb{Z}/2\mathbb{Z}) \simeq \mathbb{Z}/2\mathbb{Z}[w_2, \dots, w_n]$ via the sequence above.

7.6 Complex Vector Bundles and Chern Class

- A complex structure on a $2n$ -dimensional v.b. can be viewed as a $\mathbb{R}$ -linear map $J : E(\xi) \rightarrow E(\xi)$ such that $J(J(v)) = -v$ for each v . This in particular defines the action of $\mathbb{C}$ on $E(\xi)$ via $i \cdot v = J(v)$. Let ω be a complex bundle, we denote the underlying real bundle as $\omega_{\mathbb{R}}$.
- A complex function $f : U \rightarrow U'$ where $U \subset \mathbb{C}^n$ and $U' \subset \mathbb{C}^m$ is holomorphic if it is holomorphic coordinate-wise. On the other hand, it is holomorphic iff the induced maps on tangent spaces are complex linear. Implicitly, being complex linear is equivalent to satisfying the Cauchy-Riemann equations.
- A complex manifold is a manifold with a complex structure J on TM , such that locally it is diffeomorphic to an open subset of $\mathbb{C}^n$ such that the derivative is complex linear everywhere ($dh \circ J = J_0 \circ dh$). A map between two complex manifolds is holomorphic if Df is complex linear everywhere.
- Note that a complex vector bundle automatically has an orientation (by taking a basis $(v_1, iv_1, \dots, v_n, iv_n)$ for a basis $(v_1, \dots, v_n)$), since a complex linear transformation preserves real orientation. We can therefore talk about $e(\omega_{\mathbb{R}}) \in H^{2n}(B; \mathbb{Z})$.
- A Hermitian metric is an Euclidean metric such that $|iv| = |v|$, and this defines an inner product such that $\langle v, w \rangle = \overline{\langle w, v \rangle}$.
- (Chern Class). We define a $(n-1)$ -complex bundle ω_0 over E_0 by letting the fiber over (x, v) to be the orthogonal complement of v in E_x . Now, recall that Gysin sequence gives $\dots \rightarrow H^{i-2n}(B) \rightarrow H^i(B) \rightarrow H^i(E_0) \rightarrow H^{i+1-2n}(B) \rightarrow \dots$, so in particular $\pi_0^* : H^i(B) \simeq H^i(E_0)$ if $i < 2n-1$. We now define $c_n(\omega) := e(\omega)$ and $c_i(\omega) := (\pi_0^*)^{-1}c_i(\omega_0)$. For $i > n$ $c_i(\omega) := 0$.

- It is easily shown that the Chern classes are natural and invariant under $\oplus \epsilon^k$.
- To compute the cohomology of $\mathbb{CP}^k$, note that we have Gysin sequence:

$$\dots \longrightarrow H^i(E_0) \longrightarrow H^i(\mathbb{CP}^k) \xrightarrow{\cup c_1} H^{i+2}(\mathbb{CP}^k) \longrightarrow H^{i+1}(E_0) \longrightarrow \dots$$

Since $E_0 \simeq S^{2k+1}$, we have that $H^i(\mathbb{CP}^k) \simeq H^{i+2}(\mathbb{CP}^k)$ which is 0 if i odd and $\mathbb{Z}$ (generated by $c_1^{i/2}$) if i even. In particular, $H^*(\mathbb{CP}^n; \mathbb{Z}) = \mathbb{Z}[c_1]/(c_1^{n+1})$. Let $n \rightarrow \infty$ gives $H^*(\mathbb{CP}^\infty) = \mathbb{Z}[c_1]$.

- In fact, $H^*(\text{Gr}_n) = \mathbb{Z}[c_1, \dots, c_n]$. This is proved via induction and Gysin sequence. As expected, Gr_n classifies rank n complex bundles, in a unique way up to homotopy.
- $c(\omega \oplus \eta) = c(\omega)c(\eta)$, just as in the real case.
- Let ω be a complex bundle, we can define its conjugate bundle $\bar{\omega}$ via the opposite complex structure. It is not guaranteed that $\omega \simeq \bar{\omega}$. $c(\bar{\omega}) = 1 - c_1(\omega) + c_2(\omega) - \dots \pm c_n(\omega)$, since the underlying real orientation $(v_1, iv_1, \dots, v_n, iv_n)$ and $(v_1, -iv_1, \dots, v_n, -iv_n)$ differ by a sign $(-1)^n$.

- We can also define the dual bundle $\text{Hom}(\omega, \mathbb{C})$, and if ω admits a Hermitian metric, then $\text{Hom}(\omega, \mathbb{C}) \simeq \bar{\omega}$ via sending v to $\langle -, v \rangle$.
- For example, $c(\tau^n) = (1 + a)^{n+1}$ where τ^n is the tangent bundle over $\mathbb{CP}^n$ and $a = -c_1(\gamma^1)$: Let ω^n be the orthogonal complement of γ^1 over $\mathbb{CP}^{n+1}$, we have $\tau^n \simeq \text{Hom}(\gamma^1, \omega^n)$, and adding ϵ^1 on both sides gives $\tau^n \oplus \epsilon^1 \simeq \text{Hom}(\gamma^1, (\epsilon^1)^{n+1})$, so $c(\tau^n) = c(\text{Hom}(\gamma^1, \epsilon^1)^{n+1}) = (1 - c_1(\gamma^1))^{n+1}$.

7.7 Pontryagin Class

- The complex structure on $\xi \otimes \mathbb{C}$ is via $J(x, y) = (-y, x)$. $\xi \otimes \mathbb{C}$ is canonically isomorphic to $\bar{\xi} \otimes \mathbb{C}$, with the correspondence $(x, y) \rightarrow (x, -y)$. In particular, $c_i(\xi \otimes \mathbb{C})$ is of order 2 for odd i .
- (Pontryagin Class) The i th Pontryagin class is defined to be $p_i(\xi) := (-1)^i c_{2i}(\xi \otimes \mathbb{C}) \in H^{4i}(B; \mathbb{Z})$. $p_i(\xi) = 0$ for $i > n/2$.
 - As for Chern class, $p_i(\xi) = p_i(\xi \oplus \epsilon)$.
 - $p(\xi \oplus \eta)$ is congruent to $p(\xi)p(\eta)$ modulo elements of order 2. In other words, $2(p(\xi \oplus \eta) - p(\xi)p(\eta)) = 0$.
- For a complex bundle ω , we have a natural isomorphism $(\omega_{\mathbb{R}}) \otimes \mathbb{C} \simeq \omega \oplus \bar{\omega}$. In particular, we can compute $c(\omega_{\mathbb{R}} \otimes \mathbb{C}) = c(\omega)c(\bar{\omega})$ and hence get a formula for the Pontryagin classes.
- For an oriented bundle ξ , we have an isomorphism $(\xi \otimes \mathbb{C})_{\mathbb{R}} \simeq \xi \oplus \xi$, which either preserves or reverses the orientation depending on if $n(n-1)/2$ is even or odd. Indeed, the canonical basis for the LHS is $(v_1, iv_1, \dots, v_n, iv_n)$ and for the RHS is $(v_1, \dots, v_n, iv_1, \dots, iv_n)$. In particular, this gives $p_k(\xi) = e(\xi)^2$ if ξ is of rank $2k$.
- (Oriented Grassmanian) $\tilde{\text{Gr}}_n(\mathbb{R}^\infty)$ is the collection of n -dimensional oriented subspaces of $\mathbb{R}^\infty$. It can be viewed as $BSO(n)$.
 - Let Λ be an ID containing $\frac{1}{2}$. $H^*(\tilde{\text{Gr}}_{2m+1})$ is a polynomial ring over Λ generated by $p_1(\tilde{\gamma}^{2m+1}), \dots, p_m(\tilde{\gamma}^{2m+1})$. And $H^*(\tilde{\text{Gr}}_{2m})$ is a polynomial ring over Λ generated by $p_1(\tilde{\gamma}^{2m}), \dots, p_{m-1}(\tilde{\gamma}^{2m})$ and $e(\tilde{\gamma}^{2m})$.
- (Chern Numbers). Let K^n compact complex manifold. Let $I = (i_1, \dots, i_r)$ a partition of n , then the I -th Chern number is $c_I(K^n) = c_{i_1} \dots c_{i_r}[K^n] = \langle c_{i_1} \dots c_{i_r}, \mu_{2n} \rangle$ where μ_{2n} is the fundamental class that gives the orientation.
 - Note that $H^{2n}(\text{Gr}_n)$ has basis $c_{i_1} \dots c_{i_r}$ where $(i_1, \dots, i_r)$ gives a partition. Also, the tangent bundle over K^n is classified by a map $K^n \rightarrow \text{Gr}_n$. Now, the Chern numbers are exactly $\langle c_{i_1}(\gamma^n) \dots c_{i_r}(\gamma^n), f_*(\mu_{2n}) \rangle$ which gives exactly what $f_*(\mu_{2n})$ is.
- (Pontryagin Numbers). We similarly define $p_I[M^{4n}] = \langle p_{i_1} \dots p_{i_r}, \mu_{4n} \rangle$ where μ_{4n} is the fundamental class on M^{4n} .
 - Note that the orientation of a complexified bundle does not depend on the original orientation, so Pontryagin numbers are negated if the orientation changes. In particular, if some Pontryagin numbers are non-zero, then M does not admit an orientation-reversing diffeomorphism.
 - In particular, as the case for Stiefel-Whitney numbers, if M^{4n} has some non-zero Pontryagin numbers, then it cannot be the boundary of some W^{4n+1} .
- Similar to the real case, $H^*(\text{Gr}_n; \mathbb{Z})$ is mapped isomorphically to the ring of symmetric functions in $\mathbb{Z}[a_1, \dots, a_n]$. Let s_I to be the polynomial such that $s_I(\sigma_1, \dots, \sigma_k) = \sum x_1^{i_1} \dots x_r^{i_r}$ where $I = (i_1, \dots, i_r)$ is a partition of k , then $(s_I(c_1, \dots, c_k))_I$ form a basis for $H^{2k}(\text{Gr}_n; \mathbb{Z})$.

- Denote $s_I(c)[K^n] = s_I[K^n] = \langle s_I(c(\tau^n)), \mu_{2n} \rangle \in \mathbb{Z}$. Let $K^1, \dots, K^n$ be complex manifolds with $s_k[K^k] \neq 0$. The matrix $[c_I[K^{j_1} \dots K^{j_r}]]$ where I and $J = (j_1, \dots, j_s)$ ranges over all the partitions of n is non-singular. Similar result holds for the Pontryagin numbers.

7.8 Cobordism

- M and M' lie in the same oriented cobordism class if there exists smooth compact oriented X such that $\partial X \simeq M + (-M')$. Let Ω_n be the Abelian group of all oriented cobordism classes. We have naturally $\Omega_m \times \Omega_n \rightarrow \Omega_{m+n}$. This gives Ω_* a graded ring structure and is commutative in the graded sense.
 - For each partition of k , $M^{4k} \rightarrow p_I[M]$ gives a group homomorphism $\Omega_{4k} \rightarrow \mathbb{Z}$. We can show $\text{Rank } \Omega_{4k} \geq p(k)$ by some computations with the Pontryagin numbers ($\mathbb{CP}^{2i_1} \times \dots \times \mathbb{CP}^{2i_r}$ are linearly independent in Ω_{4k}). In fact, we can show $\text{Rank } \Omega_{4k} = p(k)$.
- (Thom Space). Assume ξ admits an Euclidean metric. $T(\xi) = E(\xi)/A$, where A is the subspace consisting of all points (x, v) where $|v| \geq 1$. $T(\xi)$ has a canonical basepoint, t_0 , so that $T(\xi) - t_0 = \{(x, v), |v| < 1\}$.
 - Assume B is a CW complex, then $T(\xi)$ is a $(k-1)$ connected CW complex, with one $(n+k)$ -cell corresponding to each n -cell of B .
 - Assume ξ is oriented (so Thom isomorphism is effective), then $H_{n+k}(T(\xi), t_0) \simeq H_{n+k}(T(\xi), T_0) \simeq H_{n+k}(E, E_0) \simeq H_n(B)$.
- A map between Abelian groups is an $\mathcal{E}$ -isomorphism if the kernel and cokernel are both finite. In particular, suppose ξ^k is oriented over a finite complex, then $\pi_{n+k}(T) \rightarrow H_{n+k}(T) \simeq H_n(B)$ is an $\mathcal{E}$ -isomorphism.
- Recall that a regular value y of a smooth map $f : M \rightarrow N$ is $y \in N$ such that $T_x M \rightarrow T_y N$ is surjective for each $x \in f^{-1}(y)$. $f^{-1}(y)$ is a $(m-n)$ -submanifold if y is regular and the set of regular values is dense.
 - Note that y is a regular value iff f is transverse to y . Let $Y \subset N$ be an $(n-k)$ -submanifold, then f is transverse to Y if $T_x M + T_y Y \rightarrow T_y N$ is surjective, and if so then $f^{-1}(Y)$ is a $(m-k)$ -submanifold. Normal bundles of Y and $f^{-1}(Y)$ can be identified.
- Every continuous $f : S^m \rightarrow T(\xi)$ is homotopic to some g which is smooth on $g^{-1}(T - t_0)$, and is transverse to the zero-section B . The oriented cobordism class of $g^{-1}(B)$ depends only on the homotopy type of g . In particular, there is a map $\pi_m(T(\xi), t_0) \rightarrow \Omega_{m-k}$.
 - (Thom). For $k > n+1$, $\pi_{n+k}(T(\tilde{\gamma}^k), t_0) \simeq \Omega_n$ canonically. Similarly, $\pi_{n+k}(T(\gamma^k), t_0) \simeq \mathcal{N}_n$ canonically, where $\mathcal{N}_n$ is the unoriented cobordism group.
 - Ω_n is finite if $4 \nmid n$, and is finitely generated of rank $p(k)$ if $n = 4k$. In particular, $\Omega_* \otimes \mathbb{Q}$ is a polynomial algebra over $\mathbb{Q}$ generated by $(\mathbb{CP}^{2i})$.

7.9 Multiplicative Sequences

- (Multiplicative Sequences). Let A^* be a graded Λ -algebra, and A^Π be the ring of formal series in A . A multiplicative sequence is defined as a sequence of polynomials $K_1(x_1), K_2(x_1, x_2), \dots$ such that each K_n is homogeneous of degree n , and that if $K(a) = 1 + K_1(a_1) + \dots$, then $K(ab) = K(a)K(b)$. We give some examples:
 - Recall $c(\bar{\omega}) = 1 - c_1(\omega) + c_2(\omega) - \dots \pm (-1)^n c_n(\omega)$. The polynomials $K_n(x_1, \dots, x_n) = (-1)^n x_n$ gives a multiplicative sequence for this.
 - Recall that if $\eta \oplus \omega = 1$, then $p(\eta)p(\omega) = 1$ (or $c(\eta)c(\omega)$, or $w(\eta)w(\omega)$). The polynomials $K_n = \sum_{i_1+2i_2+\dots+ni_n=n} \frac{(i_1+\dots+i_n)!}{i_1! \dots i_n!} (-x_1)^{i_1} \dots (-x_n)^{i_n}$ characterizes such $(K(a) = a^{-1})$.

- Recall that we have $1 - p_1 + p_2 - \dots \pm p_n = c(\omega)c(\bar{\omega})$. This is characterized by $K_{2n+1} = 0$ and $K_{2n} = x_n^2 - 2x_{n-1}x_{n+1} + \dots \mp 2x_1x_{2n-1} \pm 2x_n$.
- (Hirzebruch). Let $A^* = \Lambda[t]$. For a formal power series $f(t) = 1 + \lambda_1 t + \lambda_2 t^2 + \dots$, there exists a unique multiplicative sequence (K_n) such that $K(1+t) = f(t)$.
- Let K be a multiplicative sequence in $\mathbb{Q}$ -coefficients. For a compact oriented smooth manifold M^m , we define $K([M]) = 0$ if $4 \nmid m$ and $K([M]) = \langle K_n(p_1, \dots, p_n), \mu_{4n} \rangle$ if $4 \mid m$. This gives a ring homomorphism $\Omega_* \rightarrow \mathbb{Q}$.
- (Signature). The signature for a compact oriented smooth manifold M^m is 0 if $4 \nmid m$ and the number of positive entries on the diagonal matrix (after selecting the correct basis (a_i) of $H^{2n}(M^{4n}; \mathbb{Q})$) on $[\langle a_i \cup a_j, \mu \rangle]$ if $4 \mid m$.
 - $\sigma(M + M') = \sigma(M) + \sigma(M')$, $\sigma(M \times M') = \sigma(M)\sigma(M')$. If $M = \partial W$, then $\sigma(M) = 0$.
 - (Signature Theorem). Let (L_k) be the multiplicative sequence belonging to the power series $\sqrt{t}/\tanh \sqrt{t}$. Then $\sigma(M^{4n}) = L([M])$. This only depends on the oriented homotopy type of M .
- Given a multiplicative sequence K with coefficient in Λ , an ID with 2 invertible, we can define $k_n(\xi) = K_n(p_1(\xi), \dots, p_n(\xi)) \in H^{4n}(B; \Lambda)$. This satisfies naturality and that $k(\eta \oplus \omega) = k(\eta)k(\omega)$. On the other hand, multiplicative characteristic classes k_n satisfying such properties come from some unique multiplicative sequence K .